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MIL-STD-001553B(USAF)
w/CHANGE 5
8 February 2006
USED IN LIEU OF
MIL-STD-1553B
w/CHANGE 4
15 January 1996

DEPARTMENT OF DEFENSE
INTERFACE STANDARD

DIGITAL TIME DIVISION COMMAND/RESPONSE MULTIPLEX DATA BUS

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FOREWORD

This notice contains requirements for an Extended Waveform digital time division command/response multiplex data bus for use in systems integration that is capable of coexisting with the 1553B multiplex data bus. Even with the use of this standard, differences may exist between multiplex data buses in different system applications due to particular application requirements and the designer options allowed in this standard. The system designer must recognize this fact and design the multiplex bus controller hardware and software to accommodate such differences.

It should be noted that verification of requirements provides an important part of this notice.

These designer selected options must exist to allow the necessary flexibility in the design of specific multiplex systems in order to provide for the control mechanism, architectural redundancy, degradation concept and traffic patterns peculiar to the specific application requirements.

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1. SCOPE

1.1 Scope.

This notice establishes the performance, interface, and verification requirements for an extended digital waveform multiplexing capability (Extended 1553) that retains backwards compatibility with the 21 September 1978 release and notices 1 through 4.

The overarching requirement is to enable common operability between vendors providing extended digital waveform capability at the Line Replaceable Unit (LRU) or Weapon Replaceable Assembly (WRA) level by providing sufficient information to implement a compliant interface.

1.2 Application.

When invoked in a specification or statement of work, these requirements will apply to the extended digital waveform multiplexing capability (Extended 1553) and associated equipment as installed in the LRU and WRA developed either alone or as a portion of a weapon system or subsystem. The contractor is responsible for invoking all the requirements of this notice on any and all subcontractors he may employ unless otherwise required by the procuring agency.

Extended 1553B Waveform consists of:

a. A broadband Multi Carrier Waveform (MCW) to support real-time applications such as audio, video and non-real time traffic e.g., file transfers.

b. A baseband Filtered Waveform (FW) to support communications currently established by Notice 4 1553B devices and enables the full performance of the Multi Carrier Waveform to be achieved as specified herein.

c. A baseband Trapezoidal Waveform (TW) that is identical to the MIL-STD 1553B (to Notice 4). (see section 5.5).

Communication on the MCW network is similar to that of TW in that messages are controlled by a designated Bus Controller (BC) with all other nodes being designated as Remote Terminals (RT)s. Transmission on both A and B buses with reception on either is supported. FW and TW are to be separate and independent from MCW, including media access functions pertaining to MCW that is not to be initialized, set up, or controlled through FW or TW.

If a terminal is communicating with MCW, the terminal may only transmit FW simultaneously. Either FW or TW may be received simultaneously with MCW.

The MCW frequency band may include the baseband normally assigned to FW or TW if it is unused.

2. APPLICABLE DOCUMENTS

2.1 General.

The documents specified in sections 3, 4 or 5 of this standard are not summarized here since they form the technical basis for performance, interface, and verification. While every effort has been made to ensure the completeness of the list in this section, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed in this section.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461 Electromagnetic Interference Characteristics, Requirements for Equipment

MIL-STD-464 Electromagnetic Environmental Effects Requirements for Systems

Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

2.3 Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Multi Carrier Waveform (MCW) and applicable Open Systems Interconnection (OSI) definitions.

3.1.1 Acknowledgment.

Notification sent from one network device to another to acknowledge that some event occurred (for example, the receipt of a message).

3.1.2 Basic rate (BR).

Lowest rate of data communications used for broadcast and sending control information.

3.1.3 Baseband channel.

Trapezoidal Waveform (TW) or Filtered Waveform (FW) contained in a 0 - 5 MHz Frequency band.

3.1.4 Bus Command Frame (BCF).

A frame broadcasted from the BC that is used to convey MCW commands to all terminals residing on the bus. The set of commands issued by the BC is contained in the payload. (Reference 5.3.4.2.2.2 on page 82).

3.1.5 Bit Error Ratio (BER).

The ratio of the number of bits incorrectly received to the total number of bits transmitted.

3.1.6 Broadband channel.

Multi Carrier Waveform (MCW) contained in a 5 to 40 MHz frequency band that may also extend into the baseband channel band if FW or TW is unused.

3.1.7 Compound Transaction.

A Multi Carrier Waveform (MCW) transaction that consists of a set of commands, terminal responses, and terminal acknowledgements.

3.1.8 Connection oriented protocol.

A means of transmitting data in which the devices at end points use a procedure to establish an end-to-end (virtual) connection before any data is sent. Connection-oriented protocol service guarantees that data will arrive in the proper sequence.

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3.1.9 Connectionless protocol.

A means of transmitting data that does not require a connection to be established before data is transmitted. Messages are transmitted to their destination (end system) independently of any other messages.

3.1.10 Constellation.

A representation of a digital modulation scheme in the complex plane, being a measure of the information carried by a sub-carrier using QAM modulation through choosing a set of complex numbers to represent the modulation symbols.

3.1.11 Cyclic Redundancy Check (CRC).

A method of detecting errors in the serial transmission of data. A CRC for a block of data is calculated before it is sent, and is then sent along with the data. A new CRC is calculated on the received data. If the new CRC does not match the one that has been sent along with the data then an error has occurred.

3.1.12 Data Symbol.

Contained in the payload of a PHY frame coded into blocks of 8 OFDM symbols. Data symbols carry information using Basic Rate or Variable Bit Rate modulation.

3.1.13 Data Link Layer (DLL).

Layer 2 of the OSI reference model that provides data transfer across a physical link. At this layer, data packets are encoded and decoded into bits to be transmitted and received across the physical layer (PHY). The IEEE has further divided this layer into two sublayers: the MAC and LLC sublayers.

3.1.14 Differential Binary Phase Shift Keying (DBPSK).

A method of digital communication in which the phase of a transmitted signal is varied to convey information. It uses two opposite signal phases (0 and 180 degrees). The digital signal is broken up timewise into individual bits (binary digits). The state of each bit is determined according to the state of the preceding bit. DBPSK is used to transmit Frame Control (FC).

3.1.15 Effective stub length increase.

Group delay converted to equivalent increase in cable length.

3.1.16 End system.

Non-routing host or node in an OSI network, which acts as a source of, and/or destination for, data traffic carried on the network.

3.1.17 Encapsulation.

Appending protocol information, in the form of a header and perhaps a footer, to an SDU.

3.1.18 Filtered Waveform (FW).

Waveform referenced in 5.1.3 of this document.

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3.1.19 Frame.

A Multi Carrier Waveform (MCW) Protocol Data Unit (PDU) at the PHY level. (Referenced in 5.3.4.2.2 on page 80).

3.1.20 Frame Control (FC).

Symbols that are used to carry PHY frame information, modulated in such a way that demodulation can be achieved by any receiver without requiring a tone map.

3.1.21 Frame error rate.

The ratio of the number of frames incorrectly received to the total number of frames transmitted.

3.1.22 Frame Sync.

Section of a MCW frame used for synchronization between terminals. (Referenced in 5.3.4.2.2.1.1 on page 80).

3.1.23 Frame Payload.

Section of a MCW frame that contains any data being transmitted between terminals. (Referenced in 5.3.4.2.2.1.3 on page 82).

3.1.24 Header.

Protocol information placed before frame payload data when encapsulating that data in preparation for network transmission.

3.1.25 In-band signaling.

Transmission of protocol management information (e.g., connection establishment handshaking) within a frequency range normally used for data transmission.

3.1.26 Interconnection.

The provision of data communication paths between stations.

3.1.27 Interface node.

A Bus Controller or Remote Terminal transceiver that utilizes the MCW protocol.

3.1.28 Interface Data Unit (IDU).

Data that crosses an OSI interface from one layer to the next layer and is composed of a Service Data Unit and an Interface Control Information.

3.1.29 Local area network (LAN).

A local area network (LAN) is a group of computers and associated devices that share a common communications line or wireless link.

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3.1.30 Logical Link Control (LLC).

IEEE-defined upper functional part of a data link layer that supports initiation of control signal interchange, framing and MAC-sublayer addressing, fragmentation and defragmentation, interpretation of received commands and generation of appropriate responses and actions taken regarding the error/flow control and recovery in the communication links.

3.1.31 Medium Access Control (MAC).

IEEE-defined lower functions part of a data link layer that supports: frame delineation and recognition, performing addressing and error checking, handling access to shared media, scheduling and acquiring the right to use the underlying physical medium.

3.1.32 Multi Carrier Waveform (MCW).

A bandwidth configurable technology to enable the transmission of high data rate signals over existing MIL-STD-1553B cables and couplers.

3.1.33 Network Layer (NL).

Layer 3 of the OSI reference model. This layer provides connectivity, routing and path selection as well as logical paths (virtual connections) for transmitting data between two end systems.

3.1.34 Network management.

Generic term used to describe systems or actions that help maintain, characterize, or troubleshoot a network.

3.1.35 OFDM Symbol.

A unit of transmission that contains the sum of the bits carried by all the available sub carriers. Data on each sub-carrier is independently modulated using Quadrature Amplitude Modulation (QAM) or Differential Binary Phase Shift Keying (DBPSK) modulations.

3.1.36 Open System Interconnection (OSI).

The international standardization program created by ISO and ITU-T to develop data networking standards that facilitate multi vendor equipment interoperability. OSI forms the basic communications model for MCW Waveform. Each layer is a logical subsystem and each subsystem is a server for the layer above it and a client of the layer below it. Each subsystem has a set of methods and attributes. Adjacent subsystems communicate through their common boundary. Peer to Peer communications is always indirect through connections provided by the server sub-layer below.

3.1.37 Open System Interconnection (OSI) peer to peer interfaces.

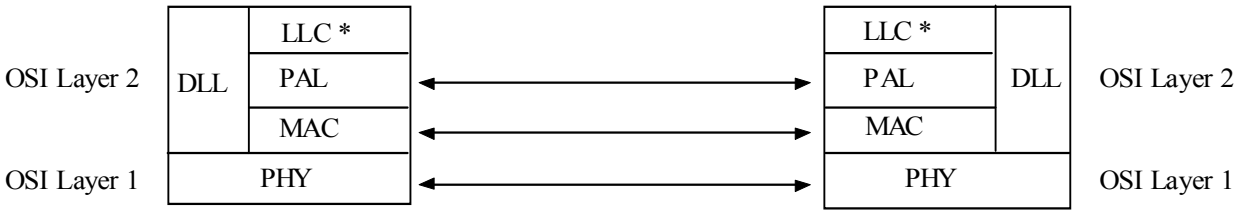
MCW Peer to Peer OSI interfaces are listed below and illustrated in FIGURE 1.

a. PHY-PHY.

b. MAC-MAC.

c. PAL-PAL.

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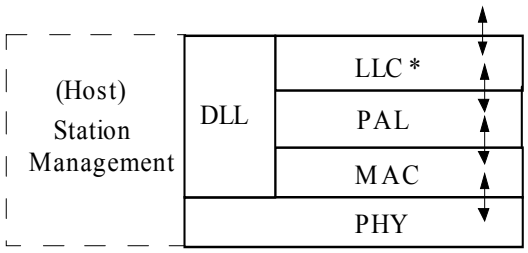


* The Logical Link Control (LLC) Sublayer is not defined herein.

FIGURE 1. MCW OSI peer to peer interfaces.

3.1.38 OSI stacked level interfaces.

SDU Interfaces between MCW Stacked layers/sublayers are shown as two way arrows with the OSI layers included for reference in FIGURE 2. Station Management, which is part of the Host, is shown in dotted lines.



* The Logical Link Control (LLC) Sublayer is not defined herein.

FIGURE 2. MCW OSI stacked layered interfaces.

3.1.39 Orthogonal frequency-division multiplexing (OFDM).

A method of digital modulation in which a signal is split into several narrowband channels at different frequencies.

3.1.40 Packet.

Synonym for Protocol Data Unit (PDU). In this document the PDU within the PHY layer is called a Frame.

3.1.41 Packet Adaptation Layer (PAL).

A sublayer above the MAC that allows transmission of arbitrary length packets over MAC frames.

3.1.42 Payload.

A synonym for Service Data Unit (SDU).

3.1.43 Peer protocol.

The sequence of message exchanges between two entities in the same layer that utilize the services of the underlying layers to effect the successful transfer of data and/or control information from one location to another.

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3.1.44 Peer-to-peer.

A communications model in which each party has the same capabilities and either can initiate a communication session.

3.1.45 Performance measures.

Metrics that provide quantitative or qualitative measures of the performance of systems or subsystems of interest.

3.1.46 Physical layer (PHY).

Layer 1 of the OSI reference model. The physical layer defines the electrical, mechanical, procedural, and functional specifications for activating, maintaining, and deactivating the physical link between end systems.

3.1.47 Protocol.

A set of rules and formats (semantic and syntactic) which determines the communication behavior of entities within a system to allow interoperability between vendors.

3.1.48 Protocol Data Unit (PDU).

The combination of data passed from the adjacent higher layer protocol (the Service Data Unit (SDU) or payload) and protocol information for this layer, also known as a "packet" or, at the PHY layer, a "frame."

3.1.49 Puncturing.

Elimination or omission of some encoded bits in a transmitter and correspondingly inserting a dummy "zero" metric into the convolutional decoder on the receiver side in place of the eliminated bits.

3.1.50 Quadrature Amplitude Modulation (QAM).

A method of combining two amplitude-modulated (AM) signals into a single channel, thereby doubling the effective bandwidth.

3.1.51 Reference terminal

A terminal defined by the custodian agency that conforms to the requirements of this standard.

3.1.52 Service Data Unit (SDU).

Unit of information from an upper-layer protocol that defines a service request to the adjacent lower-layer protocol. Following the Service request, the SDU is encapsulated by protocol information at the lower layer; thus the SDU becomes the "payload" for the lower-layer protocol packet.

3.1.53 Sync Bus Command Frame (SCBF).

The Sync Bus Command Frame (SCBF) is sent periodically by the BC to allow RTs to coordinate timing with the BC.

3.1.54 Sync Symbols.

Contained in the PHY frame preamble and generated using an approximation to a linear FM waveform. Used in order for a receiver to detect the start of the frame, calculate OFDM symbol timing and estimate channel response to demodulate modulated VBR data.

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3.1.55 Terminal Acknowledge Frame (TAF).

A frame used to indicate whether a TRF was received successfully. (Reference 5.3.4.2.2.4)

3.1.56 Terminal Interface.

The bus interface on the physical MCW hardware that is isolated from the Line Replaceable Unit (LRU) for the purposes of MCW testing.

3.1.57 Terminal Response Frame (TRF).

Contains the MCW transmitted data that was requested by a single command from the BC.

3.1.58 Throughput.

Rate of information arriving at, and possibly passing through, a particular point in a network system.

3.1.59 Time Division Multiple Access (TDMA).

A means for multiple terminals to share access to a common physical layer medium, whereby one terminal transmits on the medium at a time.

3.1.60 Training.

A means of determining the data encoding parameters between two Extended 1553 terminals.

3.1.61 Trapezoidal Waveform (TW).

A waveform equivalent to the MIL-STD 1553B (to Notice 4). (Reference 5.5).

3.1.62 Variable bit rate modulation (VBR).

A coherent modulation scheme that uses a negotiated “optimum” set of modulation parameters for communicating data between MCW interface nodes for any particular channel.

3.2 Trapezoidal Waveform (TW) definitions.

The definitions pertaining to the TW are contained in 5.5.3.

3.3 Acronyms.

ADC: Analog to Digital Converter

AFE: Analog Front End

ATF: Arbitrary Test Frame

AWGN: Additive White Gaussian Noise

BN: Bus Notification

BC: Bus Controller

BCF: Bus Command Frame

BER: Bit Error Ratio

BR: Basic Rate

CRC: Cyclic Redundancy Check

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DAC: Digital to Analog Converter

DBPSK: Differential Binary Phase Shift Keying

DC: Direct Current

DLL: Data Link Layer

FE: Frame Exchange

FEC: Forward Error Correction

FW: Filtered Waveform

IDFT: Inverse Discrete Fourier Transform

IEC - International Electro technical Commission

IEEE: Institute of Electrical and Electronics Engineers

ISO: International Standards Organization

LAN: Local Area Network

LCI: Local Connection Identifier

LGS: Long Guard Slots

LLC: Logical Link Control

LRU: Line Replaceable Unit

LSB: Least Significant Bit

MAC: Media Access Control

Mbps: Megabits per second

MCW: Multi Carrier Waveform

MHz: Megahertz

MSB: Most Significant Bit

MSDU: MAC Service Data Unit

NL: Network Layer

OSI: Open Systems Interconnection

PER: Packet Error Rate

POC: Proof of Compliance

PSD: Power Spectral Density

PHY: Physical Layer

PDU: Protocol Data Unit

QAM: Quadrature Amplitude Modulation

RMS: Root Mean Square

RS: Reed Solomon

SDU: Service Data Unit

SGS: Short Guard Slots

SNR: Signal to Noise Ratio

TAF: Terminal Acknowledge Frame

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TBD: To be Determined

TF: Training Frame

TRF: Terminal Response Frame

TW: Trapezoidal Waveform

UUT: Unit Under Test

VCRM: Verification Compliance Reference Matrix

VBR: Variable Bit Rate

VC: Virtual Connection

VCE: Virtual Connection End Point

WRA: Weapon Replaceable Assembly

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4. GENERAL REQUIREMENTS

4.1 Notice 5 certification.

In order to prove performance and interoperability between vendor systems installed in LRUs and WRAs, certification to this standard requires verification by each vendor of the requirements for the Extended 1553 capability contained herein. Requirements verification is based on completion of the Verification Cross Reference Matrix (VCRM) forms and approval of the specific vendor proof of compliance evidence referenced by these forms. The VCRM template forms are provided as TABLE IV in 5.2.1.3.

It is in the interest of the government that in order to prove interoperability between systems, a vendor's extended 1553 hardware capability may have to be proven interoperable with hardware known to meet the requirements of this standard.

To this end, the custodian agency reserves the right to require a set of interoperability tests to prove conformance to this notice 5. Certification would require test data to be provided showing interoperability in accordance with these tests and may entail samples of a terminal unit to be submitted to the custodian agency to be certified by actual interoperability testing with a compliant unit.

4.2 MIL-STD-1553B (to Notice 4).

The extended capabilities of Notice 5 are cognizant of and backwards compatible with the requirements of MIL-STD-1553B (to Notice 4). Trapezoidal Waveform (TW) contained in 5.5 is identical to MIL-STD-1553B (to Notice 4).

For reference, all subsections in 5.5 correspond to the equivalent section in MIL-STD-1553B (to notice 4) and all figures, tables, references and formats are unchanged as explained in 5.5.

4.3 Extended waveform referenced to OSI layers.

Trapezoidal Waveform (TW) requirements (section 5.5) can be hypothetically mapped to the corresponding Physical (PHY) layer and Media Access Control (MAC) sub layer of the OSI model. Similarly, MCW protocols are also implemented primarily on the PHY layer and MAC sub layers of the OSI model. Because these lower OSI layers are inherently suitable to the implementation of Base Band and Broadband components of Extended 1553B waveforms in Line Replaceable Units (LRU)s and Weapon Replaceable Assemblies (WRA)s, OSI will be the principle means by which different vendor implementations of extended 1553 are to be adopted.

The following Open System Interconnection (OSI) specifications are provided for background and information:

4.3.1 ISO/IEC 10022:1996 Open Systems Interconnection Physical Service Definition.

ISO/IEC 2382-26:1996 Information technology — Vocabulary — Part 26: Open Systems Interconnection

ISO/IEC 8348:2002 Information technology — Open Systems Interconnection — Network service definition.

ISO/IEC 10731:1994 Information technology — Open Systems Interconnection — Basic Reference Model — Conventions for the definition of OSI services

ISO/IEC 8886: 1996 Information technology — Open Systems Interconnection — Data Link Service Definition

ISO/IEC 10028:1993 Information technology — Telecommunications and information exchange between systems — Definition of the relaying functions of a Network layer intermediate system

Copies of these documents are available online at: International Organization for Standardization: http://www.iso.org/iso/en/prods-services/ISOstore/store.html or from International Organization for Standardization
International Organization for Standardization (ISO)
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5. DETAILED REQUIREMENTS

An Extended 1553B terminal shall be capable of communication using Multi Carrier Waveform (MCW), Filtered Waveform (FW) and Trapezoidal Waveform (TW) either individually or simultaneously.

If a terminal is to communicate using MCW as well as FW or TW, then the terminal shall:

a. Transmit FW but not TW.

b. Be capable of receiving FW or TW or both.

FW and Trapezoidal Waveform (TW) shall be separate and independent from MCW including media access functions that pertain to MCW. MCW functions shall not be initialized, set up or controlled by a terminal through FW or TW.

The electrical and performance requirements in this section shall be subject to verification as described in 5.2. Each requirement in this section corresponds to its associated test referenced in 5.2.

5.1 Electrical and Performance Requirements.

5.1.1 General.

The performance requirements stated below shall not degrade by more than 10 % over the full range of environmental conditions that the system is required to operate as required by the procuring agency.

5.1.2 MCW requirements.

Unless otherwise indicated, the requirements stated below apply to terminals with both transformer coupled and direct coupled stubs.

5.1.2.1 Not applicable.

5.1.2.2 Not applicable.

5.1.2.3 Output characteristics.

5.1.2.3.1 Output voltage levels.

The output voltage at the terminal bus interface when transmitting a MCW signal occupying the 5 - 35 MHz frequency band shall be a nominal 300 mV_{rms} into 75 Ω. Higher voltage levels may be used provided RF emissions constraints as specified in MIL-STD-461E are satisfied.

Verification shall be in accordance with the methodologies shown in TABLE IV and the test configuration and test procedure described in section 5.2.2.3.1.

5.1.2.3.2 Power Spectral Density (PSD).

The nominal PSD of the output MCW signal shall be -74 dBm/Hz into 75 Ω within the configured frequency band. The PSD of the spurious emissions transmitted outside the band should be no larger than as referenced in TABLE I. Higher PSD levels may be used provided that RF emissions constraints as specified in MIL-STD-461E are satisfied. A typical shape of the 5 - 35 MHz MCW PSD is illustrated in FIGURE 3.

Verification shall be in accordance with the methodologies shown in TABLE IV and the test configuration and test procedure contained in 5.2.2.3.2.

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TABLE I. MCW out-of-band spurious emissions.

Frequency band	≥ 45MHz	Starting at 4MHz above and below the configured frequency band.
Maximum out-of band PSD w.r.t. in-band PSD	-70 dB	-45 dB

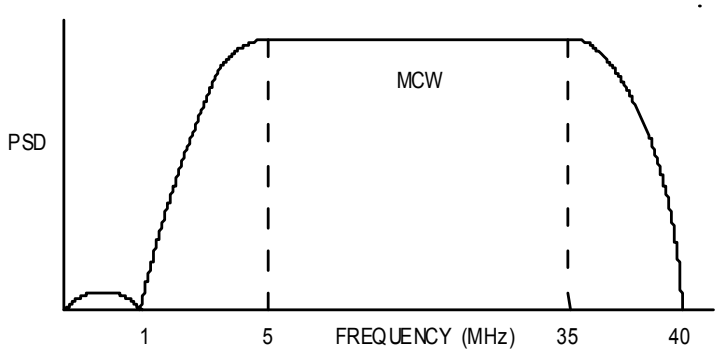


FIGURE 3. MCW Power Spectral Density.

5.1.2.3.3 Transmit spectral flatness.

The output MCW signal shall have no sub-carrier with a power that differs from the average power of all the sub-carriers by more than ± 2 dB for Basic Rate (BR) modulation, and ± 3 dB for Variable Bit Rate (VBR) modulation.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.3.3.

5.1.2.3.4 Output signal fidelity.

The output MCW signal, for all data modulation modes, shall have a signal fidelity of 45 dB minimum SNR per sub-carrier.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.3.4.

5.1.2.3.5 Output noise.

Any noise transmitted when the terminal is receiving or has power removed, shall not exceed -140 dBm/Hz within the frequency band configured for MCW communication.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.3.5.

5.1.2.3.6 Output transients.

From the time beginning 2.5µs after the transmission of the last MCW sample of a data frame, the maximum voltage shall be no greater than ± 5mV peak.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.3.6.

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5.1.2.3.7 Clock frequency accuracy.

The clock frequency of the MCW signal shall be 80 MHz ± 50 ppm with the Bus Controller being the source of the system clock for the network. All RTs will synchronize their local clock to this system clock.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.3.7.

5.1.2.3.8 Output timing accuracy.

Within a compound transaction, the output MCW signal shall arrive to the bus interface of the transmitting terminal within ± 500 ns of the nominal transmit time. The nominal transmit time shall be derived based on the bus schedule contained within the Bus Command frame.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.3.8.

5.1.2.4 Input characteristics.

5.1.2.4.1 Input signal compatibility.

The terminal shall be capable of receiving and operating with the incoming MCW signals specified herein and shall accept valid MCW signals as specified in the Interface Requirements in 5.3. The terminal shall respond to an input signal whose level is within the range of 500 μV_{rms} to 130 mV_{rms}.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.4.1.

5.1.2.4.2 Common mode rejection.

The MCW receiver shall have a common mode rejection of at least 30 dB from direct current (DC) to 40 MHz.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.4.2.

5.1.2.4.3 Susceptibility to noise and interference.

The terminal shall exhibit a performance specified as maximum error rate and minimum throughput as stated in the impairment tests described below.

5.1.2.4.3.1 Susceptibility to Additive White Gaussian Noise (AWGN).

The terminal shall exhibit a maximum BER of 10^{-12} at the nominal throughputs specified in TABLE II, with a 15% maximum allowance (nominal minus 15%) to account for implementation variations, over the 5 – 35 MHz frequency band when operating in the presence of AWGN at the identified SNR values.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.4.3.1.

TABLE II. AWGN susceptibility.

SNR [dB]	∞	35	23
Throughput [Mbps]	200	145	70

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5.1.2.4.3.2 Susceptibility to a single narrow-band interference source.

The terminal shall exhibit a maximum BER of 10^{-12} and a nominal throughput of 70 Mbps, with a 15 percent maximum allowance (nominal minus 15 percent) to account for implementation variations, over the 5 – 35 MHz frequency band when operating in the presence of a sinusoidal in-band interference source with a total signal-to-interference ratio of -20 dB.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.4.3.2.

5.1.2.4.3.3 Susceptibility to TW interference.

The terminal shall exhibit a maximum BER 10^{-12} and a nominal throughput of 10 Mbps, with a 15 percent maximum allowance (nominal minus 15 percent), to account for implementation variations, over the 5 - 35 MHz frequency when operating in the presence of a 2 V peak-to-peak TW interference.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.4.3.3.

5.1.2.4.3.4 Susceptibility to FW interference.

The terminal shall exhibit a maximum BER 10^{-12} and a nominal throughput of 200 Mbps, with a 15 percent maximum allowance (nominal minus 15 percent), to account for implementation variations over the 5 – 35 MHz frequency band when operating in the presence of a 20 V peak-to-peak FW interference.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.4.3.4.

5.1.2.5 Test networks.

MCW performance shall also be assessed using the test networks described below.

5.1.2.5.1 Test network #1 (20 foot bus).

This network consists of a 20 foot bus connecting 2 terminals that utilize single stub couplers. The length of each stub is 20 feet for the transformer coupled bus and 2 feet for the direct coupled bus. The performance baseline for this configuration shall be a nominal data throughput of 200 Mbps, with a 15 percent maximum allowance (nominal minus15 percent) to account for implementation variations at a BER of 10^{-12} over the 5 - 35 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.5.1.

5.1.2.5.2 Test network #2 (328 foot bus).

This network consists of a 328 foot bus connecting 2 terminals that utilize single stub couplers. The length of each stub is 20 feet for the transformer coupled bus and 2 feet for the direct coupled bus. The performance baseline for this configuration shall be a nominal data throughput of 200 Mbps, with a 15 percent maximum allowance (nominal minus15 percent) to account for implementation variations), at a BER of 10^{-12} over the 5 - 35 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.5.2.

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5.1.2.5.3 Test network #3.

This network consists of a 300 foot bus between End A and End B with 30 idle TW terminals that utilize both single-stub and multi-stub couplers. The spacing between the couplers varies from 5 to 70 feet, with the length of each stub being 20 feet for the transformer coupled bus and 2 feet for the direct coupled bus. A terminal shall be able to communicate using MCW signaling over each section and the entire length of the network with a BER of 10^{-12} over the 5 - 35 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.5.3.

5.1.2.5.4 Mixed networks.

A terminal shall be able to communicate using MCW signaling over each section and end to end of a mixed bus containing both transformer and direct coupled stubs at a BER of 10^{-12} over the 5 - 35 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.5.4.

5.1.2.6 Terminal diagnostics.

5.1.2.6.1 Digital loop-back.

The terminal shall be capable of initiating a digital loop-back test at the terminal as installed in the LRU under operational diagnostic conditions. The digital loop-back test shall demonstrate a minimum throughput of 235 Mbps with a Bit Error Rate (BER) of 10^{-12} over the 5 - 35 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.6.1.

5.1.2.6.2 Analog loop-back.

The terminal shall be capable of initiating an analog loop-back test and demonstrate a minimum throughput of 200 Mbps with a Bit Error Rate (BER) of 10^{-12} over the 5 - 35 MHz frequency band when isolated from the LRU.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.6.2.

5.1.2.6.3 Data logging.

In order to supplement performance verification and diagnostics, the terminal shall provide statistical data logging and recording capability. As a minimum, it will be possible for the following data metrics to be logged and recorded.

a. Number of synchronization failures (tracking failures) per peer terminal.

b. Number of received PHY frames per peer terminal.

c. Number of Frame Control (FC) CRC failures per peer terminal.

d. Number of payload CRC failures per peer terminal.

e. Number of retransmissions per peer terminal.

f. Minimum, maximum and current throughput (based on the instantaneous tone maps) per peer terminal.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.6.3.

5.1.2.7 Redundant data bus.

If redundant data buses are used, the following requirements shall apply to those data buses.

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5.1.2.7.1 Electrical isolation.

All terminals shall have a minimum of 45 dB isolation between data buses for all frequencies in the frequency range of 1 kHz to 40 MHz.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.7.1.

5.1.2.7.2 Dual redundant data bus.

The terminal shall transmit the same signal on both buses simultaneously. When receiving, the terminal shall receive on one bus at a time as specified by the bus command.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.2.7.2.

5.1.3 Filtered Waveform (FW) requirements.

Filtered Waveform (FW) shall support and be compliant with communications currently established by 1553B (to Notice 4) while simultaneously enabling the full performance of Multi Carrier Waveform (MCW) to be achieved as stated in 5.1.2.

Unless otherwise noted, the requirements stated below shall apply to terminals with both transformer coupled and direct coupled stubs.

5.1.3.1 Output characteristics.

5.1.3.1.1 Output voltage levels.

The output voltage at the terminal bus interface when transmitting a FW signal shall conform to the TW specifications in 5.5.4.5.2.1.1 for transformer coupled stubs and 5.5.4.5.2.2.1.1 for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.3.1.1.

5.1.3.1.2 Power Spectral Density (PSD).

The maximum power spectral density in the 0 – 1 MHz frequency band of the output FW signal shall be -28 dBm/Hz into 75 Ω for transformer coupled stubs, and -37 dBm/Hz for direct coupled stubs. A typical shape of the PSD is illustrated in FIGURE 4. A terminal shall not generate spurious emissions outside of the 0 – 1 MHz frequency band above the levels specified in TABLE III.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.3.1.2.

TABLE III. FW out-of-band emissions.

Frequency band	2 – 5 MHz	5 – 10 MHz	> 10 MHz
Maximum out-of-band PSD w.r.t. in-band PSD	-35 dB	-58 dB	-80 dB

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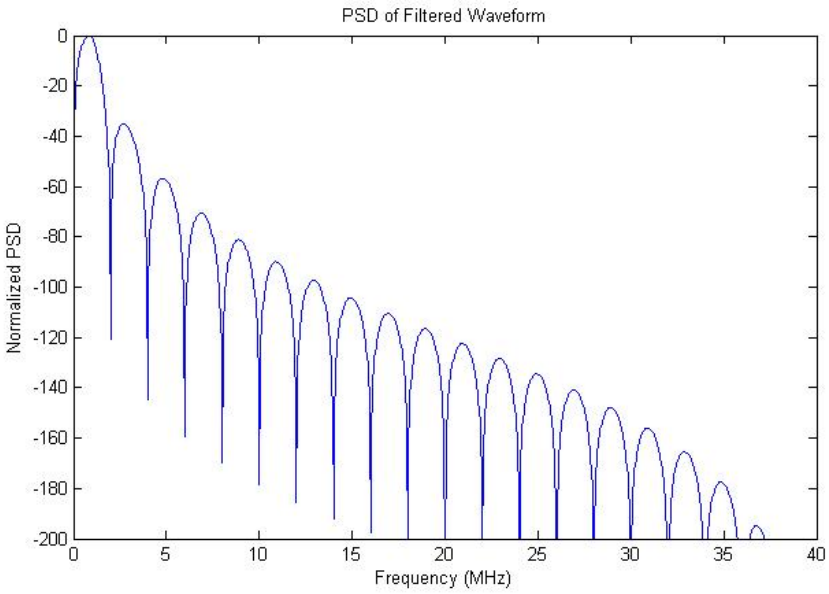


FIGURE 4. FW Power Spectral Density (PSD).

5.1.3.1.3 Output waveform.

The transmitted filtered waveform shall have:

a. Zero crossing deviations not exceeding ± 25.0 ns from the ideal crossing point, measured with respect to the previous zero crossing.

b. Rise and fall time between 100 and 300 ns when measured from levels of 10 to 90 percent of full waveform peak-to-peak.

c. Waveform distortion not exceeding 900 mV peak, including overshoot and ringing, for transformer coupled stubs, and 300 mV peak for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.3.1.3.

5.1.3.1.4 Output noise.

Any noise transmitted in the 0 – 5 MHz frequency band when the terminal is receiving or has power removed, shall not exceed a value as specified for the TW in 5.5.4.5.2.1.1.3 for transformer coupled stubs, and in 5.5.4.5.2.2.1.3 for direct coupled stubs. The noise transmitted in the frequency band configured for MCW communication shall conform to the corresponding MCW requirement in 5.1.2.3.5.

Verification shall be in accordance with the methodologies shown in TABLE IV and the test configuration and test procedure contained in 5.2.3.1.4.

5.1.3.1.5 Output symmetry (transients).

From the time beginning 2.5µs after the mid-bit crossing of the parity bit of the last word transmitted by a terminal, the maximum voltage shall be no greater than as specified for the TW in 0 for transformer coupled stubs, and in 5.5.4.5.2.2.1.4 for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.3.1.5.

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5.1.3.1.6 Clock frequency accuracy.

The clock frequency of the FW signal shall be 1 MHz ± 50 ppm.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.3.1.6.

5.1.3.2 Input characteristics.

5.1.3.2.1 Input impedance.

When the terminal has FW transmit capability, the magnitude of the input impedance within the 75 kHz to 1 MHz frequency range, when the RT is not transmitting or has the power removed, shall be as specified in 5.5.4.5.2.1.2.3 for transformer coupled stubs, and in 5.5.4.5.2.2.2.3 for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE IV, and the test configuration and test procedure contained in 5.2.3.2.1.

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5.2 Verification of Electrical and Performance Requirements.

5.2.1 General.

Verification methodology of each performance requirement is identified in the VCRM TABLE IV.

Verification for each performance requirement shall consist of vendor supplied Proof of Compliance (POC) reports identified and cross referenced for each of the methodologies identified in TABLE IV.

5.2.1.1 Proof of compliance reports.

For each requirement, as indicated in TABLE IV, proof of compliance reports shall include as a minimum the following:

a. Analysis and simulation data as required.

b. Formal test reports that provide detailed accounts of actual test configurations, specific test equipment and procedures.

c. Data presentation as referenced in each test.

The formal test reports referenced in b. shall include an analysis that shows equivalence between the actual test configurations used and the generic test configurations contained in 5.2.2.

5.2.1.2 Verification methodology.

Verification methods are defined as follows:

a. Analysis (A): provides a proof that a requirement can be met prior to an actual physical simulation or test.

b. Simulation (S): provides proof that a requirement can be physically met prior to build and test. Simulation may be used in lieu of a test where it is impractical to conduct a physical test. Presentation of simulation results shall include proof that the simulation conditions and the obtained results correspond to what is to be expected had a test using real hardware been conducted.

c. Demonstration (D): Demonstration provides proof that a requirement can be met through the observation of a suitable event(s) that does not entail parametric measurements.

d. Test (T): Involves parametric measurement within stated limits generally involving a test configuration and specific test equipment.

Analysis and Simulation as referenced in a. and b. shall provide proofs that include consideration of the wide ranging environmental conditions such as temperature, EMI/EMC and other conditions that the UUT is required to operate in as required by the procuring agency.

Testing as referenced in d. shall require adherence to the test conditions and test procedures specified in 5.2.2 and 5.2.3. The tests shall be conducted under the environmental conditions over which the UUT is required to operate, and shall be performed on a UUT of the same configuration to which simulation and analysis were conducted prior to testing.

5.2.1.3 Verification Cross-Reference Matrix (VCRM).

TABLE IV provides the verification methodology for the performance requirements of 5.1. The table also serves as a template for which POC documentation is referenced for each requirement.

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TABLE IV. Performance VCRM.

Section Reference	Performance Heading	Verification Method	POC Report Reference (Vendor Supplied)	
			Transformer coupled	Direct coupled
5.1.1	General	N/A: Header	N/A	N/A
5.1.2	MCW requirements	N/A: Header	N/A	N/A
5.1.2.3	Output characteristics	N/A: Header	N/A	N/A
5.1.2.3.1	Output voltage levels	A, T		
5.1.2.3.2	Power Spectral Density (PSD)	A, T		
5.1.2.3.3	Transmit spectral flatness	S, T		
5.1.2.3.4	Output signal fidelity	S, T		
5.1.2.3.3	Output noise	A, T		
5.1.2.3.6	Output transients	S, T		
5.1.2.3.7	Clock frequency accuracy	A, T		
5.1.2.3.8	Output timing accuracy	A,T		
5.1.2.4	Input characteristics	N/A: Header	N/A	N/A
5.1.2.4.1	Input signal compatibility	S, T		
5.1.2.4.2	Common mode rejection	S, T		
5.1.2.4.3	Susceptibility to noise and interference	N/A: Header	N/A	N/A
5.1.2.4.3.1	Susceptibility to Additive White Gaussian Noise (AWGN)	A, T		
5.1.2.4.3.2	Susceptibility to a single narrow-band interference source	S, T		
5.1.2.4.3.3	Susceptibility to TW interference	S, T		
5.1.2.4.3.4	Susceptibility to FW interference	S, T		
5.1.2.5	Test networks	N/A: Header	N/A	N/A
5.1.2.5.1	Test network #1 (20 foot bus)	S, T		
5.1.2.5.2	Test network #2 (328 foot bus)	S, T		
5.1.2.5.3	Test network #3	S, T		

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TABLE IV. Performance VCRM (Continued).

Section Reference	Performance Heading	Verification Method	POC Report Reference (Vendor Supplied)	
			Transformer coupled	Direct coupled
5.1.2.5.4	Mixed networks	S, T		
5.1.2.6	Terminal diagnostics	N/A: Header	N/A	N/A
5.1.2.6.1	Digital loop-back	A, T		
5.1.2.6.2	Analog loop-back	A, T		
5.1.2.6.3	Data logging	D		
5.1.2.7	Redundant data bus	N/A: Header	N/A	N/A
5.1.2.7.1	Electrical isolation	T		
5.1.2.7.2	Dual redundant data bus	T		
5.1.3	Filtered Waveform (FW) requirements	N/A: Header	N/A	N/A
5.1.3.1	Output characteristics	N/A: Header	N/A	N/A
5.1.3.1.1	Output voltage levels	A,T		
5.1.3.1.2	Power Spectral Density (PSD)	A, T		
5.1.3.1.3	Output waveform	A, T		
5.1.3.1.4	Output noise	A, T		
5.1.3.1.5	Output symmetry (transients)	S, T		
5.1.3.1.6	Clock frequency accuracy	A, T		
5.1.3.2	Input characteristics	N/A: Header	N/A	N/A
5.1.3.2.1	Input impedance	S, T		

5.2.1.4 Error rate measurements.

For verification purposes, BER may be equated to a verifiable Frame Error Rate (FER) with an identifiable number of bits per frame. The FER to be measured shall be derived from the target BER using the equation:

$$FER = 1 - (1 - BER)^{BitsPerFrame} \approx BER \times BitsPerFrame$$

A frame error shall include any fault which causes an associated error bit to be set in a terminal's status response or, which causes a terminal not to respond to a valid command.

The tests shall be run continuously until the specified performance metrics are achieved or the test fails.

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5.2.2 Compliance with MCW requirements.

Unless otherwise indicated, the tests described below apply to terminals with both transformer coupled and direct coupled stubs.

A benign test environment shall be provided which ensures that interference from external ambient noise sources not included as part of the test are eliminated or minimized.

All tests shall be carried out with the terminal interface hardware isolated from the LRU.

5.2.2.1 Characteristics of reference couplers and reference cables.

Throughput and BER performance for the susceptibility tests in 5.2.2.4.3 and over the test networks described in 5.2.2.5 is to be based on the following component characteristics:

a. Reference cable: characteristic impedance Z0 of 75 Ω ±5%, maximum attenuation of 2.5 dB / 100 ft at 5 MHz, and maximum attenuation of 7.5 dB / 100 ft at 40 MHz.

b. Reference coupler: isolation resistors of 0.75·Z0 ±2%; wide-band transformer with a turns ratio of 1.41:1, insertion loss of no more than 2 dB anywhere across the 1 – 40 MHz frequency band, and 3 dB cut-off frequencies no worse than 75 kHz and 50 MHz respectively.

5.2.2.2 Test signals.

Test signals used for performance testing shall be as described below.

5.2.2.2.1 Test signal #1 (training frame).

This signal consists of a periodically repeated MCW training frame at 80 MHz / sample occupying the 5 MHz to 35 MHz frequency band with a 15 dB peak-to-rms ratio. The transmission shall contain 12.6 µs (1008 samples) of zero guard time between training frames.

5.2.2.2.2 Test signal #2 (Basic Rate frame).

This signal consist of a periodically repeated Basic Rate modulated MCW frame at 80 MHz / sample occupying the 5 MHz to 35 MHz frequency band with a 15 dB peak-to-rms ratio. The transmission shall contain 12.6 µs (1008 samples) of zero guard time between data frames.

5.2.2.2.3 Test signal #3 (ACK test).

This signal consist of a periodically repeated MCW compound transaction at 80 MHz / sample occupying the 5 MHz to 35 MHz frequency band with a 15 dB peak-to-rms ratio. The transmission shall contain a Bus Command frame (requesting an ACK) followed by a Basic Rate (BR) data frame and sufficient guard time to accommodate the acknowledgement frame.

5.2.2.2.4 Test signal #4 (training compound transaction).

This signal consist of a periodically repeated MCW compound transaction at 80 MHz / sample occupying the 5 MHz to 35 MHz frequency band with a 15 dB peak-to-rms ratio. The transmission shall contain a Bus Command frame followed by a maximum length training frame and 25.4 µs (2032 samples) of guard time.

5.2.2.2.5 Test signal #5 (BER test compound transaction).

This signal is a repeated succession of MCW compound transactions at 80 MHz / sample, occupying the 5 – 35 MHz frequency band, and with a 15 dB peak-to-rms ratio. The signal has the following structure:

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a. A training compound transaction, as specified under the construction of test signal #4.

b. A Tone Map return compound transaction, containing a Bus Command frame followed by a Tone Map message frame originating from the terminal under test and 25.4 µs (2032 samples) of guard time.

c. 10 compound transactions, each containing a Bus Command frame (not requesting ACK) followed by a maximum length data frame and 25.4 µs (2032 samples) of guard time. Modulation of the data frame is based on the Tone Map received by the BC in the Tone Map return compound transaction referenced in b. The transmitted data bits within these compound transactions are random.

5.2.2.2.6 Test signal #6 (bus command frame only compound transaction).

This signal is similar to test signal #3 except that the data frame is replaced with a guard time of matching duration.

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5.2.2.3 Output characteristics.

5.2.2.3.1 Output voltage levels.

The output voltage at the terminal bus interface when transmitting the MCW signal shall be measured based on the following test configuration and procedure.

5.2.2.3.1.1 Test objectives.

Refer to the requirements described in 5.1.2.3.1 on page 12.

5.2.2.3.1.2 Test equipment.

a. True RMS voltmeter.

b. Terminating resistor RL of 75 Ω ±2%.

5.2.2.3.1.3 Test configuration.

Refer to the test configuration in FIGURE 5.

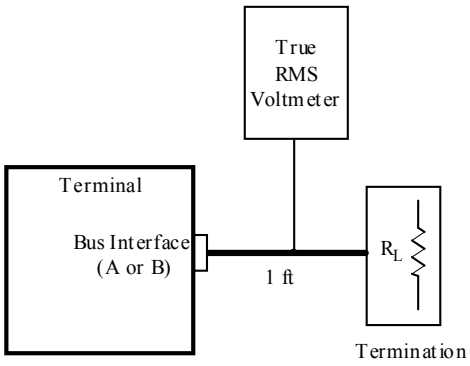


FIGURE 5. Test configuration 1.

5.2.2.3.1.4 Test procedure.

Step 1: Set up the terminal to continuously transmit test signal #1 on the bus A interface.

Step 2: Measure and record the maximum RMS voltage at the bus A interface.

Step 3: Repeat steps 1 to 2 using the bus B interface.

5.2.2.3.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The measured RMS voltage, for each bus interface.

c. Indication of compliance.

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5.2.2.3.2 Power Spectral Density (PSD).

The nominal PSD of the MCW signal shall be measured based on the following test configuration and procedure.

5.2.2.3.2.1 Test objectives.

Refer to the requirements described in 5.1.2.3.2 on page 12.

5.2.2.3.2.2 Test equipment.

a. Spectrum analyzer.

b. Calibrated 75 Ω / 50 Ω balun.

5.2.2.3.2.3 Test configuration.

Refer to the test configuration in FIGURE 6.

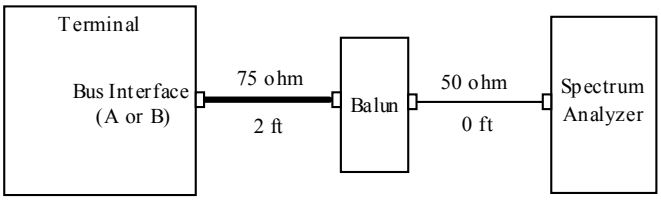


FIGURE 6. Test configuration 2.

5.2.2.3.2.4 Test procedure.

Step 1: Set up the terminal to continuously transmit test signal #1 on the bus A interface.

Step 2: Set up the spectrum analyzer to sweep the 1 kHz to 80 MHz frequency band and record the PSD levels. The insertion loss of the balun must be taken into account and compensated for if it is > 1dB anywhere within the tested frequency range.

Step 3: Repeat steps 1 to 2 using the bus B interface.

5.2.2.3.2.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Graphical data showing, for each bus interface, the measured power level versus frequency.

c. Indication of compliance.

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5.2.2.3.3 Transmit spectral flatness.

The spectral flatness of the transmitted MCW signal shall be measured based on the following test configuration and test procedure.

5.2.2.3.3.1 Test objectives.

Refer to the requirements described in 5.1.2.3.3 on page 13.

5.2.2.3.3.2 Test equipment.

Refer to the test equipment for the PSD test in 5.2.2.3.2.2 on page 26.

5.2.2.3.3.3 Test configuration.

Refer to the test configuration in FIGURE 6 on page 26.

5.2.2.3.3.4 Test procedure.

Step 1: Set up the terminal to continuously transmit test signal #2 (Basic Rate frame) on the bus A interface.

Step 2: Set up the spectrum analyzer to sweep the 5 MHz to 35 MHz frequency band and record the maximum and minimum PSD level. Half the difference between the recorded maximum and minimum levels shall not exceed the maximum allowed spectral ripple.

Step 3: Repeat steps 1 to 2 using the bus B interface.

Step 4: Repeat steps 1 to 3 using test signal #1 (training frame) instead of test signal #2.

5.2.2.3.3.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The recorded maximum and minimum PSD levels, for each bus interface.

c. Indication of compliance.

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5.2.2.3.4 Output signal fidelity.

The fidelity of the output MCW signal shall be measured based on the following test configuration and test procedure.

5.2.2.3.4.1 Test objectives.

Refer to the requirements described in 5.1.2.3.4 on page 13.

5.2.2.3.4.2 Test equipment.

a. Data acquisition system with 50 Ω input impedance and -130 dBm/Hz maximum noise floor.

b. Calibrated 75 Ω / 50 Ω balun.

c. Off-line digital signal processing capability.

5.2.2.3.4.3 Test configuration.

Refer to the test configuration in FIGURE 7.

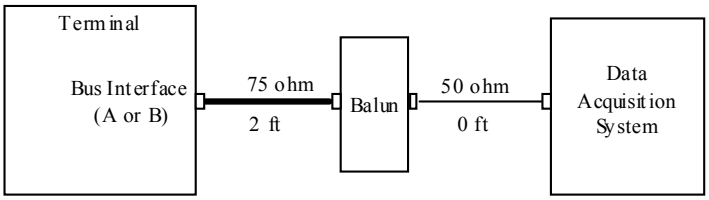


FIGURE 7. Test configuration 3.

5.2.2.3.4.3.1 Test procedure.

Step 1: Set up the terminal to continuously transmit test signal #1 on the bus A interface.

Step 2: Capture a sufficient number of frames of the test signal.

Step 3: Using off-line signal processing, derive and record the SNR for each sub-carrier.

Step 4: Repeat steps 1 to 3 using the bus B interface.

5.2.2.3.4.3.2 Data presentation.

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Graphical data showing, for each bus interface, the determined SNR versus frequency or sub-carrier index for all sub-carriers in the 5 -35 MHz frequency band.

c. Indication of compliance.

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5.2.2.3.5 Output noise.

The noise being output by the terminal when it is powered off and when it is receiving, shall be measured based on the following test configuration and procedure.

5.2.2.3.5.1 Test objectives.

Refer to the requirements described in 5.1.2.3.5 on page 13.

5.2.2.3.5.2 Test equipment.

Refer to the test equipment for the PSD test in 5.2.2.3.2.2 on page 26.

5.2.2.3.5.3 Test configuration.

Refer to the test configuration in FIGURE 6 on page 26.

5.2.2.3.5.4 Test procedure.

Set up the spectrum analyzer to sweep the 75 kHz to 40 MHz frequency band.

Step 1: Power off the terminal.

Step 2: Record the measured PSD levels. The insertion loss of the balun must be taken into account and compensated for if it is > 1dB anywhere within the tested frequency range.

Step 3: Repeat step 2 for bus B interface.

Step 4: Power on the terminal and place it into receive configuration.

Step 5: Repeat steps 2 to 3.

5.2.2.3.5.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Graphical data showing, for each bus interface, the measured noise power level versus frequency.

c. Indication of compliance.

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5.2.2.3.6 Output transients.

The output transient MCW signal level shall be measured based on the following test configuration and procedure.

5.2.2.3.6.1 Test objectives.

Refer to the requirements described in 5.1.2.3.6 on page 13.

5.2.2.3.6.2 Test equipment.

a. Oscilloscope (recording), set to 50 Ω input impedance.

b. Calibrated 75 Ω / 50 Ω balun.

5.2.2.3.6.3 Test configuration.

Refer to the test configuration in FIGURE 8.

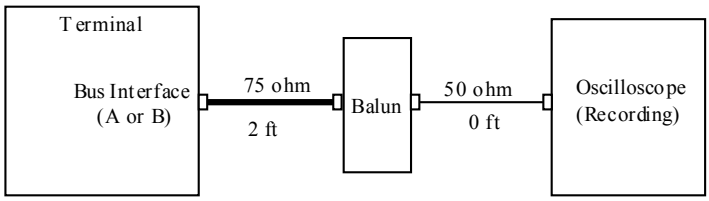


FIGURE 8. Test configuration 4.

5.2.2.3.6.4 Test procedure.

Step 1: Set up the terminal to continuously transmit test signal #1 on the bus A interface.

Step 2: Measure with the oscilloscope the maximum transient peak voltage level from 2.5 µs after the transmission of the last MCW sample of the data frame to the time immediately preceding the start of the subsequent data frame, compensating for the voltage ratio of the balun.

Step 3: Repeat steps 1 to 2 using the bus B interface.

5.2.2.3.6.5 Data presentation.

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Measured maximum transient peak voltage, for each bus interface.

c. Indication of compliance.

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5.2.2.3.7 Clock frequency accuracy.

The clock frequency of the MCW signal shall be measured based on the following test configuration and test procedure.

5.2.2.3.7.1 Test objectives.

Refer to the requirements described in 5.1.2.3.7 on page 14.

5.2.2.3.7.2 Test equipment.

a. Frequency counter.

5.2.2.3.7.3 Test configuration.

Refer to the test configuration in FIGURE 9.

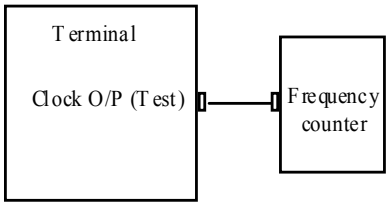


FIGURE 9. Test configuration 5.

5.2.2.3.7.4 Test procedure.

Step 1: Connect the frequency counter to the terminal MCW clock test connector or an equivalent test point.

Step 2: Record the clock frequency.

5.2.2.3.7.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The exact measured clock frequency.

c. Indication of compliance.

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5.2.2.3.8 Output timing accuracy.

The timing of the output MCW signal shall be measured based on the following test configuration and test procedure.

5.2.2.3.8.1 Test objectives.

Refer to the requirements described in 5.1.2.3.8 on page 14.

5.2.2.3.8.2 Test equipment.

a. Programmable arbitrary waveform generator.

b. Oscilloscope (recording) set to 1 MΩ input impedance.

c. Calibrated 75 Ω / 50 Ω balun.

5.2.2.3.8.3 Test configuration.

Refer to the test configuration in FIGURE 10.

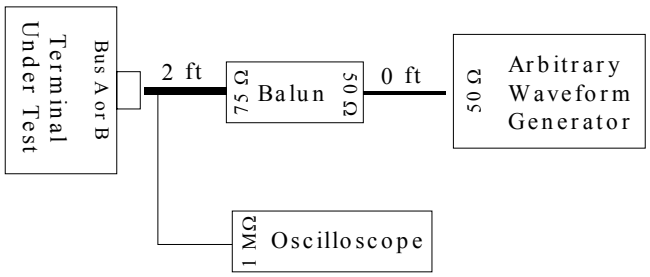


FIGURE 10. Test configuration 6.

5.2.2.3.8.4 Test procedure.

Set up the test configuration using the bus A interface. Configure the programmable arbitrary waveform generator to continuously transmit test signal #6. Configure the bus command within the compound transaction to request a training frame to be sent.

Step 1: Using the oscilloscope, observe and record the signal containing the bus command and the start of the training frame.

Step 2: Measure the time from the start of the bus command to the start of the training frame.

Step 3: Repeat steps 1 to 2 using the bus B interface.

5.2.2.3.8.5 Data presentation.

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The recorded signal, the nominal transmit time and the measured transmit time, for each bus interface.

c. Indication of compliance.

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5.2.2.4 Input characteristics.

5.2.2.4.1 Input signal compatibility.

MCW input signal compatibility shall be measured based on the following test configuration and procedure.

5.2.2.4.1.1 Test objectives.

Refer to the requirements described in 5.1.2.4.1 on page 14.

5.2.2.4.1.2 Test equipment.

a. Programmable 75 Ω attenuator.
b. Programmable arbitrary waveform generator.
c. Oscilloscope (recording), set to 1 MΩ input impedance.
d. Calibrated 75 Ω / 50 Ω balun.
e. True RMS voltmeter.

5.2.2.4.1.3 Test configuration.

Refer to the test configuration in FIGURE 11.

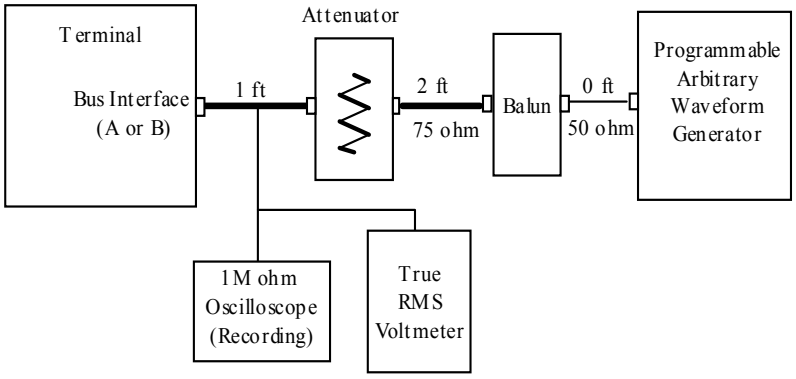


FIGURE 11. Test configuration 7.

5.2.2.4.1.4 Test procedure.

Configure the programmable arbitrary waveform generator to continuously transmit test signal #3.

Step 1: Using the bus A interface, adjust the attenuator such that the signal at the terminal is at the highest specified voltage level. Observe on the oscilloscope the presence of RT acknowledgement frames.

Step 2: Record the waveform of one compound transaction containing the acknowledgement frame.

Step 3: Repeat steps 1 to 2 for the lowest specified voltage level.

Step 4: Repeat steps 1 to 3 using the bus B interface.

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5.2.2.4.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Records of RT acknowledgements for the minimum and maximum input signal levels, for each bus interface.

c. Indication of compliance.

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5.2.2.4.2 Common mode rejection.

The common mode rejection of the MCW receiver shall be measured based on the following test configuration and procedure.

5.2.2.4.2.1 Test objectives.

Refer to the requirements described in 5.1.2.4.2 on page 14.

5.2.2.4.2.2 Test equipment.

a. Programmable 75 Ω wide-band amplifier.
b. Programmable arbitrary waveform generator.
c. Oscilloscope set to1 MΩ input impedance.
d. Data acquisition system with 1 MΩ input impedance.
e. Off-line digital signal processing capability.

5.2.2.4.2.3 Test configuration.

Refer to the test configuration in FIGURE 12.

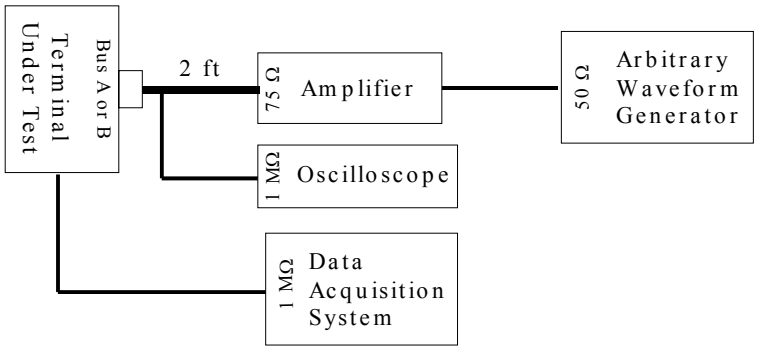


FIGURE 12. Test configuration 8.

5.2.2.4.2.4 Test procedure.

Set up the test configuration using the bus A interface. Connect the signal to the terminal as common mode excitation. Connect the data acquisition system to the A/D converter input on the MCW receive path of the terminal.

TABLE V. Common mode rejection test frequencies.

Frequency [MHz]	1	5	20	40

Step 1: Configure the arbitrary waveform generator to continuously transmit a sinusoidal signal of frequency chosen from TABLE V. Verify on the oscilloscope the shape of the signal injected into the terminal.

Step 2: Measure the signal amplitude on the oscilloscope. Adjust the amplifier until the amplitude is 10 V.

Step 3: Trigger the data acquisition.

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Step 4: Process the received samples off-line and determine the amplitude of the received signal at the test frequency. Calculate the common mode rejection.

Step 5: Repeat steps 1 to 4 for the other frequencies listed in TABLE V.

Step 6: Repeat steps 1 to 5 using the bus B interface.

5.2.2.4.2.5 Data presentation.

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Tabular data showing the determined CMR at each test frequency, for both powered off and receive configuration, and for each bus interface.

c. Indication of compliance.

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5.2.2.4.3 Susceptibility to noise and interference.

The noise and interference susceptibility tests shall be run continuously until the specified performance metrics are achieved or the test fails.

5.2.2.4.3.1 Susceptibility to Additive White Gaussian Noise (AWGN).

Data rate (throughput) and BER performance in the presence of AWGN shall be measured based on the following test configuration and procedure.

5.2.2.4.3.1.1 Test objectives.

Refer to the requirements described in 5.1.2.4.3.1 on page 14.

5.2.2.4.3.1.2 Test equipment.

a. Reference couplers (2 for transformer and 2 for direct coupled, corresponding to the desired bus coupling).
b. Reference bus segment cables.
c. Terminating resistor RL of 75 Ω ±2%.
d. Calibrated 50 Ω / 75 Ω balun.
e. Programmable 75 Ω attenuator.
f. Programmable arbitrary waveform generator.
g. True RMS voltmeter.
h. Oscilloscope, set to 1 MΩ input impedance.
i. Low-pass filter with the following characteristics: pass-band stop frequency at 40 MHz, stop-band start frequency at 50 MHz, minimum stop-band attenuation of 60 dB.
j. Error counter.
k. Reference terminal.

5.2.2.4.3.1.3 Test configuration.

Refer to the test configuration in FIGURE 13.

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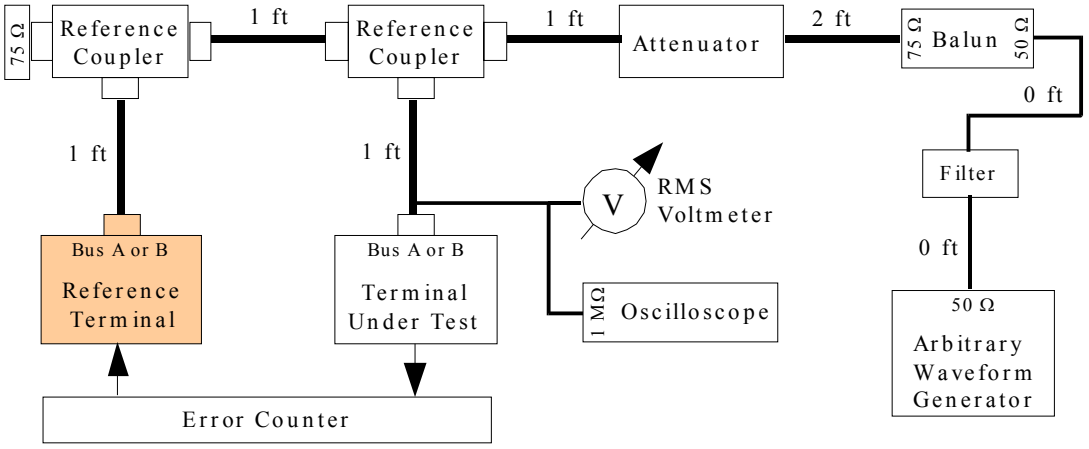


FIGURE 13. Test configuration 9.

5.2.2.4.3.1.4 Test procedure.

Set up the test configuration using the bus A interface at both terminals. Configure the reference terminal as bus controller and the terminal under test as RT. Generate independent samples of AWGN of 15 mV_{rms} at a sampling rate of at least 100 MHz / sample. Generate as many samples as to match the storage capacity of the arbitrary waveform generator.

Step 1: With the arbitrary waveform generator and the MCW terminals in the powered on state and not transmitting any signal (idle), confirm on the oscilloscope that there is no other interference present than background noise.

Step 2: Configure the reference terminal to continuously transmit test signal #4 and measure the received RMS signal level with the voltmeter.

Step 3: Reconfigure the reference terminal to idle. Set up the arbitrary waveform generator to continuously transmit the AWGN samples.

Step 4: Measure the received RMS noise level with the voltmeter. Adjust the attenuator until the measured RMS value yields an SNR value (with respect to the just measured received signal level) as specified in TABLE II. For the SNR value of ∞ do not inject any noise.

Step 5: Reconfigure the reference terminal to continuously transmit test signal #5.

Step 6: Record the measured BER and the minimum throughput over the duration of the test.

Step 7: Repeat steps 1 to 6 for the other SNR values in TABLE II.

Step 8: Repeat steps 1 to 7 using the bus B interface.

5.2.2.4.3.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Tabular data showing, for each bus interface, the measured RMS value of the received MCW signal and of the noise, for each SNR level at which the test was conducted.

c. Tabular data showing, for each bus interface, the actual data throughput and BER value, for each SNR level at which the test was conducted.

d. Indication of compliance.

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5.2.2.4.3.2 Susceptibility to a single narrow-band interference source.

Data rate (throughput) and BER performance in the presence of a single sinusoidal interference source shall be measured based on the following test configuration and procedure.

5.2.2.4.3.2.1 Test objectives.

Refer to the requirements described in 5.1.2.4.3.2 on page 15.

5.2.2.4.3.2.2 Test equipment.

Refer to the test equipment for the AWGN susceptibility test in 5.2.2.4.3.1.2 on page 37.

5.2.2.4.3.2.3 Test configuration.

Refer to the test configuration in FIGURE 13 on page 38.

5.2.2.4.3.2.4 Test procedure.

Set up the test configuration using the bus A interface at both terminals. Configure the reference terminal as bus controller and the terminal under test as RT. The test shall be performed at the frequencies given by the following equation

$$f_k = k \cdot 5\,[MHz] + 2.5\,[MHz] + \frac{78.125\,[kHz]}{2}, \quad k = 0 \ldots 7$$

Generate samples of the sinusoidal interference of amplitude 1 V at a sampling rate of at least 100 MHz / sample. For each test frequency, the samples shall be generated in such a manner that there is no phase discontinuity when the transmitted signal wraps around from the last sample to the first sample.

Step 1: With the arbitrary waveform generator and the MCW terminals in the powered on state and not transmitting any signal (idle), confirm on the oscilloscope that there is no other interference present than background noise.

Step 2: Configure the reference terminal to continuously transmit test signal #4 and measure the received RMS signal level with the voltmeter.

Step 3: Reconfigure the reference terminal to idle. Set up the arbitrary waveform generator to continuously transmit the samples of the interference signal of the lowest frequency given above.

Step 4: Measure the received RMS interference level with the voltmeter. Confirm on the oscilloscope the shape of the interference signal. Adjust the attenuator until the measured RMS value yields the specified SIR (with respect to the just measured received signal level).

Step 5: Reconfigure the reference terminal to continuously transmit test signal #5.

Step 6: Record the measured BER and the minimum throughput over the duration of the test.

Step 7: Repeat steps 1 to 6 for the other interference signal frequencies.

Step 8: Repeat steps 1 to 7 using the bus B interface.

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5.2.2.4.3.2.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Tabular data showing, for each bus interface, the measured RMS value of the received MCW signal and of the interference, for each frequency at which the test was conducted.

c. Tabular data showing, for each bus interface, the actual data throughput and BER value, for each frequency at which the test was conducted.

d. Indication of compliance.

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5.2.2.4.3.3 Susceptibility to TW interference.

Data rate (throughput) and BER performance in the presence of TW interference source shall be measured based on the following test configuration and procedure.

5.2.2.4.3.3.1 Test objectives.

Refer to the requirements described in 5.1.2.4.3.3 on page 15.

5.2.2.4.3.3.2 Test equipment.

a. Reference couplers (2 for transformer and 2 for direct coupled, corresponding to the desired bus coupling).
b. Reference bus segment cables.
c. Terminating resistor RL of 75 Ω ±2%.
d. Programmable 75 Ω wide-band amplifier.
e. Programmable arbitrary waveform generator.
f. True RMS voltmeter.
g. Oscilloscope set to 1 MΩ input impedance.
h. Low-pass filter with the following characteristics: pass-band stop frequency at 40 MHz, stop-band start frequency at 50 MHz, minimum stop-band attenuation of 60 dB.
i. Error counter.
j. Reference terminal.

5.2.2.4.3.3.3 Test configuration.

Refer to the test configuration in FIGURE 14.

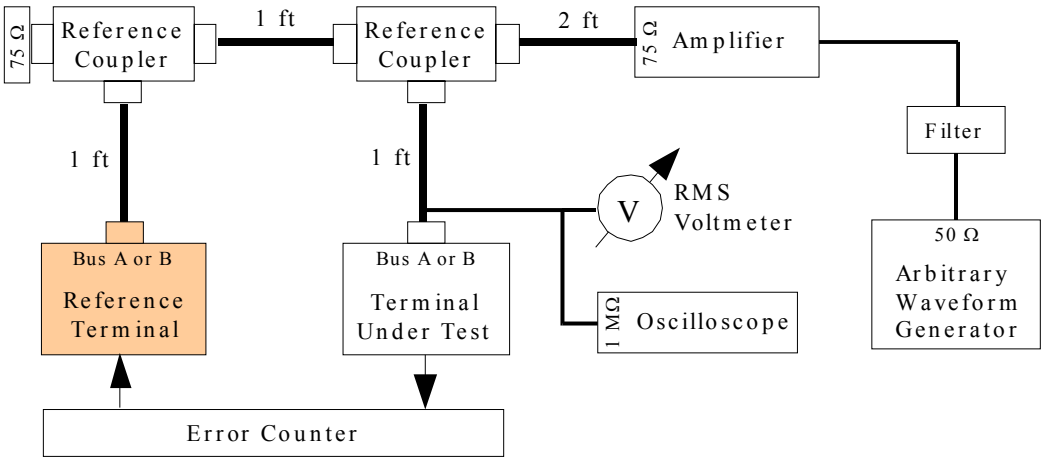


FIGURE 14. Test configuration 10.

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5.2.2.4.3.3.4 Test procedure.

Set up the test configuration using the bus A interface at both terminals. Configure the reference terminal as bus controller and the terminal under test as RT. Generate samples of TW interference of amplitude 1V at a sampling rate of at least 100 MHz / sample.

Step 1: With the arbitrary waveform generator and the MCW terminals in the powered on state and not transmitting any signal (idle), confirm on the oscilloscope that there is no other interference present than background noise.

Step 2: Configure the reference terminal to continuously transmit test signal #4 and measure the received RMS signal level with the voltmeter.

Step 3: Disconnect the RMS voltmeter. Reconfigure the reference terminal to idle. Set up the arbitrary waveform generator to continuously transmit the samples of the interference signal.

Step 4: Measure the received signal amplitude with the oscilloscope and confirm the shape of the interference signal. Adjust the amplifier until the measured value yields the specified peak-to-peak voltage.

Step 5: Reconfigure the reference terminal to continuously transmit test signal #5.

Step 6: Record the measured BER and the minimum throughput over the duration of the test.

Step 7: Repeat steps 1 to 6 using the bus B interface.

5.2.2.4.3.3.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Measured RMS value of the MCW signal.

c. The actual data throughput and BER value, for each bus interface.

d. Indication of compliance.

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5.2.2.4.3.4 Susceptibility to FW interference.

Data rate (throughput) and BER performance in the presence of FW interference source shall be measured based on the following test configuration and procedure.

5.2.2.4.3.4.1 Test objectives.

Refer to the requirements described in 5.1.2.4.3.4 on page 15.

5.2.2.4.3.4.2 Test equipment.

Refer to the test equipment for the TW susceptibility test in 5.2.2.4.3.3.2 on page 41.

5.2.2.4.3.4.3 Test configuration.

Refer to the test configuration in FIGURE 14 on page 41.

5.2.2.4.3.4.4 Test procedure.

Refer to the test procedure for the TW susceptibility test in 5.2.2.4.3.3.4 on page 42, replacing the TW interference signal with a FW interference signal.

5.2.2.4.3.4.5 Data presentation.

Refer to the data presentation for the TW susceptibility test in 5.2.2.4.3.3.5 on page 42.

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5.2.2.5 Test networks

5.2.2.5.1 Test network #1 (20 foot bus).

Data rate (throughput) and BER performance over test network #1 shall be measured based on the following test configuration and procedure.

5.2.2.5.1.1 Test objectives.

Refer to the requirements described in 5.1.2.5.1 on page 15.

5.2.2.5.1.2 Test equipment.

a. Reference terminal.

b. Error counter.

c. Reference couplers and reference cable segments for the bus and stubs.

d. 2 terminating resistors of 75 Ω ±2%.

5.2.2.5.1.3 Test configuration.

Refer to the test configuration in FIGURE 15 for the transformer coupled bus, and in FIGURE 16 for the direct coupled bus. Either a physical test network or a network simulator can be used.

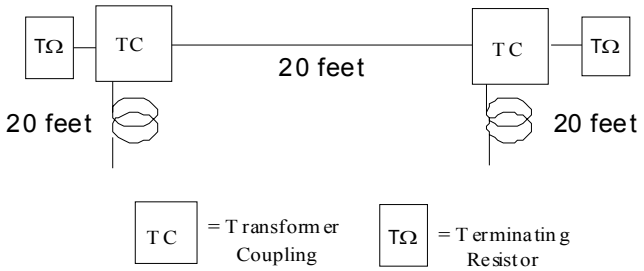


FIGURE 15. Transformer coupled test network #1.

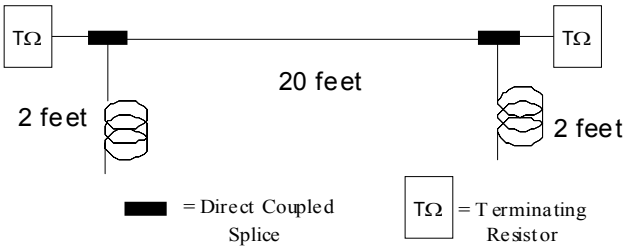


FIGURE 16. Direct coupled test network #1.

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5.2.2.5.1.4 Test procedure.

Configure the reference terminal as bus controller and the terminal under test as RT. Connect the output port on the error counter to the Data IN port of the BC, and the input port of the error counter to the Data OUT port of the RT.

Step 1: Connect the reference terminal and the terminal under test to the stubs at the two ends of the bus using the bus A interface.

Step 2: Run a BER and throughput test.

Step 3: Repeat steps 1 to 2 using the bus B interface.

5.2.2.5.1.5 Data presentation.

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The actual data throughput and BER value, for each bus interface.

c. Indication of compliance.

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5.2.2.5.2 Test network #2 (328 foot bus).

Data rate (throughput) and BER performance over test network #2 shall be measured based on the following test configuration and procedure.

5.2.2.5.2.1 Test objectives.

Refer to the requirements described in 5.1.2.5.3 on page 15.

5.2.2.5.2.2 Test equipment.

Refer to the test equipment for test network #1 in 5.2.2.5.1.2 on page 44.

5.2.2.5.2.3 Test configuration.

Refer to the test configuration in FIGURE 17 for the transformer coupled bus, and in FIGURE 18 for the direct coupled bus. Either a physical test network or a network simulator can be used.

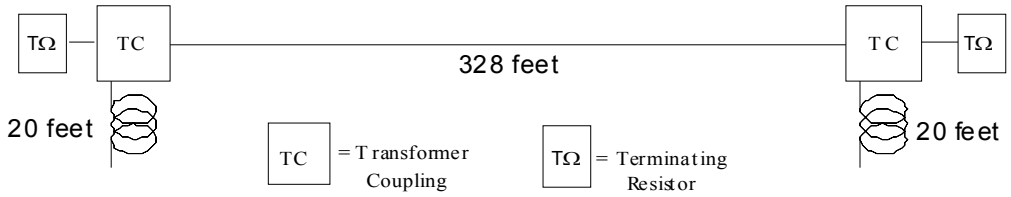


FIGURE 17. Transformer coupled test network #2.

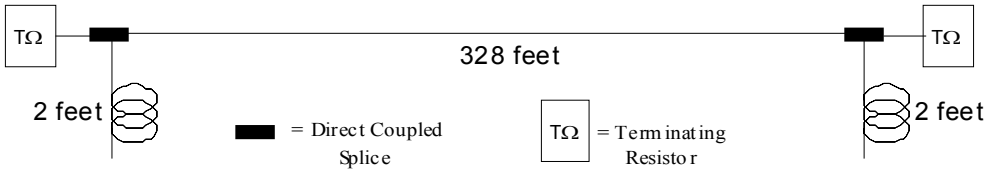


FIGURE 18. Direct coupled test network #2.

5.2.2.5.2.4 Test procedure.

Refer to the test procedure for test network #1 in 5.2.2.5.1.4 on page 45.

5.2.2.5.2.5 Data presentation.

Refer to the data presentation for test network #1 in 5.2.2.5.1.5 on page 45.

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5.2.2.5.3 Test network #3.

Ability to communicate over test network #3 shall be assessed based on the following test configuration and procedure.

5.2.2.5.3.1 Test objectives.

Refer to the requirements described in 5.1.2.5.3 on page 16.

5.2.2.5.3.2 Test equipment.

a. Oscilloscope (recording), set to 1 MΩ input impedance.

b. Reference terminal.

c. Reference couplers and reference cable segments for the bus and stubs.

d. 2 terminating resistors of 75 Ω ±2%.

e. 30 stub terminating resistors of 1 kΩ ±2% for the transformer coupled bus, and of 2 kΩ ±2% for the direct coupled bus.

5.2.2.5.3.3 Test configuration.

Refer to the test configuration in FIGURE 19 for the transformer coupled bus, and in FIGURE 20 for the direct coupled bus. All stub terminations, except at end A and end B, are of 1 kΩ for the transformer coupled bus, and of 2 kΩ for the direct coupled bus. Either a physical test network or a network simulator can be used.

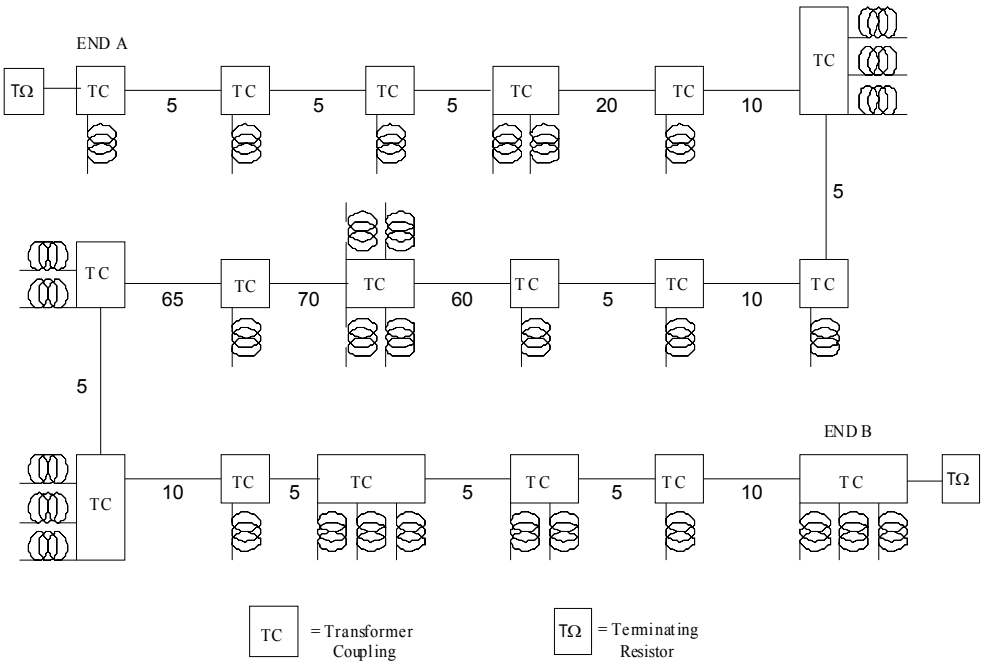


FIGURE 19. Transformer coupled test network #3.

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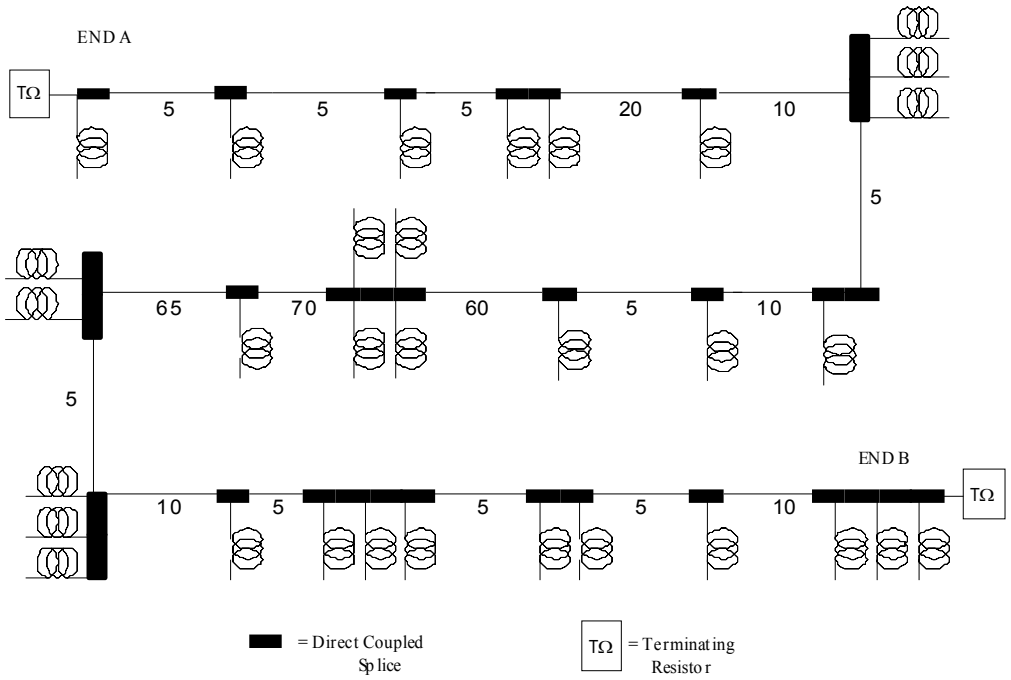


FIGURE 20. Direct coupled test network #3.

5.2.2.5.3.4 Test procedure.

Set up the test configuration using the bus A interface. Connect the oscilloscope to the same stub as the terminal under test. Configure the reference terminal as bus controller and the terminal under test as RT.

Step 1: Configure the bus controller to generate compound transactions corresponding to test signal #3.

Step 2: Observe on the oscilloscope the presence of RT acknowledgement frames and record the waveform of one compound transaction containing the acknowledgement frame.

Step 3: Interchange the location of the bus controller and the RT and repeat steps 1 to 2.

5.2.2.5.3.5 Data presentation.

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Records of RT acknowledgements.

c. Indication of compliance.

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5.2.2.5.4 Mixed networks.

Ability to communicate over mixed networks shall be assessed based on the following test configuration and procedure.

5.2.2.5.4.1 Test objectives.

Refer to the requirements described in 5.1.2.5.4 on page 16.

5.2.2.5.4.2 Test equipment.

Refer to the test equipment for test network #3 in 5.2.2.5.3.2 on page 47. The 30 stub terminations are not necessary for this test.

5.2.2.5.4.3 Test configuration.

Refer to the test configuration in FIGURE 21. This network consists of a 20 foot bus connecting 2 terminals that utilize single-coupled stubs of 20 feet for the transformer coupled stub and 2 feet for the direct coupled stub. Either a physical test network or a network simulator can be used.

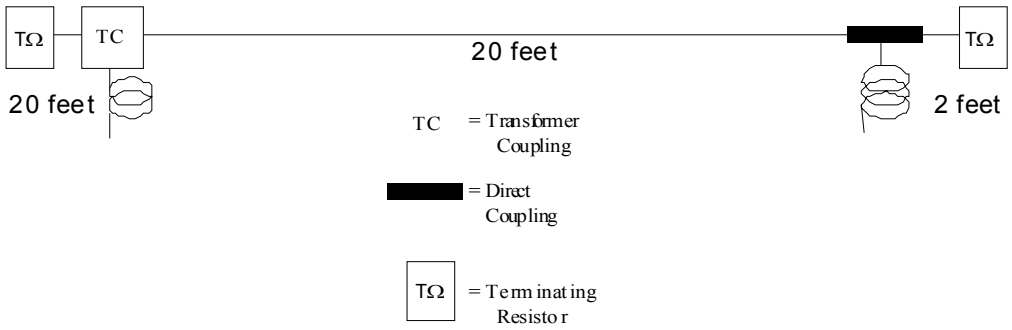


FIGURE 21. Mixed network.

5.2.2.5.4.4 Test procedure.

Refer to the test procedure for test network #3 in 5.2.2.5.3.4 on page 48.

5.2.2.5.4.5 Data presentation.

Refer to the data presentation for test network #3 in 5.2.2.5.3.5 on page 48.

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5.2.2.6 Terminal diagnostics.

5.2.2.6.1 Digital loop-back.

Data rate (throughput) and BER performance in the digital loop-back configuration shall be measured based on the following test configuration and procedure.

5.2.2.6.1.1 Test objectives.

Refer to the requirements described in 5.1.2.6.1 on page 16.

5.2.2.6.1.2 Test equipment.

a. Error counter.

5.2.2.6.1.3 Test configuration.

Refer to the test configuration in FIGURE 22.

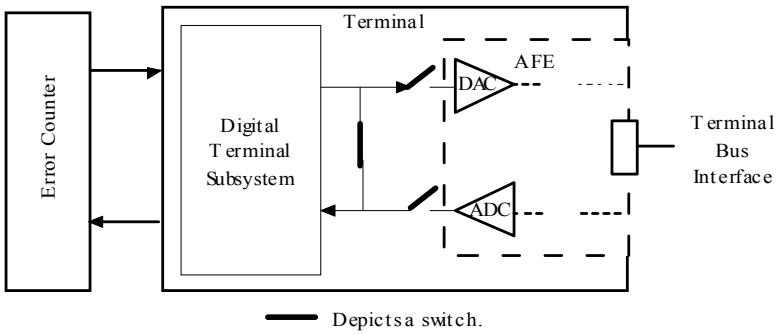


FIGURE 22. Test configuration 11.

5.2.2.6.1.4 Test procedure.

Configure the terminal to carry out a digital loop-back self-test. For this purpose, the Analog Front-End (AFE) is completely bypassed by routing the digital signal at the input of the transmitter Digital to Analog Converter (DAC) directly to the output of the receiver Analog to Digital Converter (ADC) while isolating both converters from the "digital terminal."

Step 1: Record the measured BER and the minimum throughput over the duration of the test.

5.2.2.6.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The actual data throughput and BER value.

c. Indication of compliance.

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5.2.2.6.2 Analog loop-back.

Data rate (throughput) and BER performance in the analog loop-back configuration shall be measured based on the following test configuration and procedure.

5.2.2.6.2.1 Test objectives.

Refer to the requirements described in 5.1.2.6.2 on page 16.

5.2.2.6.2.2 Test equipment.

a. Error counter.

b. Terminating resistor RL of 75 Ω ±2%.

5.2.2.6.2.3 Test configuration.

Refer to the test configuration in FIGURE 23.

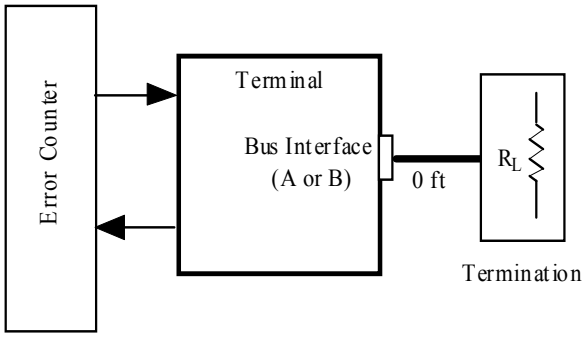


FIGURE 23. Test configuration 12.

5.2.2.6.2.4 Test procedure.

Configure the terminal to carry out an analog loop-back test.

Step 1: Set up the test configuration for testing at the terminal bus A interface.

Step 2: Record the measured BER and the minimum throughput over the duration of the test.

Step 3: Repeat steps 1 to 2 using the bus B interface.

5.2.2.6.2.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Tabular data showing, for each bus interface, the actual data throughput and BER value.

c. Indication of compliance.

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5.2.2.6.3 Data logging.

Conformance with the data logging requirements shall be assessed based on the following test configuration and procedure.

5.2.2.6.3.1 Test objectives.

Refer to the requirements described in 5.1.2.6.3 on page 16.

5.2.2.6.3.2 Test equipment.

None.

5.2.2.6.3.3 Test configuration.

Refer to the test configuration in FIGURE 22 on page 50.

5.2.2.6.3.4 Test procedure.

Configure the terminal to carry out a digital loop-back self-test. Demonstrate that the data logger is capable of monitoring and recording the data metrics referenced in 5.1.2.6.3.

5.2.2.6.3.5 Data presentation.

a. Detailed test configuration, test equipment and test procedures used for conducting the test.
b. List of logged parameters and metrics.
c. Indication of compliance.

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5.2.2.7 Redundant data bus.

5.2.2.7.1 Electrical isolation.

The electrical isolation between the bus A and bus B interface of the terminal shall be measured based on the following test configuration and procedure.

5.2.2.7.1.1 Test objectives.

Refer to the requirements described in 5.1.2.7.1 on page 17.

5.2.2.7.1.2 Test equipment.

a. Vector network analyzer.

5.2.2.7.1.3 Test configuration.

Refer to the test configuration in FIGURE 24.

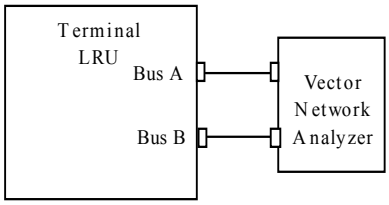


FIGURE 24. Test configuration 13.

5.2.2.7.1.4 Test procedure.

With the terminal powered and idle (not transmitting), measure the S_{21} transmission coefficient over the 1 kHz to 40 MHz frequency band and record the magnitude on a logarithmic scale.

5.2.2.7.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Graphical data showing the level of measured isolation in dB versus frequency.

c. Indication of compliance.

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5.2.2.7.2 Dual redundant data bus.

Dual bus transmission and reception on either bus will be based on the following test configuration and procedure.

5.2.2.7.2.1 Test objectives.

Refer to the requirements described in 5.1.2.7.2 on page 17.

5.2.2.7.2.2 Data transmission.

5.2.2.7.2.2.1 Test equipment.

a. Oscilloscope (recording), set to 50 Ω input impedance.
b. Calibrated 75 Ω / 50 Ω balun.

5.2.2.7.2.2.2 Test configuration.

Refer to the test configuration in FIGURE 8 on page 30.

5.2.2.7.2.2.3 Test procedure.

Set up the terminal to continuously transmit test signal #1. Using the oscilloscope, observe and record the signal at the bus A and bus B interface.

5.2.2.7.2.2.4 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.
b. Oscilloscope traces of the transmitted signal, for each bus interface.
c. Indication of compliance.

5.2.2.7.2.3 Data reception.

5.2.2.7.2.3.1 Test equipment.

a. 2 externally synchronized programmable arbitrary waveform generators.
b. 2 calibrated 75 Ω / 50 Ω baluns.
c. Oscilloscope (recording), set to 1 MΩ input impedance.

5.2.2.7.2.3.2 Test configuration.

Refer to the test configuration in FIGURE 25.

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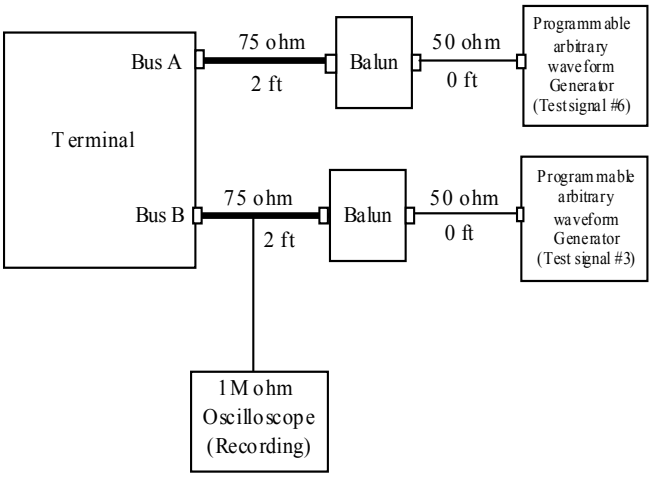


FIGURE 25. Test configuration 14.

5.2.2.7.2.3.3 Test procedure.

Step 1: Configure the arbitrary waveform generators to continuously transmit test signals #3 and #6 respectively. Configure the bus command within both compound transactions to specify bus B for data transfer. Configure the terminal for a bus A only startup procedure.

Step 2: Connect test signal #6 to the bus A interface connector on the terminal, and test signal #3 to the bus B interface via the baluns. Connect the oscilloscope to the signal generator on bus B.

Step 3: Using the oscilloscope, observe and record the waveform of one compound transaction containing the acknowledgement frame.

Step 4: Reconfigure the bus commands within test signals #3 and #6 to specify bus A for data transfer. Configure the terminal for a bus B only startup procedure.

Step 5: Connect test signal #6 to the bus B interface connector on the terminal, and test signal #3 to the bus A interface via the baluns. Connect the oscilloscope to the signal generator on bus A.

Step 6: Using the oscilloscope, observe and record the waveform of one compound transaction containing the acknowledgement frame.

5.2.2.7.2.3.4 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Records of RT acknowledgements.

c. Indication of compliance.

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5.2.3 Compliance with Filtered Waveform (FW) requirements.

Unless otherwise indicated, the tests described below apply to terminals with both transformer coupled and direct coupled stubs.

A benign test environment shall be provided which ensures that interference from external ambient noise sources not included as part of the test are eliminated or minimized.

5.2.3.1 Output characteristics.

5.2.3.1.1 Output voltage levels.

The output voltage at the terminal bus interface when transmitting the FW signal shall be measured based on the following test configuration and procedure.

5.2.3.1.1.1 Test objectives.

Refer to the requirements described in 5.1.3.1.1 on page 17.

5.2.3.1.1.2 Test equipment.

a. Oscilloscope (recording), set to 50 Ω input impedance.
b. Calibrated 75 Ω / 50 Ω balun.
c. Programmable 75 Ω attenuator.

5.2.3.1.1.3 Test configuration.

Refer to the test configuration in FIGURE 26.

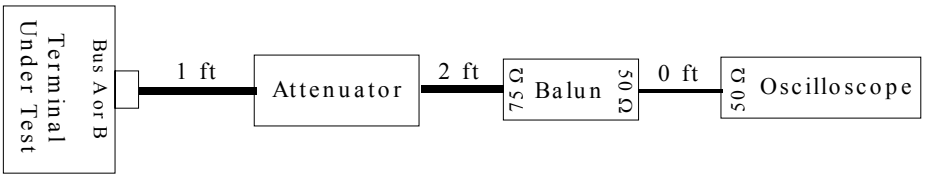


FIGURE 26. Test configuration 15.

5.2.3.1.1.4 Test procedure.

Step 1: Dial in 20 dB attenuation on the programmable attenuator.

Step 2: Set up the terminal to continuously transmit a periodic signal on the bus A interface.

Step 3: Using the oscilloscope, measure and record the maximum peak-to-peak voltage.

Step 4: Repeat steps 2 to 3 using the bus B interface.

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5.2.3.1.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Measured maximum peak-to-peak voltage, for each bus interface.

c. Indication of compliance.

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5.2.3.1.2 Power Spectral Density (PSD).

The PSD of the FW signal shall be measured based on the following test configuration and procedure.

5.2.3.1.2.1 Test objectives.

Refer to the requirements described in 5.1.3.1.2 on page 17.

5.2.3.1.2.2 Test equipment.

a. Spectrum analyzer.

b. Calibrated 75 Ω / 50 Ω balun.

c. Programmable 75 Ω attenuator.

5.2.3.1.2.3 Test configuration.

Refer to the test configuration in FIGURE 27.

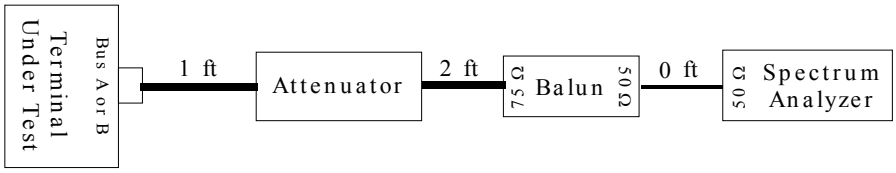


FIGURE 27. Test configuration 16.

5.2.3.1.2.4 Test procedure.

Step 1: Dial in 30 dB attenuation on the programmable attenuator.

Step 2: Set up the terminal to continuously transmit a periodic signal on the bus A interface.

Step 3: Set up the spectrum analyzer to sweep the 1 kHz to 40 MHz frequency band and record the PSD levels. The insertion loss of the balun must be taken into account and compensated for if it differs by more than 1dB across the tested frequency range.

Step 4: Repeat steps 2 to 3 using the bus B interface.

5.2.3.1.2.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The maximum measured PSD level in the 0-1 MHz frequency band.

c. Graphical data showing, for each bus interface, the measured power level in dB versus frequency. The dB scale shall be with respect to the maximum measured PSD level presented under b. above.

d. Indications of compliance.

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5.2.3.1.3 Output waveform.

The time domain characteristics of the transmitted FW signal shall be measured based on the following test configuration and procedure.

5.2.3.1.3.1 Test objectives.

Refer to the requirements described in 5.1.3.1.3 on page 18.

5.2.3.1.3.2 Test equipment.

Refer to the test equipment for the output voltage level test in 5.2.3.1.1.2 on page 56.

5.2.3.1.3.3 Test configuration.

Refer to the test configuration in FIGURE 26 on page 56.

5.2.3.1.3.4 Test procedure.

Step 1: Dial in 20 dB attenuation on the programmable attenuator.

Step 2: Set up the terminal to continuously transmit a periodic signal on the bus A interface.

Step 3: Measure with the oscilloscope the zero crossing deviation, rise and fall time, and waveform distortion. For the latter measurement, compensate for the voltage ratio of the balun. Refer to the TW requirements for definitions of the above quantities.

Step 4: Repeat steps 2 to 3 using the bus B interface.

5.2.3.1.3.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. For each bus interface, the measured zero crossing deviation, rise and fall time, and waveform distortion.

c. Indication of compliance.

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5.2.3.1.4 Output noise.

The noise being output by the terminal when it is powered off and when it is receiving, shall be measured based on the following test configuration and procedure.

5.2.3.1.4.1 Test objectives.

Refer to the requirements described in 5.1.3.1.4 on page 18.

5.2.3.1.4.2 Test equipment.

a. True RMS voltmeter.

b. Terminating resistor RL of 75 Ω ±2%.

c. Low-pass filter with the following characteristics: pass-band stop frequency at 4 MHz, stop-band start frequency at 5 MHz, minimum stop-band attenuation of 60 dB.

Also, refer to the test equipment for the MCW output noise test in 5.2.2.3.5.2 on page 29.

5.2.3.1.4.3 Test configuration.

Refer to the test configuration in FIGURE 28.

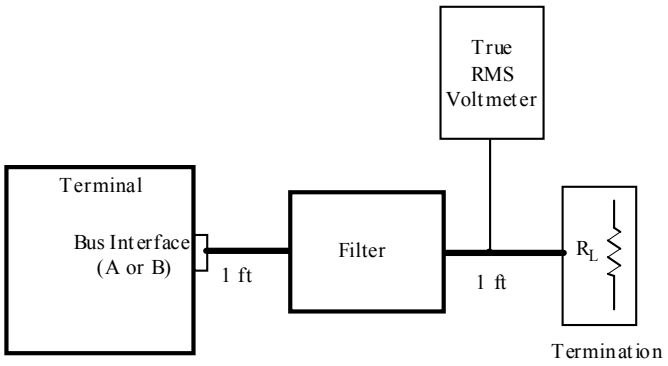


FIGURE 28. Test configuration 17.

5.2.3.1.4.4 Test procedure.

Step 1: Power off the terminal.

Step 2: Measure and record the maximum RMS voltage at the bus A interface.

Step 3: Repeat step 2 for bus B interface.

Step 4: Power on the terminal and place it into receive configuration.

Step 5: Repeat steps 2 to 3.

Step 6: Carry out the MCW output noise test in 5.2.2.3.5 on page 29.

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5.2.3.1.4.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The measured RMS voltage for the two configurations, for each bus interface.

c. Graphical data showing, for each bus interface, the measured noise power level versus frequency.

d. Indication of compliance.

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5.2.3.1.5 Output symmetry (transients).

The output transient FW signal level shall be measured based on the following test configuration and procedure.

5.2.3.1.5.1 Test objectives.

Refer to the requirements described in 5.1.3.1.5 on page 18.

5.2.3.1.5.2 Test equipment.

Refer to the test equipment for the output voltage level test in 5.2.3.1.1.2 on page 56.

5.2.3.1.5.3 Test configuration.

Refer to the test configuration in FIGURE 26 on page 56.

5.2.3.1.5.4 Test procedure.

Step 1: Dial in 20 dB attenuation on the programmable attenuator.

Step 2: Set up the terminal to continuously transmit a periodic signal on the bus A interface.

Step 3: Measure with the oscilloscope the maximum transient peak voltage level from 2.5 µs after the mid-bit crossing of the parity bit of the last word within the message to the time immediately preceding the start of the subsequent data message, compensating for the voltage ratio of the balun.

Step 4: Repeat steps 2 to 3 using the bus B interface.

5.2.3.1.5.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.
b. Measured maximum transient peak voltage, for each bus interface.
c. Indication of compliance.

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5.2.3.1.6 Clock frequency accuracy.

The clock frequency of the FW signal shall be measured based on the following test configuration and test procedure.

5.2.3.1.6.1 Test objectives.

Refer to the requirements described in 5.1.3.1.6 on page 19.

5.2.3.1.6.2 Test equipment.

a. Frequency counter.

5.2.3.1.6.3 Test configuration.

Refer to the test configuration in FIGURE 9 on page 31.

5.2.3.1.6.4 Test procedure.

Step 1: Connect the frequency counter to the terminal FW clock test connector or an equivalent test point.

Step 2: Record the clock frequency.

5.2.3.1.6.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The exact measured clock frequency.

c. Indication of compliance.

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5.2.3.2 Input characteristics.

5.2.3.2.1 Input impedance.

The input impedance of the terminal shall be measured based on the following test configuration and procedure.

5.2.3.2.1.1 Test objectives.

Refer to the requirements described in 5.1.3.2.1 on page 19.

5.2.3.2.1.2 Test equipment.

a. Vector network analyzer.

b. Calibrated 75 Ω / 50 Ω balun.

5.2.3.2.1.3 Test configuration.

Refer to the test configuration in FIGURE 29.

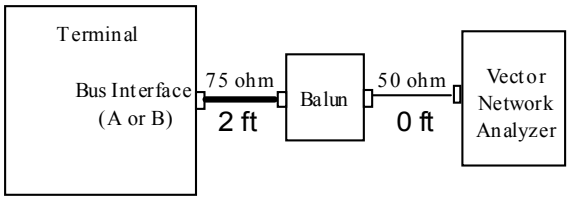


FIGURE 29. Test configuration 18.

5.2.3.2.1.4 Test procedure.

Perform an one-port calibration of the network analyzer at the 75 Ω port of the balun.

Step 1: Power off the terminal.

Step 2: Measure at the bus A interface the S_{11} reflection coefficient over the 75 kHz - 1 MHz frequency band and record the magnitude of the corresponding input impedance.

Step 3: Repeat step 2 for bus B interface.

Step 4: Power on the terminal and place it into receive configuration.

Step 5: Repeat steps 2 to 3.

5.2.3.2.1.5 Data presentation.

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. For each bus interface, graphical data showing the magnitude of the input impedance versus frequency, for both powered off and receive configuration.

c. Indication of compliance.

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5.3 Interface Requirements.

5.3.1 General.

Multi Carrier Waveform (MCW) PHY and MAC interface requirements contained in this section shall provide interoperability with Extended 1553 systems as stated in 5.4.

All interface requirements as specified herein shall be valid over the environmental conditions which the multiplex data bus system shall be required to operate over.

Interface requirements pertaining to FW and TW are identical to MIL-STD-1553B provided in 5.5 as TW requirements.

MCW signaling defines a synchronous Time Division Multiple Access (TDMA) communication system.

5.3.2 Independence of MCW from Filtered Waveform (FW) and Trapezoidal Waveform (TW).

FW and TW are to be separate and independent from MCW, including media access functions pertaining to MCW, which shall not be initialized, set up, or controlled through FW or TW.

5.3.3 MCW interface.

If a terminal is communicating with MCW, the terminal may only transmit FW simultaneously with the MCW. Either FW or TW may be received simultaneously with the MCW.PHY interface.

5.3.3.1 PHY interface.

5.3.3.2 Overview.

A terminal utilizing MCW is a bandwidth configurable transceiver capable to use 1 to 7 contiguous sub-bands spanning the 0 - 40 MHz frequency band. Each sub-band shall be of 5 MHz bandwidth with 64 tones per sub-band. In addition, the first sub-band from 0 to 5 MHz may be used if there is no simultaneously assigned FW or TW signaling.

The block diagram of the MCW transmitter is provided in FIGURE 30.

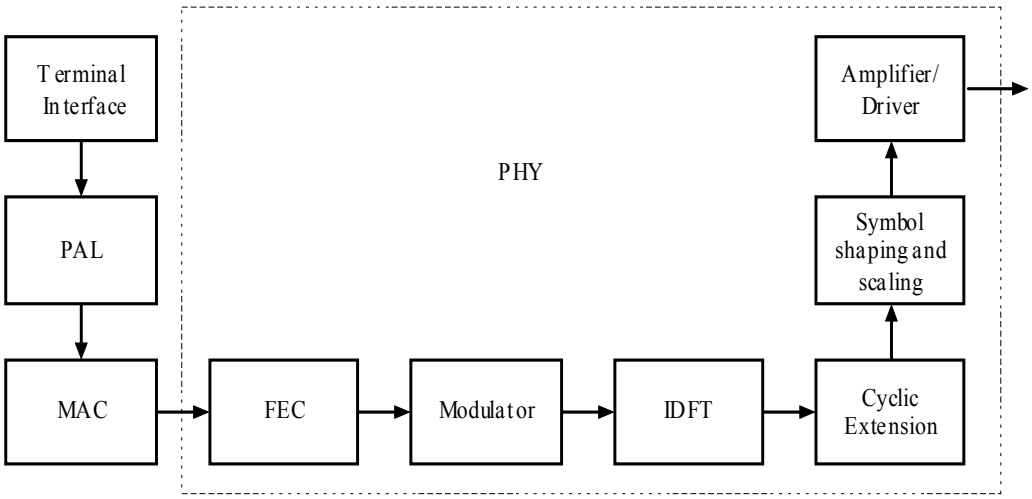


FIGURE 30. MCW PHY transmitter functional block diagram.

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The PHY shall include Forward Error Correction (FEC) and two modulation types known as Basic Rate (BR) and Variable Bit Rate (VBR) modulation. DBPSK shall be used for low rate BR modulation which is employed with Frame Control information and BR communications (such as bus command frames). Coherent QAM up to 4096 point constellations (12 bits / tone) shall be used for higher rate VBR modulation.

The 512 (maximum) tones over the 0 - 40 MHz will be processed through 1024 point Inverse Discrete Fourier Transform (IDFT) and Discrete Fourier Transform (DFT).

TABLE VI specifies the sub-band tone number and frequency assignment.

TABLE VII lists the basic PHY parameters.

TABLE VI. Sub-band tone number and frequency assignment.

Sub-band	Tone Number	Frequency [MHz]
0	0 – 63 (FW/TW Band)	0 - 4.93
1	64 - 127	5 - 9.92
2	128 - 191	10 - 14.92
3	192 - 255	15 - 19.92
4	256 - 319	20 - 24.92
5	320 - 383	25 - 29.92
6	384 - 447	30 - 34.92

TABLE VII. Basic PHY parameters.

Parameter	Value
IDFT / DFT size (N_{dft})	1024
Tone spacing [kHz]	78.125
Sub-band size	64 tones 5 MHz
Max. number of sub-bands	7

5.3.3.3 Frame structure and modulation modes.

The PHY frame structure is shown in FIGURE 31 and shall consist of a synchronization preamble (composed of 5 Sync Symbols), 8 Frame Control (FC) symbols, and a Data/Training Payload (PL) of multiples of 8 OFDM symbols.

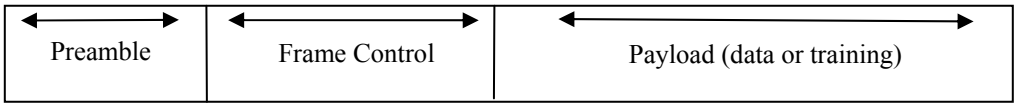


FIGURE 31. PHY frame structure.

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TABLE VIII contains the PHY frame parameters. As a minimum, a PHY frame shall contain the Preamble and the FC.

TABLE VIII. PHY frame parameters.

Parameter	Units	Value
Preamble size	Sync Symbols	5
PHY block size	OFDM symbols	8
FC size	PHY Blocks	1
PHY Frame Size (FC +PL)	PHY Blocks	1 to 31

5.3.3.3.1 Preamble.

The preamble is necessary to detect the start of frame, calculate symbol timing and estimate the channel response. Five Sync Symbols shall be used in the preamble.

5.3.3.3.1.1 Sync Symbols.

Sync Symbols shall be generated as described next. If the transmission band spans N_t consecutive tones, then these tones, indexed $k_0 \ldots k_0 + N_t - 1$, are modulated with S_{k_0+k} given below:

$$S_{k_0+k} = \exp\left[j\frac{\pi k^2}{N_t} \right],\ k = 0 \ldots N_t - 1$$

5.3.3.3.1.2 Time-reversed Sync Symbols.

In order to distinguish between the bus command frame and the RT payload frame within a MAC compound transaction, the RT Sync Symbols shall be time-reversed as described next. If the bus command frame Sync Symbol samples are generated by modulating tone k with S_k as given in 5.3.3.3.1.1, then the RT Sync Symbol samples are generated by modulating tone k with the complex conjugates S_k^*.

5.3.3.3.2 Frame Control (FC).

Eight Frame Control symbols shall be used to carry PHY frame information. FC data shall be protected with a 16-bit Cyclic Redundancy Check (CRC), Forward Error Correction (FEC) encoded, and transmitted using DBPSK modulation.

5.3.3.3.3 Data.

Data symbols shall be coded into blocks of 8 OFDM symbols, termed a PHY block, and carry information using Basic Rate (BR) or Variable Bit Rate (VBR) modulation. Both BR and VBR data shall be protected with a 32-bit CRC and encoded using FEC.

5.3.3.3.3.1 Basic Rate (BR) modulation.

Basic Rate uses DBPSK modulation to allow for low rate communications to occur before clock recovery enables higher rate coherent VBR modulation.

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5.3.3.3.3.2 Variable Bit Rate (VBR) modulation.

Variable Bit Rate modulation (VBR) is a coherent modulation scheme that uses a negotiated, "optimum" tone map for data communication over any particular channel. Each tone shall be assigned a bit capacity of 0, 1, 2, 4, 6, 8, 10 or 12 bits and QAM modulated.

5.3.3.3.4 Training.

The training symbol structure shall be the same as for data symbols except that training symbols are 4-QAM modulated with pseudo-random data. Training symbols are used to stimulate the channel with the same statistical signaling as data symbols. There is no CRC protection or FEC encoding for training symbols.

5.3.3.4 FEC.

The FEC encoder for FC and Basic Rate shall consist of a scrambler, Reed-Solomon and convolutional encoders, interleaver and repetition coding (diversity). The encoder for the VBR payload shall consist of scrambler, Reed-Solomon and convolutional encoder, puncturing, and interleaver. TABLE IX summarizes the usage of the individual encoders and the corresponding code rates.

TABLE IX. Encoders and code rates.

Modulation Type	Convolutional Code	Reed-Solomon Code	Diversity Order
FC	1/2	≤ 239/255	not more than the number of configured sub-bands
Basic Rate	1/2	≤ 239/255	
VBR	1/2 or 3/4 or 7/8	≤ 239/255	1

5.3.3.4.1 Scrambler.

The data field shall employ a scrambler of length 127. The input bytes are converted into a serial bit stream with the most significant bit first and least significant bit last. The scrambler shall use the polynomial:

$$S(x) = x^7 + x^4 + 1$$

The scrambler shall be of the form shown in FIGURE 32. The initial value loaded into the shift registers is 0x7F for PL and 0x40 for FC. The scrambler's state is reinitialized at the beginning of each PHY block.

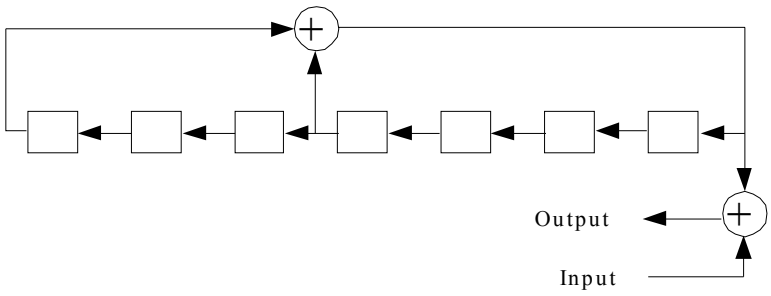


FIGURE 32. Data scrambler.

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5.3.3.4.2 Reed-Solomon encoder.

A (255, 239) Reed-Solomon code defined over the GF(256) Galois field (RS symbol length of 8 bits) is used in order to provide some level of burst error correction capability. Based on the number of bits carried in each PHY block, it may be necessary to use a shortened version of the RS code, i.e., a rate $(239-s)/(255-s)$ code, where s can be an integer between 0 and 225. The number of redundant RS symbols is 16 and the code can correct errors of up to 8 RS symbols (bytes). The field generator polynomial is $f(x) = x^8 + x^4 + x^3 + 1$ (435 octal). In forming the RS symbols, the most significant bit first and least significant bit last convention is used.

When more than one Reed-Solomon code word is contained within one PHY block, all code words must be of equal length. Denoting by b the total number of bytes available to load within one PHY block with Reed-Solomon code words (calculated based on tone map and puncture rate), the number N_{RS} and length n of the code words is found as given below:

$$N_{RS} = \text{floor}\left(\frac{b+239}{255}\right)$$

$$n = \min\left\{254, \text{floor}\left(\frac{b}{N_{RS}}\right)\right\}$$

5.3.3.4.3 Convolutional encoder.

The data field shall be encoded with a convolutional encoder with coding rate R = 1/2. The convolutional encoder shall use the generator polynomials $Y = 133_8$ and $X = 171_8$ as shown in FIGURE 33. The bit denoted as "X" shall be output from the encoder before the bit denoted as "Y." Higher data rates are derived from the basic code by puncturing.

The RS bytes are converted into a serial bit stream with the most significant bit first and least significant bit last. The convolutional encoder works on a PHY block. At the end of each PHY block there shall be 8 flush bits (8 zeros) inserted at the input in order to return the convolutional encoder to the all zero state.

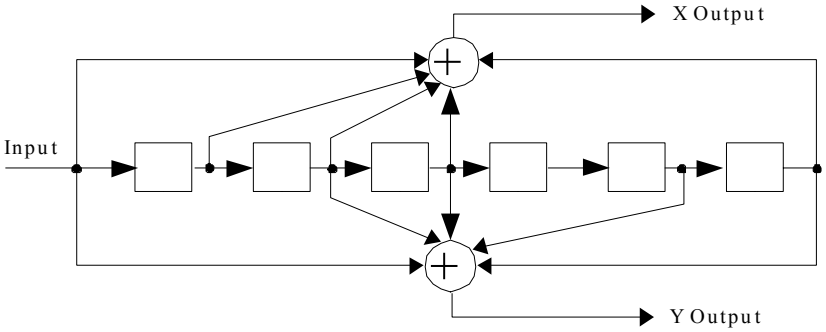


FIGURE 33. Convolutional encoder.

5.3.3.4.4 Puncturing.

In order to increase the maximum possible bit rate with VBR modulation, puncturing may be used to create rate 7/8 and rate 3/4 convolutional codes. The puncturing matrix for the rate 7/8 code is shown in FIGURE 34 with the punctured bits highlighted in gray. The corresponding output bit stream of the puncture block is shown in FIGURE 35. Bits X0 and Y0 are the first ones out of the convolutional encoder.

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X0	X1	X2	X3	X4	X5	X6
Y0	Y1	Y2	Y3	Y4	Y5	Y6

FIGURE 34. Puncturing matrix for code rate 7/8.

X0	Y0	Y1	Y2	Y3	X4	Y5	X6

FIGURE 35. Output bit stream for code rate 7/8.

The puncturing matrix for the rate 3/4 code is shown in FIGURE 36, while the output bit stream is shown in FIGURE 37.

X0	X1	X2
Y0	Y1	Y2

FIGURE 36. Puncturing matrix for code rate 3/4.

X0	Y0	Y1	X2

FIGURE 37. Output stream for code rate 3/4.

5.3.3.4.5 Zero padding.

Frequently, the length of the (optionally punctured) convolutional code word is not exactly the size of the interleaver matrix. In this case, zero padding shall be performed before interleaving in order to fill the gap.

5.3.3.4.6 Interleaver.

The purpose of the interleaver is to spread out time-localized error bursts to the whole PHY block in order to present more random-like errors to the convolutional decoder.

The interleaver matrix shall hold all the bits within one PHY block (corresponding to 8 OFDM symbols). Each OFDM symbol contains all the bits carried by all the available tones. The number of bits per tone is decided by the modulation of each tone. The interleaver parameters are provided in TABLE X.

TABLE X. Interleaver parameters.

Parameter	**Value**
Interleaver matrix width (W)	8
Column-to-column shift (S)	6

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FIGURE 38 shows the order of writing to the interleaver matrix. The gray cells represent the first cell that is written to in each row.

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39
40	41	42	43	44	45	46	47
48	49	50	51	52	53	54	55
56	57	58	59	60	61	62	63

FIGURE 38. Writing to the interleaver matrix.

FIGURE 39 shows the order of reading from the interleaver matrix. Gray cells represent the first cell which is read from in each column.

0	10	20	30	32	42	52	62
1	11	21	31	33	43	53	63
2	12	22	24	34	44	54	56
3	13	23	25	35	45	55	57
4	14	16	26	36	46	48	58
5	15	17	27	37	47	49	59
6	8	18	28	38	40	50	60
7	9	19	29	39	41	51	61

FIGURE 39. Reading from the interleaver matrix.

The number of bits per OFDM symbol when N_t tones are used for transmission shall be calculated as:

$$Bits_per_OFDMsymbol = \sum_{k=1}^{N_t} b_k$$

where b_k is the number of bits carried on the k-th of the N_t tones. Then, the number of bits in one PHY block is:

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$$Bits_per_PHYblock = 8 \cdot Bits_per_OFDMsymbol$$

In practice, the data is stored as a vector instead of a matrix. In this case, the element at $index_{out}$ in the output vector is obtained by reading out the element at $index_{IN}$ in the input vector, as given by the relation:

$$index_{IN} = W \cdot index_{OUT} - (N - 1 - S \cdot W) \cdot \text{floor}\left(\frac{W \cdot index_{OUT}}{N}\right) \quad \text{modulo } N$$

In the equation above $index_{OUT}$ is from 0 to $N-1$, N is equal to $Bits_per_PHYblock$, W is the interleaver width and S is the column-to-column shift.

5.3.3.4.7 Repetition coding (diversity).

Repetition coding (diversity) of Basic Rate and FC shall be the same. The diversity order is a network-wide programmable parameter with values from 1 to a maximum that is equal to the number of sub-bands configured for communication. When the number of available tones is less than 128 (two sub-bands), the diversity order shall be 1. When the number of available tones N_t is at least 128, the diversity order D shall be chosen to be a divisor of N_t. The bits sent over the lower $M = N_t/D$ available tones shall be repeated on the higher frequency tones. Denoting by b_k, $k = 0 \ldots M-1$, the bits sent over the lower M tones, the repetition shall be according to the relation:

$$b_{M \cdot p+k} = b_k, \quad p = 1 \ldots D-1, \quad k = 0 \ldots M-1$$

The total number of bytes available to load within one PHY block with Reed-Solomon code words (denoted by b in 5.3.3.4.2) is calculated according to the relation:

$$b = \text{floor}\left(\frac{M}{2}\right) - 1$$

The division by 2 accounts for the convolutional code rate and the "-1" for the flush byte.

5.3.3.5 Training generator.

This is a pseudo-random number generator used to generate bits which then select points in a 4-QAM constellation during training. The training generator uses two feed-back shift registers as shown in FIGURE 40. At the start of each training frame, the shift registers shall be initialized to the values shown below:

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$S_1(x) = X^7 + X^4 + 1$, initial value = 0x7F

$S_2(x) = X^6 + X^3 + 1$, initial value = 0x3F

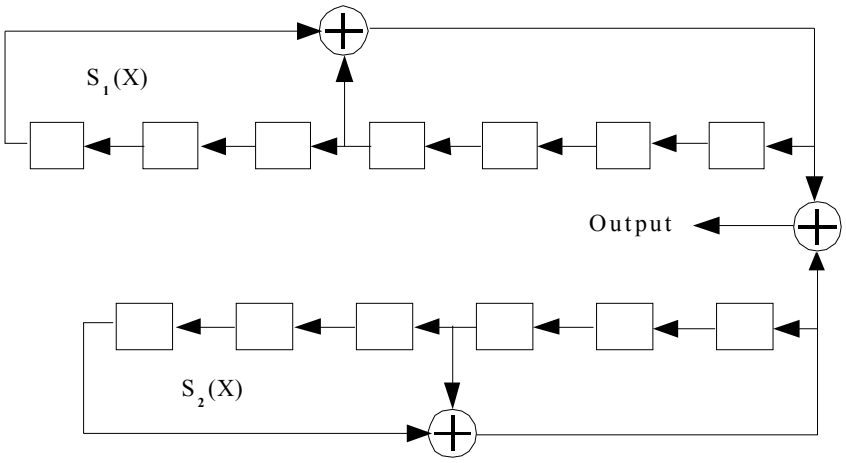


FIGURE 40. Training generator.

5.3.3.6 OFDM symbol generation.

5.3.3.6.1 Data formatter.

The Data Formatter shall organize the bits from the FEC into a format that is compatible with the QAM modulator. For mapping to a 2^{2M}-QAM constellation, the first M bits are mapped to the in-phase component while the next M bits to the quadrature component. In both cases, the last bit in is placed on the least significant position.

5.3.3.6.2 Spectral mask.

In order to reduce the interference caused by the MCW signal at frequencies other equipment shows EMI susceptibility to, a programmable spectral mask shall be used. The mask shall prevent transmission of the restricted tones. These tones shall be selected as part of the network configuration process.

5.3.3.6.3 Modulator.

The modulator shall perform QAM, BPSK and DBPSK modulation as well as sync phase modulation.

5.3.3.6.3.1 QAM modulator.

The modulator shall accept coded bits from the Data Formatter and map them to complex constellation points scaled such that all constellations have almost the same average energy. The result is a complex number for each tone.

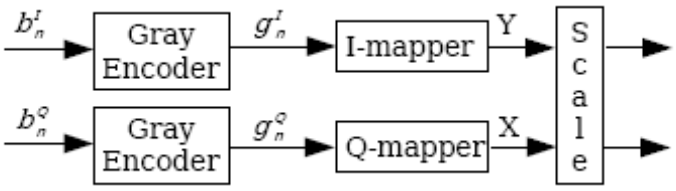


FIGURE 41. QAM modulator.

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FIGURE 41 shows the block diagram of the QAM modulator. The Gray Encoder blocks shall encode the b_n^I in-phase bits and the b_n^Q quadrature bits according to the relation:

$$g_{M-1} = b_{M-1,}$$
$$g_k = g_{k+1} \text{ xor } b_k, 0 \le k \le M-2$$

where the value of M depends on the constellation size (e.g. $M = 6$ for 2^{12}-QAM). The mapper blocks shall translate the Gray coded bits to X and Y signal coordinates in the QAM constellation. In particular, taking the decimal value of $\{g_n^I\}$, $n \in \{0,\ldots,M-1\}$ the in-phase component is mapped with zero corresponding to the most negative and the full scale to the most positive coordinate. For the quadrature component the mapping is the exact mirror of the in-phase mapping: zero corresponds to the most positive and the full scale to the most negative coordinate. This mapping is according to the relations:

$$X = -\left(2^M - 1\right) + 2 \cdot g^I$$
$$Y = +\left(2^M - 1\right) - 2 \cdot g^Q$$

where

$$g^I, g^Q \in \left\{0,\ldots,2^M - 1\right\}$$

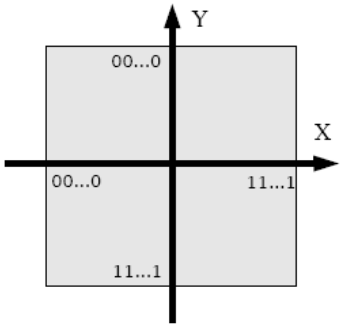


FIGURE 42. I/Q mapping.

The scaler is common to both in-phase and quadrature channels and assures the approximately same average constellation energy. The scaling values are given in TABLE XI.

TABLE XI. Constellation scaling values.

	BPSK	4-QAM	16-QAM	64-QAM	256-QAM	1024-QAM	4096-QAM
Scale Factor	1	1/2	1/4	1/8	1/16	1/32	1/64

5.3.3.6.3.2 BPSK modulator.

The BPSK modulator performs the 0→1 and 1→ -1 mapping as shown in TABLE XII, where I refers to the in-phase and Q the quadrature component of the output.

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TABLE XII. BPSK modulation.

Input bit	I-out	Q-out
0	1	0
1	-1	0

5.3.3.6.3.3 DBPSK modulator.

The DBPSK modulator changes the phase of the output by π when the input bit is 1, as shown in TABLE XIII. The initial phase of the output is zero. Modulation of the Basic Rate payload continues that of the FC, without resetting the memory of the DBPSK modulator.

TABLE XIII. DBPSK modulation.

Input bit	Output phase change
0	0
1	π

5.3.3.6.3.4 Sync phase modulation.

Sync phase modulation is a per-tone multiply of the complex constellation point (BPSK, QAM, DBPSK) with the complex sync phasor of the corresponding tone. The complex numbers modulating the tones are the ones given in 5.3.3.3.1.1. Sync phase modulation is carried out for the FC as well as the data and training payloads.

5.3.3.6.4 IDFT.

After a constellation point for each tone is chosen, complex conjugates of these values shall be input to the upper frequency half of the IDFT in a manner that ensures the time domain samples are all real. Any residual imaginary component shall be discarded. The IDFT inputs corresponding to the DC and the Nyquist frequency tones shall be set to zero, i.e. $X_0 = X_{N_{dft}/2} = 0$.

5.3.3.6.5 Time domain processing (TDP).

FIGURE 43 illustrates the TDP operation for FC and payload. FIGURE 44 illustrates the TDP operation for the Sync Preamble. These operations shall consist of cyclic extension, windowing, overlap with previous symbol and scaling.

To provide transmit spectral shaping and thus reduce spectrum splatter, the OFDM symbol is multiplied by a raised-cosine window to more quickly reduce the power of out-of-band tones. The amount of roll-off is fixed and tied to the transmit window size. The raised-cosine shaping uses the asymmetric window given by the equation below:

$$y(n) = \left\{1 - \cos\left(\frac{n \cdot \pi}{n_{win}}\right)\right\}, \quad n = 0 \ldots n_{win}$$

The TDP scaling unit is used to scale the time-domain IDFT output signal to the desired transmit level in order to control both the amount of clipping and the overall transmit power.

The default values and the allowed range of the TDP parameters are provided in TABLE XIV.

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TABLE XIV. TDP parameters.

Parameter	Number of Samples
Tx window size (n_win_tx)	16
CP+CS size	configurable; default = 272 range = 128 * k + n_win_tx, 1 ≤ k ≤ 8
CP size (n_prefix)	configurable; default = 240
CS size (n_suffix)	configurable; default = 32
1st SYNC Symbol size	n_dft + n_win_tx = 1040
2nd to 4th Sync Symbol size	n_dft = 1024
samples borrowed by 5th Sync Symbol from 1st FC symbol (n_borrow)	configurable; default = 64 range = 2 … 64
5th Sync Symbol size	configurable; default = 1088 range = n_fft + n_borrow
1st FC symbol size	configurable, default = 1232 range = n_dft + n_prefix + n_suffix – n_borrow
2nd to 8th FC symbol and PL symbol size	n_dft + n_prefix + n_suffix; default = 1296
symbol overlap size	n_win_tx
TDP scaler	configurable, default = 0dB Range: - 20 dB ... +20 dB (in 0.5 dB steps) -66 dB … -20 dB and 20 dB … 66 dB (in variable step size)

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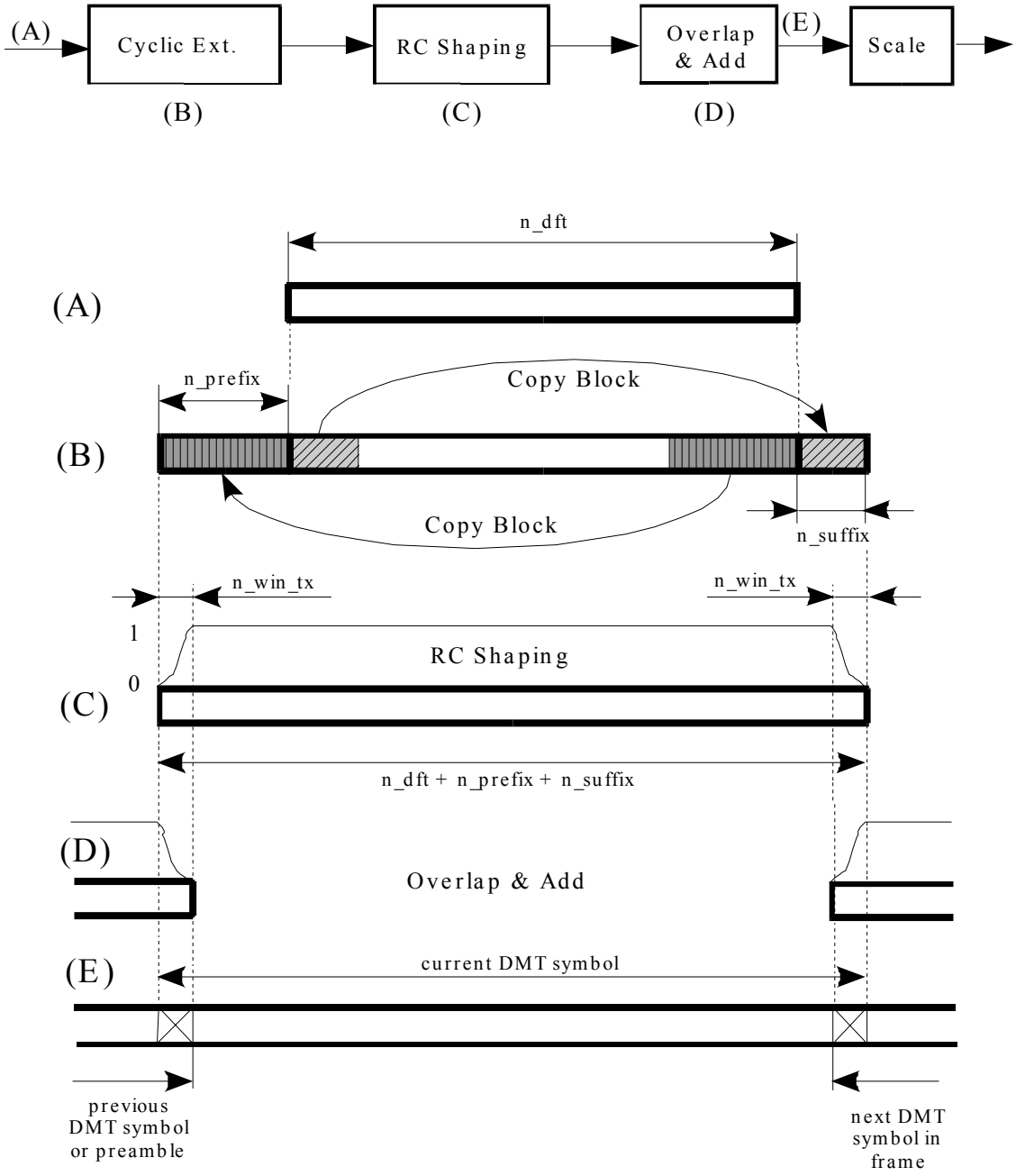


FIGURE 43. Time domain processing of FC and Payload.

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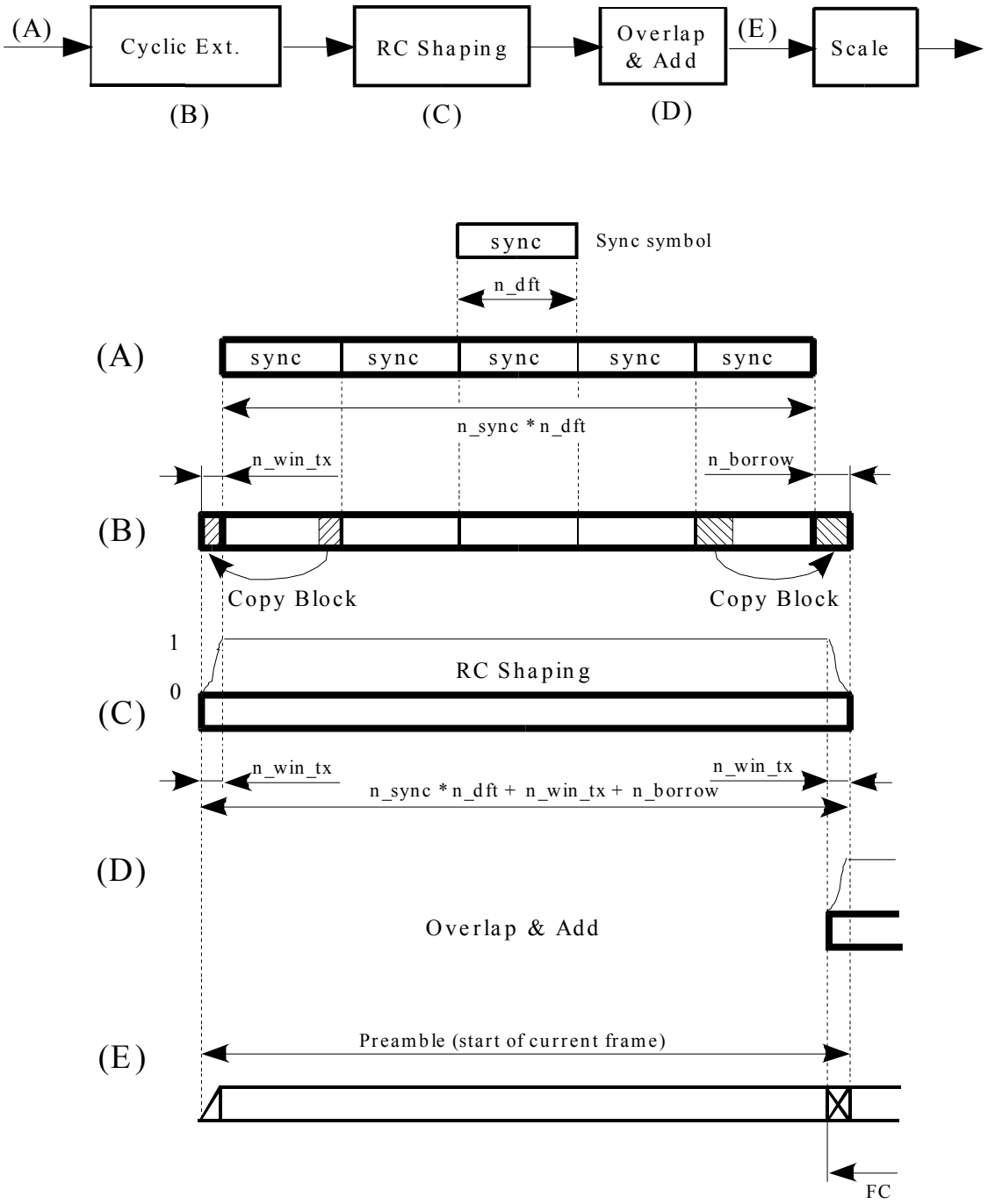


FIGURE 44. Time domain processing of Preamble.

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5.3.4 MAC protocol.

5.3.4.1 General.

Each terminal on a bus has a configurable unique address, assigned during system configuration. These addresses are used at the PHY layer for communications between terminals. The PHY layer controls the timing and length of MAC layer transmissions as directed from the Bus Controller, through a configured bus schedule.

5.3.4.2 Command-Response protocol.

While the command-response protocol has some similarities to the TW protocol, the two protocols are not the same. In particular, while a TW transaction consists of one command-response message exchange, a MCW transaction may consist of a set of several command-response message exchanges. This section describes the MCW command-response protocol under the following headings:

a. MCW compound transactions.

b. MCW frames.

5.3.4.2.1 Compound Transaction.

As referenced above, MCW uses a Compound Transaction, which consists of a set of commands, the corresponding terminal responses, and their terminal acknowledgements. The structure of a Compound Transaction is shown in FIGURE 45.

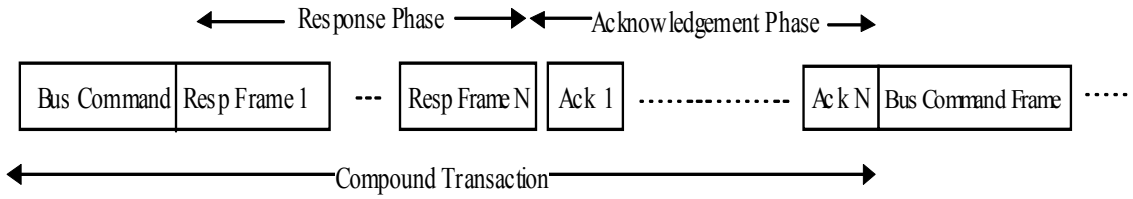


FIGURE 45. Compound transaction structure.

A Compound Transaction consists of the following message components:

a. A Bus Command Frame (BCF) containing the set of commands from the BC, consisting of Frame Exchange (FE) commands and Bus Notification (BN) commands.

b. A Terminal Response Frame (TRF) for each individual command during the response phase.

c. Zero or one Terminal Acknowledgement Frame (TAF) for each Terminal Response Frame (TRF) during the acknowledgement phase.

The TRFs, which commonly contain transmitted data, training data or test data, are generated by the transmitting terminals in the same order as the corresponding BCF exchange commands. All TRFs are transmitted before any terminal acknowledgement frames are transmitted.

The Terminal Acknowledgement Frames (TAF)s are transmitted by the terminals which received the (TRF)s if requested to do so by the Frame Exchange (FE) command. The first TAF is transmitted after the last TRF. The acknowledgement frames are transmitted in the same order as the response frames.

For each unsuccessfully received response frame, the BC may decide to include a retry command for that frame in a subsequent Compound Transaction.

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In addition, the following frame types may also be transmitted during the Response Phase:

a. Zero or more MAC Management Frames.

b. Zero or more Training Frames (TF)s.

c. Zero or more Arbitrary Test Frames (ATF)s.

5.3.4.2.2 Frames.

At the command-response protocol level control of the MAC sub-layer of the DLL, the basic unit of MCW information exchange between terminals is a PHY frame. The different types of MCW frames are discussed under the following headings:

a. General MCW frame structure.

b. Bus Command Frame (BCF).

c. Sync Bus Command Frame (SBCF).

d. Terminal Response Frame (TRF).

e. Terminal Acknowledge Frame (TAF).

f. MAC Management Frame (MMF).

g. Training Frame (TF).

h. Arbitrary Test Frame (ATF).

5.3.4.2.2.1 General MCW frame structure.

The general structure of a MCW frame is shown in FIGURE 46.

Sync (Preamble)	Frame Control	Payload (optional)

FIGURE 46. Frame structure.

A MCW frame contains three sections:

a. A Sync (preamble) section used for synchronization between terminals.

b. A Frame Control (FC) section, which contains terminal address information, the length of the frame, etc.

c. A Payload section, which contains any data being transmitted between the terminals. The payload section is optional depending on the frame type.

5.3.4.2.2.1.1 Sync section.

This section is added by the PHY layer (referenced in 5.3.3.3.1.1).

5.3.4.2.2.1.2 Frame Control (FC) section.

The 15-byte Frame Control (FC) contains information to determine the type and size of the data encoded in the payload section of the frame, as shown in TABLE XV. The reserved bytes are set to zeroes.

Transmission is by DBPSK modulation and it is protected with 16 bit CRC and encoded with a Hamming code, Reed Solomon Code and unpunctured convolutional code.

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TABLE XV. Frame Control (FC) fields.

Field	Byte	Bit Number	Number of Bits
Frame Type	0	7-5	3
Frame Size		4-0	5
Variable Field [31-24]	1	7-0	32
Variable Field [23-16]	2	7-0	
Variable Field [15-8]	3	7-0	
Variable Field [7-0]	4	7-0	
Reserved	5	7-0	8
…	…	…	…
Reserved	12	7-0	8
CRC-16 [15-8]	13	7-0	16
CRC-16 [7-0]	14	7-0	

5.3.4.2.2.1.2.1 Frame Type (FT) field.

The 3-bit Frame Type (FT) field specifies the type of data encoded in the payload section of the frame as shown in TABLE XVI.

TABLE XVI. Frame Type (FT) field.

Value	Name
0	Data Response
1	Bus Command
2	Sync Bus Command
3	MAC Management
4	Acknowledgement
5	Training
6	Arbitrary Test Frame
7	Reserved

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5.3.4.2.2.1.2.2 Frame Size field.

The Frame Size field specifies the size of the payload in 8-symbol blocks. Valid values of this field are 1- 30 except for Acknowledgement Frames that must have this field set to 0.

5.3.4.2.2.1.2.3 Variable field.

The use of the Variable field depends upon the frame type. See the specific sections on Frame Type for specific use.

5.3.4.2.2.1.2.4 CRC-16 field.

The CRC-16 Field is used to detect errors in the FC section and has a generator polynomial $G(X) = X^{16} + X^{12} + X^5 + 1$. For CRC-16 calculations, each byte is sent least significant bit (LSB) first. Bytes 0-4 are included in the calculation in network order (Byte 0 first). The initial value of the calculation is all ones (0xFFFF).

5.3.4.2.2.1.3 Payload section.

The optional payload section referenced in FIGURE 47 consists of a sequence of bytes followed by a 4-byte CRC-32 field for error detection. The usage of the data portion of the payload depends on the frame type and is described in detail in the corresponding sections. The length of the payload section is in multiples of PHY blocks.

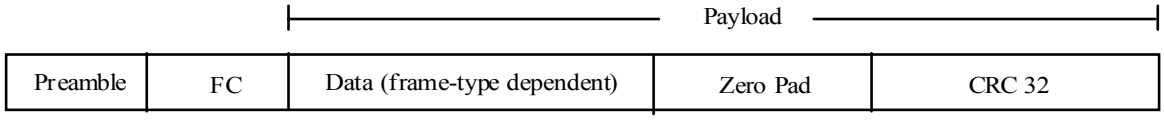


FIGURE 47. Payload section.

5.3.4.2.2.1.3.1 Zero pad.

If the entire usable space of the payload is not used, the unused bytes are filled with zeroes.

5.3.4.2.2.1.3.2 CRC-32 field.

The CRC field always occupies the last 4 bytes of the payload section. It has a generator polynomial

$$G(X) = X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1$$

For CRC-32 calculations, each byte is sent least significant bit (LSB) first. All bytes from the header and data sections are included in the calculation in network order (Byte 0 first). The initial value of the calculation is all ones (0xFFFFFFFF).

5.3.4.2.2.2 Bus Command Frame (BCF).

The BCF is used to convey commands from the Bus Controller (BC). It is broadcast to all terminals on the bus by the BC with the issued set of commands contained in the payload. The BCF can transmit from 1 to 16 commands that define the bus schedule of the compound transaction and because it is broadcast must use BR modulation that can be easily understood by all terminals.

The order of the commands determines the transmission order of the terminals during the compound transfer. Each Bus command contains information regarding:

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a. Which terminal is to transmit.

b. How much data is to be transmitted.

c. The destination terminal for the transmission.

d. How long the transmission is to last, and

e. Whether the transmission is to be a retry of an earlier unsuccessful transmission.

Each Bus Command is 4 bytes long. The first command is contained in the variable field of the FC section with additional commands, if present, contained in the payload. The structure of the BCF is shown in FIGURE 48.

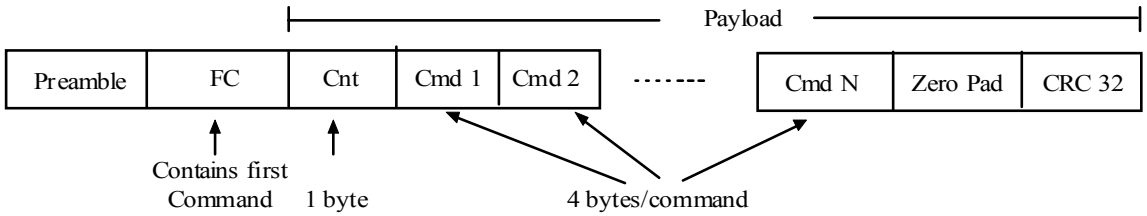


FIGURE 48. Bus Command Frame (BCF) structure.

The size of the BCF is configured at startup and remains fixed during operation. The payload size is in multiples of basic rate PHY blocks and is at least one PHY block.

5.3.4.2.2.2.1 Count field.

The Count field indicates how many additional commands are contained in the payload. It is followed by additional 4-byte commands.

5.3.4.2.2.2.2 Command Type field.

The Command Type Field defines the action to be executed as follows:

a. Command Types 0-5 are Frame Exchange commands and cause a frame exchange to be scheduled.

b. Command Type 6 is a notification command and does not cause a frame exchange to be scheduled. It is used to convey information contained within the command from the BC to the RTs. Valid values are shown in TABLE XVII.

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TABLE XVII Bus Command Frame (BCF) command categories.

Command Type	Definition	Description
0	Critical Data	Source RT transmits a critical data frame to Destination RT.
1	Non-Critical Data	Source RT transmits a non-critical data frame to Destination RT.
2	Training	Source RT transmits a Training Frame (TF) to Destination RT.
3	Tone Map	Source RT transmits a Tone Map to Destination RT.
4	MAC Management	Source RT transmits a MAC Management frame to Destination RT.
5	Arbitrary Test Frame	Source RT transmits a frame containing arbitrary test data in the payload section to Destination RT. The payload content is vendor specific.
6	Notification	BC notification.
7	Reserved	Ignored by RTs.

5.3.4.2.2.2.2.1 Frame Exchange (FE) commands.

Command Type 0-5 commands are Frame Exchange (FE) commands and have the format described in TABLE XVIII.

TABLE XVIII. Frame Exchange (FE) command fields.

Field	Byte	Bit Number	Bits	Applies to Command Type
Command Type	0	7-5	3	N/A
PHY Block Count		4-0	5	0,1,2,3,4,5
Rx Bus	1	7	1	0,1,2,3,4,5
Acknowledgement Flag		6	1	0,1,2,3,4,5
Source RT		5-0	6	0,1,2,3,4,5
Destination RT	2	7-2	6	0,1,2,3,4,5
Tone Map Index		1-0	2	0,1,3,4
Queue ID	3	7-1	7	0,1
Retry Flag/ Clock Sync Only		0	1	0,1 / 2

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5.3.4.2.2.2.2.1.1 PHY Block Count field.

The PHY Block Count field specifies the number of 8-symbol PHY blocks in the payload section of the scheduled frame. Valid values are 1-30.

5.3.4.2.2.2.2.1.2 Rx Bus Field.

The Rx Bus field specifies the bus used by the destination RT to receive the frame. A value of 0 indicates the A bus, 1 indicates the B bus.

5.3.4.2.2.2.2.1.3 Acknowledgement flag field.

The Acknowledgement flag field is set to 1 to command the destination RT to transmit a terminal acknowledgement frame indicating the receive status of a scheduled frame. No acknowledgement is sent if this field is set to 0. This bit is set to 0 for broadcast frames.

5.3.4.2.2.2.2.1.4 Source RT field.

The Source RT field specifies the source terminal of the response frame. Valid values are 0-62.

5.3.4.2.2.2.2.1.5 Destination RT field.

The Destination RT field specifies the destination terminal of the response frame. Valid values are 0-62 for unicast frames. A value of 63 (0x3f) is the broadcast address, indicating that all RTs shall process the received frame. The Basic Rate Tone Map must be used and the Acknowledgement Flag must be set to 0 for broadcast frames.

5.3.4.2.2.2.2.1.6 Tone Map Index field.

The Tone Map Index field indicates the tone map to be used by the source RT. Valid values are in TABLE XIX.

TABLE XIX. Tone Map index field.

Tone Map Index	Definition	Description
0	Basic Rate	Use Basic Rate Tone Map. This is understood by all RTs and must be used for broadcast frames.
1	VBR 1	Use VBR Tone Map 1 configured for the Source-Destination RT pair.
2	VBR 2	Use VBR Tone Map 2 configured for the Source-Destination RT pair.
3	VBR 3	Use VBR Tone Map 3 configured for the Source-Destination RT pair.

5.3.4.2.2.2.2.1.7 Queue ID field.

The Queue ID field indicates which queue to use for sourcing data. Valid values are 0-127.

5.3.4.2.2.2.2.1.8 Retry/Clock Sync Only flag.

The Retry/Clock Sync Only flag serves two purposes depending on the frame type. For data frames (types 0 and 1), this is a Retry flag. A value of 1 indicates that scheduled frame is a retry of a previous transmission. For TFs (type

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2), this is a Clock Sync Only flag. A value of 1 indicates that the TF is used for synchronization only and not for tone map generation.

5.3.4.2.2.2.2.2 Notification commands.

Command Type 6 are Notification Commands and have the format described in TABLE XX.

TABLE XX. Notification command fields.

Field	Byte	Bit Number	No. of Bits	Definition
Command Type	0	7-5	3	Set to 6 (110)
Notification Subtype		4-0	5	See TABLE XXI
Target RT	1	7-2	6	Command target
Reserved		1-0	2	
Variable Guard Slot [15-8]	2	7-0	16	Variable Guard Slots
Variable Guard Slot [7-0]	3	7-0		

5.3.4.2.2.2.2.2.1 The Notification Subtype field.

The Notification Subtype field defines the action, as described in TABLE XXI.

TABLE XXI. Notification Subtype field.

Notification Subtype	Definition	Description
0	Null Event	This notification is used when the BC has no other command to broadcast. The BC must periodically Broadcast Sync frames (BCF)s to maintain clock synchronization among RTs.
1	Variable Guard Slot	This notification directs the RTs to replace the last guard slot with a variable length guard slot specified in the Variable Guard Slot field. This notification is used to perform precision adjustments between Sync Bus Command Frames (SBCF)s. The duration is configurable to 1024k + (1024 – 16) samples, where $0 \leq k \leq 63$.
2 – 31	Reserved	

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5.3.4.2.2.3 Sync Bus Command Frame (SBCF).

The Sync Bus Command Frame (SCBF) is sent periodically at a precisely programmed rate by the BC. The SBCF indicates the start of a group of compound transactions and allows RTs to coordinate operations with the BC. Its format is identical to a normal Bus Command Frame but with the Frame Type set to 2.

a. Terminal Response Frame (TRF).

The Terminal Response Frame (TRF) contains the transmitted data that was requested by a single command from the BC. Its structure is shown in FIGURE 49.

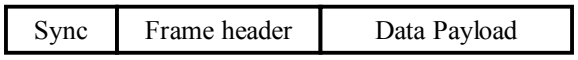


FIGURE 49. Terminal Response Frame (TRF) structure.

5.3.4.2.2.3.1.1 Terminal Response Frame (TRF) Structure.

5.3.4.2.2.3.1.1.1 Frame Header.

The frame header of the Terminal Response Frame (TRF) contains the source and destination terminal addresses, the length of frame, etc. The data being transmitted is contained in the payload section.

The Variable Field of the Frame Control (FC) header is unused and set to 0. The bus schedule determines when a source RT transmits and when a destination RT receives data frames and which queue is used to source the payload data. Communication occurs between a given source-destination RT pair as a result of the bus schedule; the frame does not otherwise have any addressing information.

5.3.4.2.2.3.1.1.2 Data Payload.

The data to be transmitted is contained in the payload section. This data is pulled at the source RT from the Queue ID specified by the BC and transmitted on the bus. The destination RT listens on the bus concurrently to receive the frame.

Terminal Response Frames may be Critical or Non-Critical. The BC controls which frames are serviced by setting the Command Type to 0 or 1 accordingly.

5.3.4.2.2.3.2 Unicast Frames.

When a Terminal Response Frame (TRF) is destined for a unique destination, the BC may command the destination RT to send a TAF during the Acknowledgement phase of the Compound Transaction. This is achieved by setting the Acknowledgement Flag to 1 in the FE command.

The TRF contains data that is being sent between two terminals on the network. The optimum encoding between those two terminals is determined by the unique network characteristics between those terminals. Thus, the data encoding used in the TRF may be specific to the source-destination pair between which data is being exchanged as defined by the tone map used.

5.3.4.2.2.3.2.1 Broadcast Frames.

When a Terminal Response Frame (TRF) is destined for the broadcast address (63 or 0x3F), the BC will suppress the generation of Terminal Acknowledgement Frames (TAF)s during the Acknowledgement phase of the Compound Transaction. This is achieved by setting the Acknowledgement Flag to 0 in the Frame Exchange command. The Terminal Response Frame shall use the Basic Rate Tone Map to ensure that all RTs can receive the broadcast frame.

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5.3.4.2.2.4 Terminal Acknowledgement Frame (TAF).

The Terminal Acknowledgement Frame (TAF) is used to indicate whether a TRF was received successfully. As shown in FIGURE 50, the TAF has no payload.



FIGURE 50. Terminal Acknowledgement Frame (TAF) structure.

5.3.4.2.2.4.1 Frame Header Variable field.

The Variable field is used to convey status to the source RT and the BC. Valid values are described in TABLE XXII.

TABLE XXII. Terminal acknowledge variable field status code.

	Byte	Bit Number	Bits	Definition
Response Type	0	7-5	3	See TABLE XXIII
Reserved		4-0	5	
Reserved	1	7-0	8	
Reserved	2	7-0	8	
Reserved	3	7-0	8	

TABLE XXIII provides the Terminal acknowledge response types.

TABLE XXIII. Terminal Acknowledge Response types.

Response Type	Name	Description
0	ACK	The Terminal Response Frame was received successfully by the destination RT.
1	NACK	The Terminal Response Frame was not received unsuccessfully due to a CRC error.
2	FAIL	The Terminal Response Frame had a correct CRC but the destination RT did not currently have the resources to process the frame.
3	NO_RX	The destination RT was not expecting the frame.
4-7	Reserved	

5.3.4.2.2.5 MAC Management Frame (MMF).

MAC Management Frames (MMF)s are handled in a similar way to Terminal Response Frames (TRF)s but are used for MAC management communications instead of application communications. The payload section contains a MAC Management Message, which is forwarded to the destination MAC Management entity.

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MAC Management Messages are used to exchange management and diagnostic messages between stations. The format of the MAC Management Message is shown in FIGURE 51.

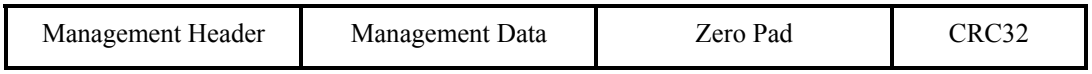


FIGURE 51. MAC Management header format.

5.3.4.2.2.5.1 Management header fields

The 4-byte message header is described in TABLE XXIV.

TABLE XXIV. MAC Management header fields.

Field	Byte	Bit Number	Bits	Definition
Reserved	0	7-0	8	Set to zeroes.
Message Version	1	7-6	2	Message Version
Message Type		5-0	6	Message Type
Message Data Length	2	7-0	8	Message Data Length [15-8]
	3	7-0	8	Message Data Length [7-0]

5.3.4.2.2.5.1.1 Message Version field.

The Message Version field is used to differentiate between this version of the protocol and other past or future versions. It is currently set to 0.

5.3.4.2.2.5.1.2 Message Type field.

The Message Type field indicates the contents of the message. Message types are described in TABLE XXV.

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TABLE XXV. MAC Management message types.

Message Type	Name	Description
0	Tone Map	Tone Map to use for subsequent transmissions to the RT that sent the message.
1	Sync Bus Command Frame Count	The count of the number of Sync Bus Command Frames transmitted by the BC. This is used by RTs to detect a loss of sync.
2	RT Queue Configuration	Configuration of egress queues in the active Bus Schedule.

5.3.4.2.2.5.1.2.1 Tone Map Message.

A Tone Map Message is a variable length message which specifies the modulation parameters of the reverse path, to be used by the destination RT the next time it sends frames back to the source RT.

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TABLE XXVI. Tone map message.

Field	Byte	Bit Number	Bits	Definition
RESERVED	0	7-4	4	Set to zeroes.
PUNC		3-2	2	Puncturing Rate: 0=1/2 1=3/4 2=7/8 3=Reserved
TMI		1-0	2	Tone Map Index: 0=Basic Rate 1=Reserved 2=VBR1 3=VBR2
TONMOD	1	7-6	2	Tone 2 Modulation [1-0]
		5-3	3	Tone 1 Modulation [2-0]
		2-0	3	Tone 0 Modulation [2-0]
TONMOD	2	7	1	Tone 5 Modulation [0]
		6-4	3	Tone 4 Modulation [2-0]
		3-1	3	Tone 3 Modulation [2-0]
		0	1	Tone 2 Modulation [2]
TONMOD	3	7-5	3	Tone 7 Modulation [2-0]
		4-2	3	Tone 6 Modulation [2-0]
		1-0	2	Tone 5 Modulation [2-1]
TONMOD	N	...	...	...
		...	...	...
		...	...	...

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Check the source to verify that this is the current version before use.

5.3.4.2.2.5.1.2.2 Tone Modulation field (TONMOD).

Each tone is defined by a 3-bit Tone Modulation field. These fields are packed into as many bytes as required, starting from the LSB of Byte 1 of the message. Eight tones are thus defined every three bytes. The first tone and the number of consecutive tones used are defined at RT configuration time and are the same for all RTs on the same bus. The Tone 0 Modulation field applies to the first tone used.

The Tone Modulation field defines the modulation type as shown in TABLE XXVII.

TABLE XXVII. Tone modulation types.

TONMOD Value	Bits/Tone	Description
0	0	Tone is unused
1	1	Coherent BPSK
2	2	Coherent 4-QAM
3	4	Coherent 16-QAM
4	6	Coherent 64-QAM
5	8	Coherent 256-QAM
6	10	Coherent 1024-QAM
7	12	Coherent 4096-QAM

5.3.4.2.2.5.1.2.3 Sync Bus Command Frame (SBCF) count.

This 4-byte message contains a count of the number of SBCFs transmitted by the BC as referenced in TABLE XXVIII.

TABLE XXVIII. Sync Bus Command Frame (SCBF) count field.

Field	Byte	Bit Number	Bits	Definition
Sync Bus Command Frame Count	0	7-0	8	Count [31-24]
	1	7-0	8	Count [23-16]
	2	7-0	8	Count [15-8]
	3	7-0	8	Count [7-0]

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5.3.4.2.2.5.1.2.4 RT Queue Configuration.

This variable length message contains the configuration for all egress queues for all RTs in the active Bus Schedule. This message is broadcast periodically by the BC and forwarded by the RT to higher protocol layers upon reception.

Each RT occupies 18 bytes of configuration data in the message, as shown in TABLE XXIX.

5.3.4.2.2.5.1.2.5 RT Address field.

The RT Address field indicates the RT that the Queue Enable field applies to.

5.3.4.2.2.5.1.2.6 Queue Enable field.

The Queue Enable field is a 128-bit bit field where each bit corresponds to a queue. Enabled queues have their corresponding bits set to 1.

TABLE XXIX. Queue configuration message.

Field	Byte	Bit Number	Bits	Definition
RT Address 1	0	7-0	8	Address [15-8]
	1	7-0	8	Address [7-0]
Queue Enable 1	2	7-0	8	Queue Enable [7-0]
	3	7-0	8	Queue Enable [15-8]
	4	7-0	8	Queue Enable [23-16]
	...	...	...	...
	17	7-0	8	Queue Enable [127-120]
RT Address 2	18	7-0	8	Address [15-8]
	19	7-0	8	Address [7-0]
Queue Enable 2	20	7-0	8	Queue Enable [7-0]
	...	...	...	...
	35	7-0	8	Queue Enable [127-120]

5.3.4.2.2.6 Training Frame (TF).

Training Frames (TF)s are handled in a similar way to Terminal Response Frames (TRF)s but are used to aid in determining optimum tone maps for a given source-destination RT pair. The payload section contains special data that modulates to predetermined signals.

Transmission of TFs does not require corresponding Terminal Acknowledgement Frames (TAF)s so the Acknowledgement Flag is always set to 0 in the command.

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5.3.4.2.2.7 Arbitrary Test Frame (ATF).

Arbitrary Test Frames (ATF)s are used to send arbitrary (vendor specific) test data in the payload portion of the PHY frame. The entire payload section, whose length is indicated in the Frame Control (FC) section, is available for use including the space reserved for the CRC-32 field.

Transmission of ATFs does not require corresponding Terminal Acknowledgement Frames. The Acknowledgement flag is therefore always set to 0 in the command.

5.3.4.2.3 Guard slots.

Guard slots are minimum idle periods between frames that are required to accommodate channel and processing delays between terminals. Two types of Guard Slots are defined.

5.3.4.2.3.1 Short Guard Slot (SGS).

Required in order to cope with the unequal channel delays from different RTs. The duration is configurable to $1024k$ + (1024 -16) samples, where $0 \leq k \leq 63$. Short Guard Slots are used except for situations described in 5.3.4.2.3.2.

5.3.4.2.3.2 Long Guard Slot (LGS).

Required in order to accommodate receiver processing latencies within RTs. The duration is configurable to $1024k$ + (1024 -16) samples, where $0 \leq k \leq 63$. LGS shall be used instead of SGS under the following conditions:

a. Between a response frame and its associated acknowledgement frame if they are adjacent in the compound transaction.

b. Between a BCF and the first response frame if its associated FE command is contained in the payload section of the BCF.

c. Between the last non-BCF and the next BCF.

5.3.4.2.4 Packet Adaptation Layer (PAL) services.

5.3.4.2.4.1 Overview of PAL services.

The Packet Adaptation Layer (PAL) provides peer Logical Link Control (LLC) entities with the ability to exchange PAL Service Data Units (PAL-SDUs) and uses the underlying services offered by the Physical Layer (PHY). This layer performs segmentation and reassembly between the MAC Sublayer and the LLC sublayer.

FIGURE 52 shows how PAL-SDUs are typically mapped into MAC frames.

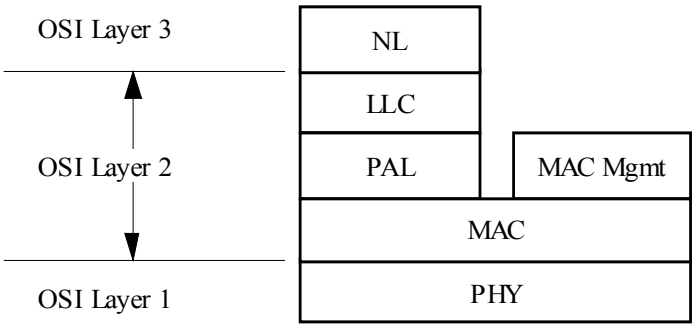


FIGURE 52. Packet adaptation layer context.

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Note: The Logical Link Control (LLC) and Network Layer (NL) are not specified herein.

This sublayering is necessary because the Logical Link Control (LLC) layer expects to be able to transmit and receive arbitrary length packets of its choosing but the MAC layer imposes a rigid set of constraints over the timing and length of the transmission as dictated by the BC.

PAL Service Data Units (PAL-SDUs) are transmitted using the payload section Critical or Non-Critical Terminal Response Frames (TRF)s. The meaning of Critical and Non-Critical data is vendor specific.

PAL operation is coordinated with Sync Bus Command Frame (SBCF) reception. Upon arrival of a SBCF, any incomplete segmentation and reassembly operations are discarded and restarted.

5.3.4.2.4.2 Egress operation.

The Packet Adaptation Layer (PAL) sublayer provides the means to queue PAL Service Data Units (PAL-SDU) data asynchronously to the bus and allows the MAC layer to demand data from a specified queue when a scheduled frame exchange arrives. It also provides the means to retry and abort PAL-SDUs already queued as required. The PAL sublayer takes PAL-SDUs, segments them if necessary, and fits them into MAC Service Data Units (MAC-SDUs). The PAL sublayer optimizes bus utilization by attempting to fill the MAC-SDU to maximum capacity.

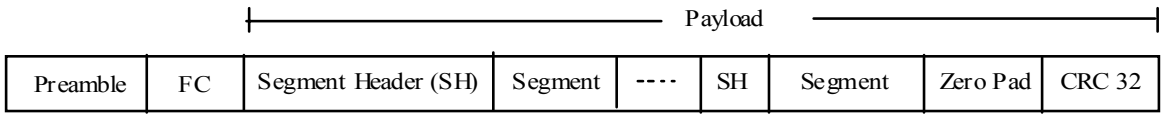


FIGURE 53. MAC-SDU structure.

A MAC-SDU may contain, in order:

a. The end segment of the previous PAL-SDU (optional).

b. One or more full PAL-SDUs (optional).

c. The start segment of the next PAL-SDU (optional).

A Response Frame may also consist solely of one middle segment of a PAL-SDU.

The open ends of PAL-SDU segments shall not abut other PAL-SDUs or PAL-SDU segments within a Response Frame.

The MAC sublayer commands the PAL sublayer to output MAC-SDUs from a given queue, as determined by the bus schedule. They are passed to the PHY layer as Data Response Frames. The MAC sublayer then awaits an Acknowledgement Frame from the destination RT to determine the status of the transmission.

In the event of unsuccessful transmission, the MAC sublayer may command the PAL sublayer to either retry or abort the previous transmission.

5.3.4.2.4.2.1 Segmentation Headers (SH).

Each segment in a MAC-SDU is prepended with a Segmentation Header on egress to allow reassembly at the destination. The PAL sublayer must generate the SH at the time of transmission as it needs to consider the size of the current MAC-SDU in the calculation. The SH may have to be regenerated to allow for changes in MAC-SDU size between retries.

The SH is four bytes long containing the fields defined in TABLE XXX.

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TABLE XXX. Segmentation header.

Field	Byte	Bit Number	Bits	Definition
Last Header Flag	0	7	1	Last Header in this MAC-SDU
Last Segment Flag	0	6	1	Last Segment of this PAL-SDU
PAL-SDU Number	0	5-0	6	Sequence Number of this PAL-SDU
Segment Number	1	7-4	4	Sequence Number of this segment
Segment Length	1	3-0	4	Segment Length [11-8]
	2	7-0	8	Segment Length [7-0]
Reserved	3	7-0	8	Reserved

5.3.4.2.4.2.1.1 Last Header Flag.

The Last Header flag is set if this segment is the last or only one in the MAC-SDU. Any remaining space in the MAC-SDU is padded with zeroes.

5.3.4.2.4.2.1.2 Last Segment Flag.

The Last Segment flag is set if this segment is the last or only one in the PAL-SDU. If all previous segments of the PAL-SDU were received successfully, this segment completes the PAL-SDU.

5.3.4.2.4.2.1.3 PAL-SDU Number.

The PAL-SDU Number is a 6-bit number incremented for each PAL-SDU. This number differentiates among PAL-SDUs to prevent inadvertent reassembly of unrelated segments. This number wraps around from 63 to 0.

5.3.4.2.4.2.1.4 Segment Number.

The Segment Number is a 4-bit number incremented for each segment of the current PAL-SDU. The number restarts at zero for the first or only segment of the PAL-SDU and wraps around from 15 to 0.

5.3.4.2.4.2.1.5 Segment Length.

The Segment Length indicates the length of the data portion of this segment in bytes. A segment can be up to 2048 bytes long. The length must be a multiple of 4 bytes except when the Last Segment flag is set, in which case the length indicates the number of valid bytes in the segment.

Segment Headers are always aligned to 4-byte boundaries relative to the start of the payload section. Segments must be padded to 4-byte boundaries with zeroes if necessary to ensure that subsequent Segment Headers are 4-byte aligned.

Depending on the size of the payload section, additional pad bytes past the last segment of the MAC-SDU may be required to fill unused space before the CRC-32 field. It should be noted that the CRC-32 field is not necessarily 4-byte aligned relative to the start of the payload section.

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5.3.4.2.4.3 Ingress operation.

The MAC sublayer receives Response Frames from the destination RT and passes the resulting MAC-SDU to the PAL sublayer. The PAL sublayer strips and reassembles received MAC-SDUs back into PAL-SDUs. Complete PAL-SDUs are passed up to the LLC for processing.

The ingress operation is illustrated in FIGURE 54.

Note: the Logical Link Control (LLC) layer is not specified herein.

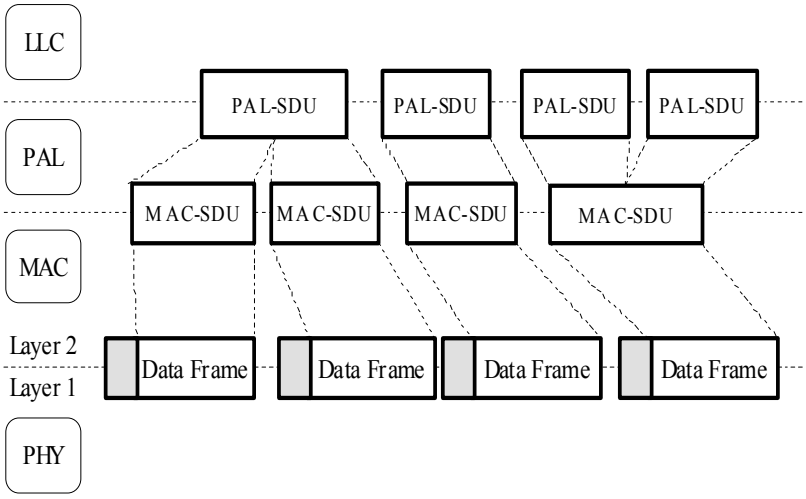


FIGURE 54. PAL sublayer functions.

The PAL provides the means to interface to standard networking stacks. However, direct or alternate use of the MAC sublayer is possible for time critical low latency applications.

5.3.4.2.4.4 PAL limits.

The PAL has the limits shown in TABLE XXXI.

TABLE XXXI. PAL maximum limits.

Parameter	Value
Maximum payload size	32,768 bytes
Maximum MAC-SDUs/PAL-SDU	32
Maximum MAC-SDU length	2048 bytes
Maximum PAL-SDUs/MAC-SDU	32

5.3.4.2.4.5 Packet Adaptation Layer (PAL).

The Packet Adaptation Layer (PAL) receives PAL-SDUs from the LLC and passes MAC-SDUs to the MAC sublayer in the egress direction, and receives MAC-SDUs from the MAC sublayer and passes PAL-SDUs to the LLC in the ingress direction. At this layer, the timing of PAL-SDU transfers is not tied to the BC.

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5.3.4.2.4.6 Packet Adaptation Layer (PAL) Data Services.

The following PAL data service primitives are provided to the LLC:

a. PAL-SEND.request.

b. PAL-SEND.indication.

c. PAL-RECEIVE.indication.

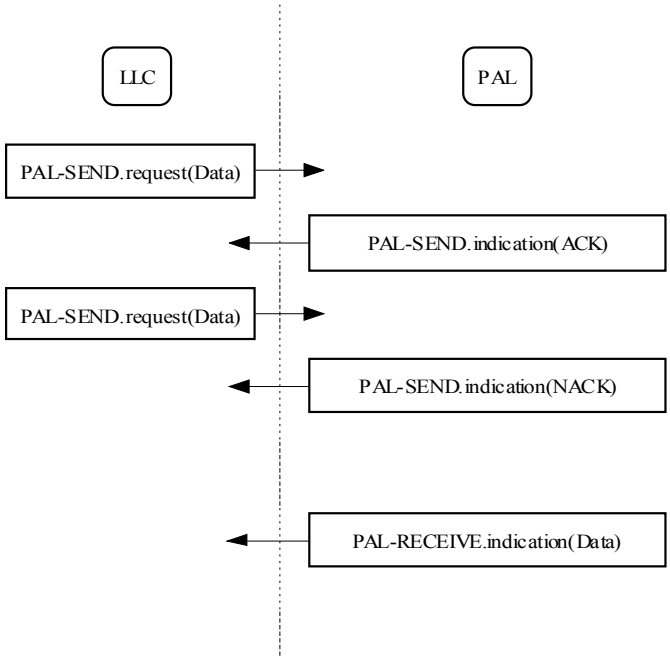


FIGURE 55. MAC data service primitives.

5.3.4.2.4.6.1 PAL-SEND.request.

5.3.4.2.4.6.1.1 Function.

This primitive requests the transfer of a PAL-SDU from the LLC layer to the specified destination LLC layer.

5.3.4.2.4.6.1.2 Semantics of the service primitive.

PAL-SEND.request (
QueueID,
PAL-SDU Data
)

The QueueID parameter indicates the queue on which the PAL-DSU will be sent. VCs are mapped to QueueIDs during connection setup.

The PAL-SDU Data parameter contains the PAL-SDU packet data to be transmitted. The length must not exceed MTU bytes.

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5.3.4.2.4.6.1.3 When generated.

This primitive is generated by the LLC layer entity whenever a PAL-SDU is to be transferred to a peer LLC layer entity.

5.3.4.2.4.6.1.4 Effect of Receipt.

The receipt of this primitive causes the PAL sublayer to queue PAL-SDU data to the specified queue if there is room for the entire PAL-SDU. A PAL-SEND.indication is sent to the LLC.

5.3.4.2.4.6.2 PAL-SEND.indication.

5.3.4.2.4.6.2.1 Function.

This primitive indicates status information from the MAC to the LLC.

5.3.4.2.4.6.2.2 Semantics of the service primitive.

PAL-SEND.indication (
QueueID,
Status
)

The QueueID parameter indicates the queue that the status indication applies to.

The Status parameter contains a status code.

TABLE XXXII. PAL-SEND.indication status codes.

Status Code	Description
ACK	The PAL-SDU has been accepted into the queue.
NACK	The PAL-SDU has not been accepted into the queue because the queue is full or invalid.

5.3.4.2.4.6.2.3 When generated.

This primitive is generated by the PAL entity whenever status information needs to be conveyed to the LLC layer entity.

5.3.4.2.4.6.2.4 Effect of Receipt.

Action based on the receipt of this primitive is implementation dependent.

5.3.4.2.4.6.3 PAL-RECEIVE.indication.

5.3.4.2.4.6.3.1 Function.

This primitive indicates the availability of PAL-SDU data from the PAL to the LLC.

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5.3.4.2.4.6.3.2 Semantics of the service primitive.

PAL-RECEIVE.indication (
QueueID,
PAL-SDU Data
)

The QueueID parameter indicates the queue that the PAL-SDU data was received on.

The PAL-SDU Data parameter contains the PAL-SDU data received.

5.3.4.2.4.6.3.3 When generated.

This primitive is passed by the SAR entity to the LLC entity whenever a complete data packet has been reassembled successfully.

5.3.4.2.4.6.3.4 Effect of receipt.

The LLC processes the PAL-SDU.

5.3.4.2.5 MAC sublayer.

Operation of the MAC sublayer (MAC) is closely coupled with the BC. MAC sublayer services are divided into two types:

a. MAC Data Services – provides the means to transfer application data to and from upper layers.
b. MAC Management Services – provides the means to configure the operation of the bus.

5.3.4.2.5.1 MAC Data Services.

The following MAC service primitives are provided to the PAL:

a. MD-SEND.command.
b. MD-SEND.response.
c. MD-SEND.indication.
d. MD-RECEIVE.indication.
e. MD-STATUS.indication.

5.3.4.2.5.1.1 MD-SEND.command.

5.3.4.2.5.1.1.1 Function.

This primitive requests the transfer of a Data Frame from the PAL sublayer to the MAC sublayer. The PAL inserts Segmentation Headers (SH) as required to adapt PAL-SDU data into MAC-SDU Frames.

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5.3.4.2.5.1.1.2 Semantics of the service primitive.

MD-SEND.command (
QueueID,
Length,
Retry
)

The QueueID parameter indicates the queue from which Data Frame is requested. The Length parameter indicates the length of the scheduled Data Frame. The Retry parameter indicates whether this is a retransmission of a previously failed transmission.

5.3.4.2.5.1.1.3 When generated.

This primitive is generated by the MAC sublayer at the start of a scheduled Data Frame slot where this RT is the source as configured in the bus schedule.

5.3.4.2.5.1.1.4 Effect of Receipt.

The receipt of this primitive by the PAL sublayer causes it to generate a MD-SEND.response primitive.

5.3.4.2.5.1.2 MD-SEND.response.

5.3.4.2.5.1.2.1 Function.

This primitive passes Response Frame payload data from the PAL sublayer to the MAC sublayer.

5.3.4.2.5.1.2.2 Semantics of the service primitive.

MD-SEND.response (
QueueID,
Data
)

The QueueID parameter indicates the queue that the response applies to.

The Data parameter contains Response Frame payload data. The size of the payload shall be equal to the Length parameter of the previously sent MD-SEND.command primitive.

5.3.4.2.5.1.2.3 When generated.

This primitive is generated by the PAL sublayer in response to a MD-SEND.command primitive.

5.3.4.2.5.1.2.4 Effect of Receipt.

The reception of this primitive by the MAC sublayer causes it to pass the Response Frame to the PHY.

5.3.4.2.5.1.3 MD-SEND.indication.

5.3.4.2.5.1.3.1 Function.

This primitive passes status information of the last transmission from the MAC sublayer to the PAL sublayer.

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5.3.4.2.5.1.3.2 Semantics of the service primitive.

MD-SEND.indication (
QueueID,
Status
)

The QueueID parameter indicates the queue that the response applies to.

The Status parameter indicates the status of the last transmission as shown in TABLE XXXIII.

TABLE XXXIII. MD-SEND.indication status codes.

Status Code	Description
ACK	The Response Frame was sent successfully to the destination and the PAL may now remove the sent data from the queue.
NACK	The Response Frame was not sent successfully to the destination and the PAL shall prepare to retransmit data from the last successful point at the next scheduled frame exchange.
ABORT	The Response Frame was not sent successfully to the destination and the PAL shall flush the current PAL-SDU from the queue.

5.3.4.2.5.1.3.3 When generated.

This primitive is generated by the MAC sublayer to the PAL when a Response frame is received from the PHY.

5.3.4.2.5.1.3.4 Effect of Receipt.

The reception of this primitive by the PAL sublayer causes it to perform the appropriate action described in TABLE XXXIII.

5.3.4.2.5.1.4 MD-RECEIVE.indication.

5.3.4.2.5.1.4.1 Semantics of the service primitive.

MD-RECEIVE.indication (
QueueID,
Data,
)

The QueueID parameter indicates the queue that the response applies to.

The Data parameter contains the payload data received.

5.3.4.2.5.1.4.2 When generated.

This primitive is generated by the MAC sublayer to the PAL when a Data frame is received from the PHY.

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5.3.4.2.5.1.4.3 Effect of Receipt.

The reception of this primitive by the PAL sublayer causes it to queue the data for reassembly.

5.3.4.2.5.2 MAC Management Services.

MAC management services are called to configure operation of the bus.

a. MM-BCMD.command.
b. MM-BCMD.indication.
c. MM-SEND.command.
d. MM-SEND.response.
e. MM-RECEIVE.indication.
f. MM-TONE.command.
g. MM-TONE.response.
h. MM-WAVE.command.
i. MM-TONE.response.

5.3.4.2.5.2.1 MM-BCMD.command.

5.3.4.2.5.2.1.1 Function.

This primitive commands a terminal to send a BCF.

5.3.4.2.5.2.1.2 Semantics of the service primitive

MM-BCMD.command (
Type,
Count,
Command1,
Command2,
...
CommandN
)

The Type parameter indicates whether a regular (Type 1) or Sync (Type 2) Bus Command is sent.

The Count parameter indicates the number of commands to send. The range is from 1 to 16.

The Command parameters are a sequence of 4-byte Bus Commands.

5.3.4.2.5.2.1.3 When generated.

This primitive is generated by the MAC Management entity at the start of a Compound Transaction if this terminal is the BC. A Sync Bus Command is generated at the start of a Minor Period.

5.3.4.2.5.2.1.4 Effect of Receipt.

The reception of this primitive by the MAC management layer will cause it to send a BCF.

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5.3.4.2.5.2.2 MM-BCMD.indication.

5.3.4.2.5.2.2.1 Function.

This primitive indicates the reception of a BCF from the PHY.

5.3.4.2.5.2.2.2 Semantics of the service primitive.

MM-BCMD.indication(
Type,
Count,
Command1,
Command2,
...
CommandN
)

The Type parameter indicates whether a regular (Type 1) or Sync (Type 2) Bus Command was received.

The Count parameter indicates the number of commands received. The range is from 1 to 16.

The Command parameters are a sequence of 4-byte Bus Commands.

5.3.4.2.5.2.2.3 When generated.

This primitive is generated when a BCF is received from the PHY.

5.3.4.2.5.2.2.4 Effect of Receipt.

The reception of this primitive by the MAC management layer will cause it to prepare for upcoming Frame Exchange commands in the current Compound Transaction. Reception of a Sync BCF indicates the start of a Minor Period.

5.3.4.2.5.2.3 MM-SEND.command.

5.3.4.2.5.2.3.1 Function.

This primitive requests the transfer of a MAC Management Frame from the MAC Management entity to the MAC sublayer.

5.3.4.2.5.2.3.2 Semantics of the service primitive.

MM-SEND.command (
Length
)

The Length parameter indicates the length of the scheduled MAC Management Frame.

5.3.4.2.5.2.3.3 When generated.

This primitive is generated by the MAC sublayer at the start of a scheduled MAC Management Frame slot where this RT is the source as configured in the bus schedule.

5.3.4.2.5.2.3.4 Effect of Receipt.

The receipt of this primitive by the MAC Management entity causes it to generate a **MM-SEND.response** primitive.

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Check the source to verify that this is the current version before use.

5.3.4.2.5.2.4 MM-SEND.response.

5.3.4.2.5.2.4.1 Function.

This primitive passes MAC Management Frame payload data from the MAC Management entity to the MAC sublayer.

5.3.4.2.5.2.4.2 Semantics of the service primitive.

MM-SEND.response (
Data
)

The **Data** parameter contains MAC Management Frame payload data. The size of the payload shall be equal to the Length parameter of the previously sent **MM-SEND.command** primitive.

5.3.4.2.5.2.4.3 When generated.

This primitive is generated by the MAC Management entity in response to a **MM-SEND.command** primitive.

5.3.4.2.5.2.4.4 Effect of Receipt.

The reception of this primitive by the MAC sublayer causes it to pass the MAC Management Frame to the PHY.

5.3.4.2.5.2.5 MM-RECEIVE.indication.

5.3.4.2.5.2.5.1 Semantics of the service primitive.

MM-RECEIVE.indication (
Data
)

The **Data** parameter contains the payload data received.

5.3.4.2.5.2.5.2 When generated.

This primitive is generated by the MAC sublayer to the MAC Management entity when a MAC Management frame is received from the PHY.

5.3.4.2.5.2.5.3 Effect of Receipt.

The reception of this primitive by the MAC Management entity causes it to process the MAC Management message.

5.3.4.2.5.2.6 MM-TRAIN.command.

5.3.4.2.5.2.6.1 Function.

This primitive requests the transfer of a Training Frame (TF) from the MAC Management entity to the MAC sublayer.

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Check the source to verify that this is the current version before use.

5.3.4.2.5.2.6.2 Semantics of the service primitive.

MM-TRAIN.command (
Length
)

The Length parameter indicates the length of the scheduled TF.

5.3.4.2.5.2.6.3 When generated.

This primitive is generated by the MAC sublayer at the start of a scheduled TF slot where this RT is the source as configured in the bus schedule.

5.3.4.2.5.2.6.4 Effect of Receipt.

The receipt of this primitive by the MAC Management entity causes it to generate a **MM-TRAIN.response** primitive.

5.3.4.2.5.2.7 MM-TRAIN.response.

5.3.4.2.5.2.7.1 Function.

This primitive passes Training Frame (TF) payload data from the MAC Management entity to the MAC sublayer.

5.3.4.2.5.2.7.2 Semantics of the service primitive.

MM-TRAIN.response (
Data
)

The Data parameter contains TF payload data. The size of the payload shall be equal to the Length parameter of the previously sent MM-TRAIN.command primitive.

5.3.4.2.5.2.7.3 When generated.

This primitive is generated by the MAC Management entity in response to a MM-TRAIN.command primitive.

5.3.4.2.5.2.7.4 Effect of Receipt.

The reception of this primitive by the MAC sublayer causes it to pass the TF to the PHY.

5.3.4.2.5.2.8 MM-WAVE.command.

5.3.4.2.5.2.8.1 Function.

This primitive requests the transfer of an ATF from the MAC Management entity to the MAC sublayer.

5.3.4.2.5.2.8.2 Semantics of the service primitive.

MM-WAVE.command (
Length
)

The Length parameter indicates the length of the scheduled ATF.

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5.3.4.2.5.2.8.3 When generated.

This primitive is generated by the MAC sublayer at the start of a scheduled ATF slot where this RT is the source as configured in the bus schedule.

5.3.4.2.5.2.8.4 Effect of Receipt.

The receipt of this primitive by the MAC Management entity causes it to generate a **MM-WAVE.response** primitive.

5.3.4.2.5.2.9 MM-WAVE.response.

5.3.4.2.5.2.9.1 Function.

This primitive passes ATF payload data from the MAC Management entity to the MAC sublayer.

5.3.4.2.5.2.9.2 Semantics of the service primitive.

MM-WAVE.response (
Data
)

The Data parameter contains ATF payload data. The size of the payload shall be equal to the Length parameter of the previously sent MM-WAVE.command primitive.

5.3.4.2.5.2.9.3 When generated.

This primitive is generated by the MAC Management entity in response to a **MM-WAVE.command** primitive.

5.3.4.2.5.2.9.4 Effect of Receipt.

The reception of this primitive by the MAC sublayer causes it to pass the ATF to the PHY.

5.4 Interoperability.

The custodian agency reserves the right to maintain a set of interoperability tests to prove conformance to this notice 5. Certification would require test data to be provided showing interoperability in accordance with these tests and may entail samples of a terminal unit to be submitted to the custodian agency to be certified by actual interoperability testing with a compliant unit.

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5.5 Trapezoidal Waveform (TW).

Notice 5 is cognizant of the requirements of MIL-STD-1553B (to Notice 4) in that extended capabilities are added. This section contains the requirements and characteristics of the TW and is identical to MIL-STD-1553B (to notice 4).

Note: For reference, all sections in 5.5 correspond to the equivalent section in MIL-STD-1553B (to notice 4) and all figures, tables, references and formats are unchanged. (E.g., section 4.3.3.5 in MIL-STD-1553B (to notice 4) corresponds to 5.5.4.3.3.5 under section 5.5 in this document, Figure 1 in this section corresponds to Figure 1 in MIL-STD-1553B (to notice 4) and a reference to 4.3.3.5.1.7 in the text will actually refer to 5.5.4.3.3.5.1.7 within section 5.5.

5.5.1 SCOPE

5.5.1.1 Scope.

This standard establishes requirements for digital, command/response, time division multiplexing (Data bus) techniques. It encompasses the data bus line and its interface electronics illustrated on Figure 1, and also defines the concept of operation and information flow on the multiplex data bus and the electrical and functional formats to be employed.

5.5.1.2 Application.

When invoked in a specification or statement of work, these requirements shall apply to the multiplex data bus and associated equipment which is developed either alone or as a portion of a weapon system or subsystem development. The contractor is responsible for invoking all the applicable requirements of this Military Standard on any and all subcontractors he may employ.

5.5.2 REFERENCE DOCUMENTS

5.5.2.1 Issue of document.

The following document, of the issue in effect on date of invitation for bid or request for proposal, forms a part of the standard to the extent specified herein.

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-464 Electromagnetic Environmental Effects Requirements for Systems

Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

5.5.3 DEFINITIONS

5.5.3.1 Bit.

Contraction of binary digit: may be either zero or one. In information theory a binary digit is equal to one binary decision or the designation of one or two possible values or states of anything used to store or convey information.

5.5.3.2 Bit rate.

The number of bits transmitted per second.

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5.5.3.3 Pulse code modulation (PCM).

The form of modulation in which the modulation signal is sampled, quantized, and coded so that each element of information consists of different types or numbers of pulses and spaces.

5.5.3.4 Time division multiplexing (TDM).

The transmission of information from several signal sources through one communication system with different signal samples staggered in time to form a composite pulse train.

5.5.3.5 Half duplex.

Operation of a data transfer system in either direction over a single line, but not in both directions on that line simultaneously.

5.5.3.6 Word.

In this document a word is a sequence of 16 bits plus sync and parity. There are three types of words: command, status, and data.

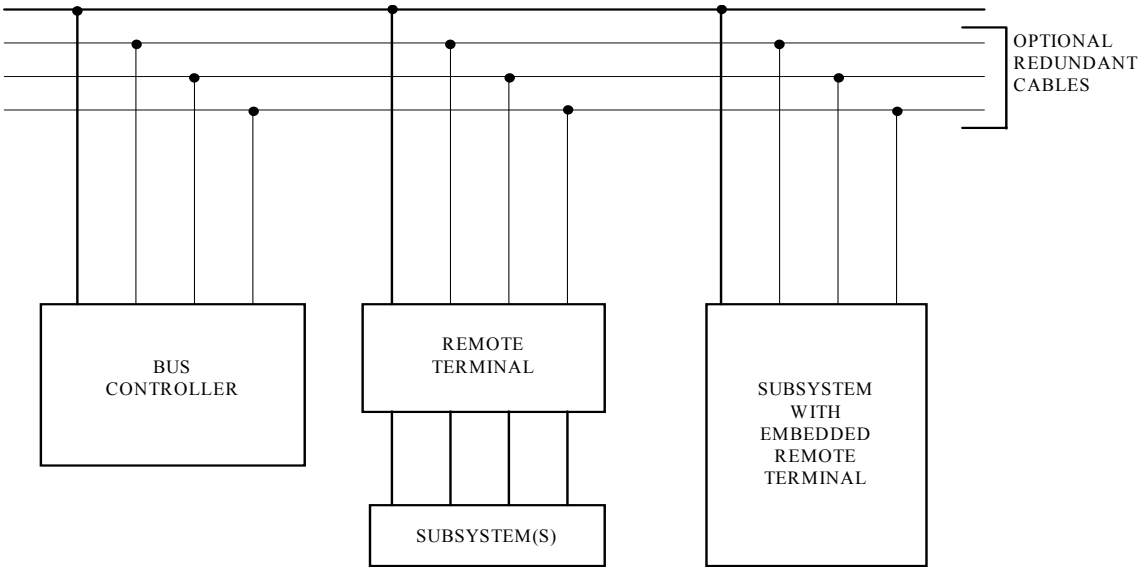


FIGURE 1. Sample multiplex data bus architecture.

5.5.3.7 Message.

A single message is the transmission of a command word, status word, and data words if they are specified. For the case of a remote terminal to remote terminal (RT to RT) transmission, the message shall include the two command words, the two status words, and data words.

5.5.3.8 Subsystem.

The device or functional unit receiving data transfer service from the data bus.

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5.5.3.9 Data bus.

Whenever data bus or bus is referred to in this document it shall imply all the hardware including twisted shielded pair cables, isolation resistors, transformers, etc., required to provide a single data path between the bus controller and all the associated remote terminals.

5.5.3.10 Terminal.

The electronic module necessary to interface the data bus with the subsystem and the subsystem with the data bus. Terminals may exist as separate line replaceable units (LRU's) or be contained within the elements of the subsystem.

5.5.3.11 Bus controller (BC).

The terminal assigned the task of initiating information transfers on the data bus.

5.5.3.12 Bus monitor (BM).

The terminal assigned the task of receiving bus traffic and extracting selected information to be used at a later time.

5.5.3.13 Remote terminal (RT).

All terminals not operating as the bus controller or as a bus monitor.

5.5.3.14 Asynchronous operation.

For the purpose of this standard, asynchronous operation is the use of an independent clock source in each terminal for message transmission. Decoding is achieved in receiving terminals using clock information derived from the message.

5.5.3.15 Dynamic bus control.

The operation of a data bus system in which designated terminals are offered control of the data bus.

5.5.3.16 Command/Response.

Operation of a data bus system such that remote terminals receive and transmit data only when commanded to do so by the bus controller.

5.5.3.17 Redundant data bus.

The use of more than one data bus to provide more than one data path between the subsystems, i.e., dual redundant data bus, tri-redundant data bus, etc.

5.5.3.18 Broadcast.

Operation of a data bus system such that information transmitted by the bus controller or a remote terminal is addressed to more than one of the remote terminals connected to the data bus.

5.5.3.19 Mode code.

A means by which the bus controller can communicate with the multiplex bus related hardware, in order to assist in the management of information flow.

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5.5.4 GENERAL REQUIREMENTS

5.5.4.1 Test and operating requirements.

All requirements as specified herein shall be valid over the environmental conditions which the multiplex data bus system shall be required to operate.

5.5.4.2 Data bus operation.

The multiplex data bus system in its most elemental configuration shall be as shown on Figure 1. The multiplex data bus system shall function asynchronously in a command /response mode, and transmission shall occur in a half-duplex manner. Sole control of information transmission on the bus shall reside with the bus controller, which shall initiate all transmissions. The information flow on the data bus shall be comprised of messages which are, in turn, formed by three types of words (command, data, and status) as defined in 4.3.3.5.

5.5.4.3 Characteristics.

5.5.4.3.1 Data form.

Digital data may be transmitted in any desired form, provided that the chosen form shall be compatible with the message and word formats defined in this standard. Any unused bit positions in a word shall be transmitted as logic zeros.

5.5.4.3.2 Bit priority

The most significant bit shall be transmitted first with the least significant bits following in descending order of value in the data word. The number of bits required to define a quantity shall be consistent with the resolution or accuracy required. In the event that multiple precision quantities (information accuracy or resolution requiring more than 16 bits) are transmitted, the most significant bits shall be transmitted first, followed by the word(s) containing the lesser significant bits in numerical descending order. Bit packing of multiple quantities in a single data word is permitted.

5.5.4.3.3 Transmission method.

5.5.4.3.3.1 Modulation.

The signal shall be transferred over the data bus in serial digital pulse code modulation form.

5.5.4.3.3.2 Data code.

The data code shall be Manchester II bi-phase level. A logic one shall be transmitted as a bipolar coded signal 1/0 (i.e., a positive pulse followed by a negative pulse). A logic zero shall be a bipolar coded signal 0/1 (i.e., a negative pulse followed by a positive pulse). A transition through zero occurs at the midpoint of each bit time (see Figure 2).

5.5.4.3.3.3 Transmission bit rate.

The transmission bit rate on the bus shall be 1.0 megabit per second with a combined accuracy and long-term stability of ± 0.1 percent (i.e., ±1000 hertz (Hz)). The short-term stability (i.e., stability over 1.0 second interval) shall be at least 0.01 percent (i.e., ±100 Hz).

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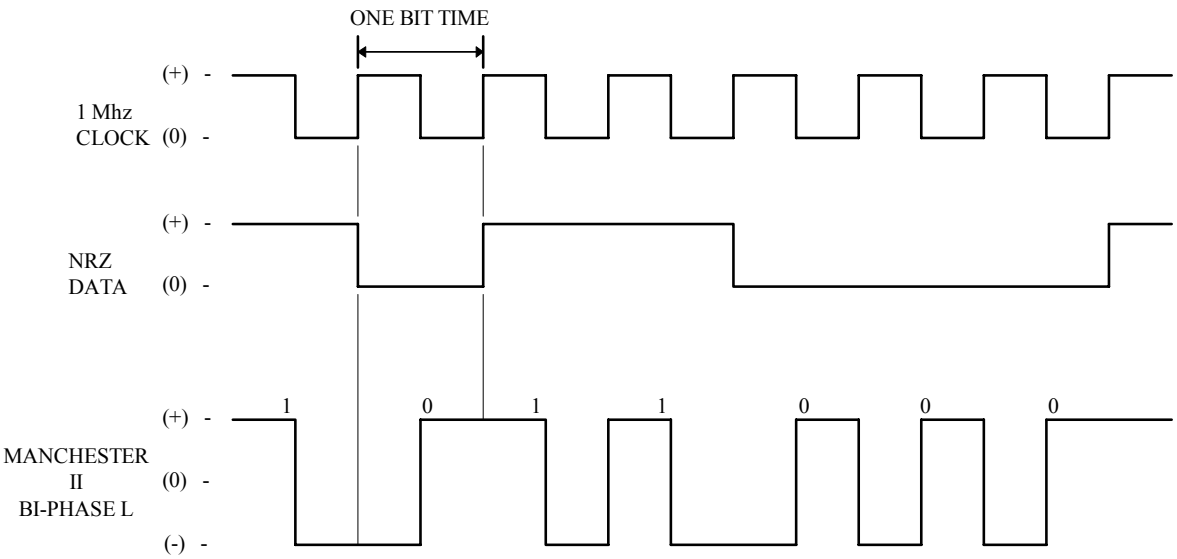


FIGURE 2. Data encoding.

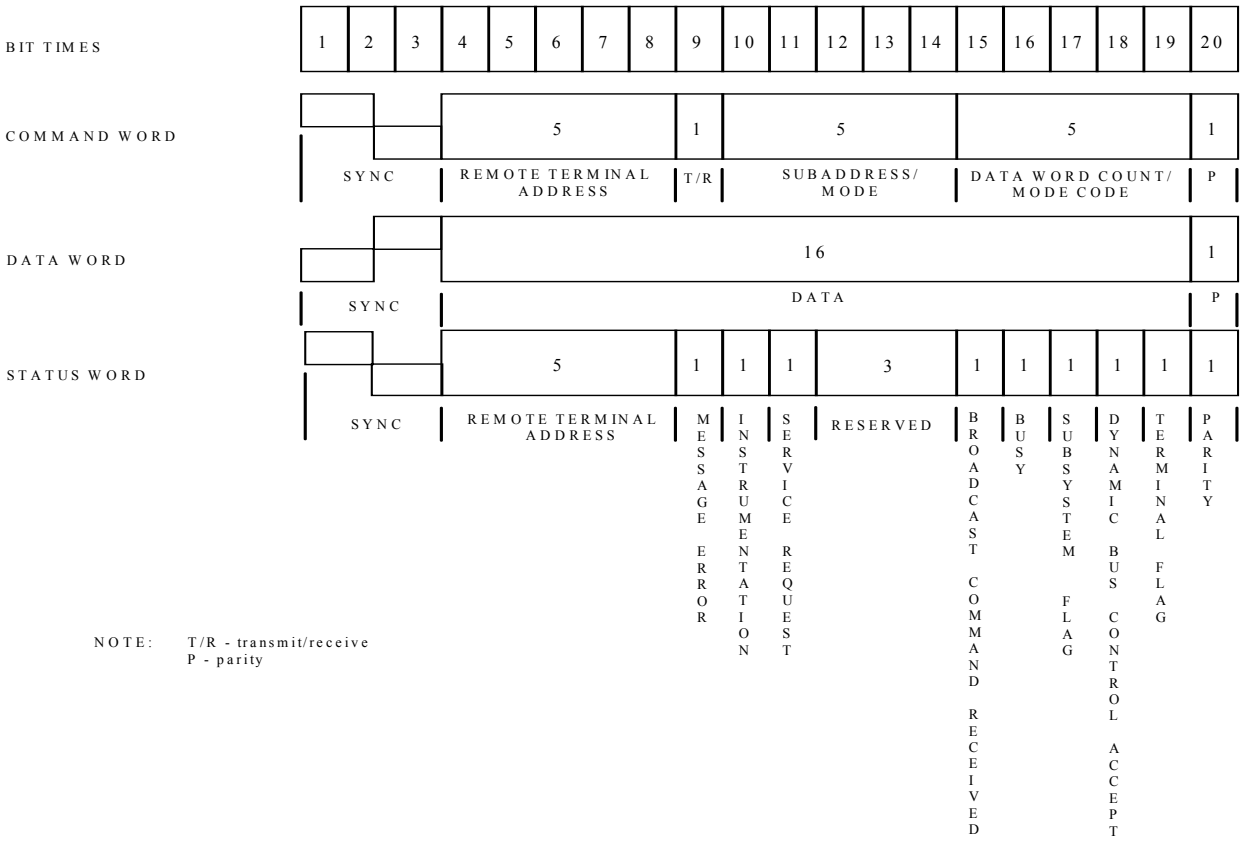


FIGURE 3. Word formats.

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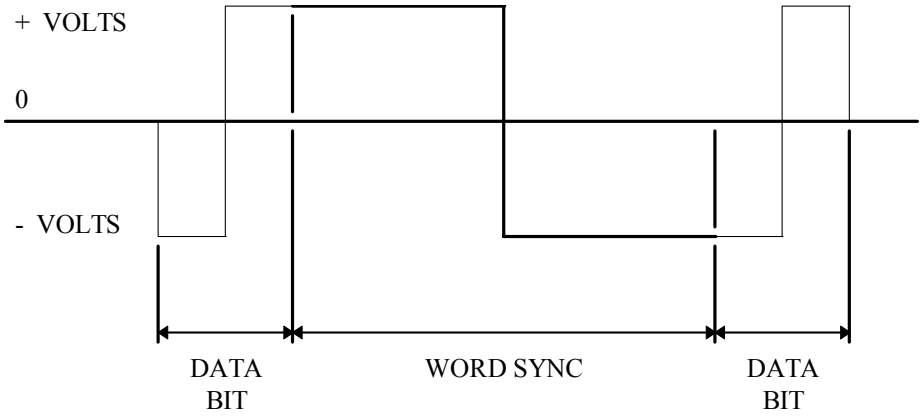


FIGURE 4. Command and status sync.

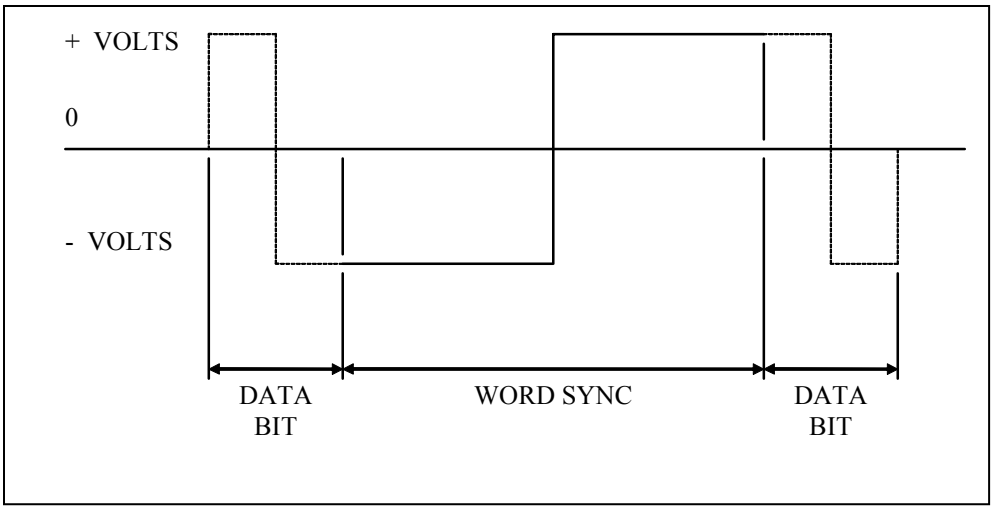


FIGURE 5. Data sync.

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5.5.4.3.3.4 Word size.

The word size shall be 16 bits plus the sync waveform and the parity bit for a total of 20 bits times as shown on Figure 3.

5.5.4.3.3.5 Word formats.

The word formats shall be as shown on Figure 3 for the command, data, and status words.

5.5.4.3.3.5.1 Command word.

A command word shall be comprised of a sync waveform, remote terminal address field, transmit/receive (T/R) bit, subaddress/mode field, word count/mode code field, and a parity (P) bit (see Figure 3).

5.5.4.3.3.5.1.1 Sync.

The command sync waveform shall be an invalid Manchester waveform as shown in Figure 4. The width shall be three bit times, with the sync waveform being positive for the first one and one-half bit times, and then negative for the following one and one-half bit times. If the next bit following the sync waveform is a logic zero, then the last half of the sync waveform will have an apparent width of two clock periods due to the Manchester encoding.

5.5.4.3.3.5.1.2 Remote terminal address.

The next five bits following the sync shall be the RT address. Each RT shall be assigned a unique address. Decimal address 31 (11111) shall not be assigned as a unique address. In addition to its unique address, a RT shall be assigned decimal address 31 (11111) as the common address, if the broadcast option is used.

5.5.4.3.3.5.1.3 Transmit/receive.

The next bit following the remote terminal address shall be the T/R bit, which shall indicate the action required of the RT. A logic zero shall indicate the RT is to receive, and a logic one shall indicate the RT is to transmit.

5.5.4.3.3.5.1.4 Subaddress/mode.

The next five bits following the R/T bit shall be utilized to indicate an RT subaddress or use of mode control, as is dictated by the individual terminal requirements. The subaddress/mode values of 00000 and 11111 are reserved for special purposes, as specified in 4.3.3.5.1.7, and shall not be utilized for any other function.

5.5.4.3.3.5.1.5 Data word count/mode code.

The next five bits following the subaddress/mode field shall be the quantity of data words to be either sent out or received by the RT or the optional mode code as specified in 4.3.3.5.1.7. A maximum of 32 data words may be transmitted or received in any one message block. All 1's shall indicate a decimal count of 31, and all 0's shall indicate a decimal count of 32.

5.5.4.3.3.5.1.6 Parity.

The last bit in the word shall be used for parity over the preceding 16 bits. Odd parity shall be utilized.

5.5.4.3.3.5.1.7 Optional mode control.

For RT's exercising this option a subaddress/mode code of 00000 or 11111 shall imply that the contents of the data word count/mode code field are to be decoded as a five bit mode command. The mode code shall only be used to communicate with the multiplex bus related hardware, and to assist in the management of information flow, and not

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to extract data from or feed data to a functional subsystem. Codes 00000 through 01111 shall only be used for mode codes which do not require transfer of a data word. For these codes, the T/R bit shall be set to 1. Codes 10000 through 11111 shall only be used for mode codes which require transfer of a single data word. For these mode codes, the T/R bit shall indicate the direction of data word flow as specified in 4.3.3.5.1.3. No multiple data word transfer shall be implemented with any mode code. The mode codes are reserved for the specific functions as specified in Table I and shall not be used for any other purpose. If the designer chooses to implement any of these functions, the specific codes, T/R bit assignments, and use of a data word, shall be used as indicated. The use of the broadcast command option shall only be applied to particular mode codes as specified in Table I.

5.5.4.3.3.5.1.7.1 Dynamic bus control.

The controller shall issue a transmit command to an RT capable of performing the bus control function. This RT shall respond with a status word as specified in 4.3.3.5.3. Control of the data bus passes from the offering bus controller to the accepting RT upon completion of the transmission of the status word by the RT. If the RT rejects control of the data bus, the offering bus controller retains control of the data bus.

5.5.4.3.3.5.1.7.2 Synchronize (without data word).

This command shall cause the RT to synchronize (e.g., to reset the internal timer, to start a sequence, etc.). The RT shall transmit the status word as specified in 4.3.3.5.3.

5.5.4.3.3.5.1.7.3 Transmit status word.

This command shall cause the RT to transmit the status word associated with the last valid command word preceding this command. This mode command shall not alter the state of the status word.

5.5.4.3.3.5.1.7.4 Initiate self test.

This command shall be used to initiate self test within the RT. The RT shall transmit the status word as specified in 4.3.3.5.3.

5.5.4.3.3.5.1.7.5 Transmitter shutdown.

This command (to only be used with dual redundant bus systems) shall cause the RT to disable the transmitter associated with the redundant bus. The RT shall not comply with a command to shut down a transmitter on the bus from which this command is received. In all cases, the RT shall respond with a status word as specified in 4.3.3.5.3 after this command.

5.5.4.3.3.5.1.7.6 Override transmitter shutdown.

This command (to only be used with dual redundant bus systems) shall cause the RT to enable a transmitter which was previously disabled. The RT shall not comply with a command to enable a transmitter on the bus from which this command is received. In all cases, the RT shall respond with a status word as specified in 4.3.3.5.3 after this command.

5.5.4.3.3.5.1.7.7 Inhibit terminal flag (T/F) bit.

This command shall cause the RT to set T/F bit in the status word specified in 4.3.3.5.3 to logic zero until otherwise commanded. The RT shall transmit the status word as specified in 4.3.3.5.3.

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TABLE I. Assigned mode codes.

T/R Bit	Mode Code	Function	Associated Data Word	Broadcast Command Allowed
1	00000	Dynamic bus Control	NO	NO
1	00001	Synchronize	NO	YES
1	00010	Transmit Status Word	NO	NO
1	00011	Initiate Self Test	NO	YES
1	00100	Transmitter Shutdown	NO	YES
1	00101	Override Transmitter Shutdown	NO	YES
1	00110	Inhibit Terminal Flag Bit	NO	YES
1	00111	Override Inhibit Terminal Flag Bit	NO	YES
1	01000	Reset Remote Terminal	NO	YES
1	01001	Reserved	NO	TBD
1	↓	↓	↓	↓
	01111	Reserved	NO	TBD
1	10000	Transmit Vector Word	YES	NO
0	10001	Synchronize	YES	YES
1	10010	Transmit Last Command	YES	NO
1	10011	Transmit BIT Word	YES	NO
0	10100	Selected Transmitter Shutdown	YES	YES
0	10101	Override Selected Transmitter Shutdown	YES	YES
1 or 0	10110	Reserved	YES	TBD
1 or 0	↓	↓	↓	↓
	11111	Reserved	YES	TBD

NOTE: To be determined (TBD)

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5.5.4.3.3.5.1.7.8 Override inhibit T/F bit.

This command shall cause the RT to override the inhibit T/F bit specified in 4.3.3.5.1.7.7. The RT shall transmit the status word as specified in 4.3.3.5.3.

5.5.4.3.3.5.1.7.9 Reset remote terminal.

This command shall be used to reset the RT to power up initialized state. The RT shall first transmit its status word, and then reset.

5.5.4.3.3.5.1.7.10 Reserved mode codes (01001 to 01111).

These mode codes are reserved for future use and shall not be used.

5.5.4.3.3.5.1.7.11 Transmit vector word.

This command shall cause the RT to transmit a status word as specified in 4.3.3.5.3 and a data word containing service request information.

5.5.4.3.3.5.1.7.12 Synchronize (with data word).

The RT shall receive a command word followed by a data word as specified in 4.3.3.5.2. The data word shall contain synchronization information for the RT. After receiving the command and data word, the RT shall transmit the status word as specified in 4.3.3.5.3.

5.5.4.3.3.5.1.7.13 Transmit last command word.

This command shall cause the RT to transmit its status word as specified in 4.3.3.5.3 followed by a single data word which contains bits 4-19 of the last command word, excluding a transmit last command word mode code received by the RT. This mode command shall not alter the state of the RT's status word.

5.5.4.3.3.5.1.7.14 Transmit built-in-test (BIT) word.

This command shall cause the RT to transmit its status word as specified in 4.3.3.5.3 followed by a single data word containing the RT BIT data. This function is intended to supplement the available bits in the status word when the RT hardware is sufficiently complex to warrant its use. The data word, containing the RT BIT data, shall not be altered by the reception of a transmit last command or a transmit status word mode code. This function shall not be used to convey BIT data from the associated subsystem(s).

5.5.4.3.3.5.1.7.15 Selected transmitter shutdown.

This command shall cause the RT to disable the transmitter associated with a specified redundant data bus. The command is designed for use with systems employing more than two redundant buses. The transmitter that is to be disabled shall be identified in the data word following the command word in the format as specified in 4.3.3.5.2. The RT shall not comply with a command to shut down a transmitter on the bus from which this command is received. In all cases, the RT shall respond with a status word as specified in 4.3.3.5.3.

5.5.4.3.3.5.1.7.16 Override selected transmitter shutdown.

This command shall cause the RT to enable a transmitter which was previously disabled. The command is designed for use with systems employing more than two redundant buses. The transmitter that is to be enabled shall be identified in the data word following the command word in the format as specified in 4.3.3.5.2. The RT shall not

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comply with a command to enable a transmitter on the bus from which this command is received. In all cases, the RT shall respond with a status word as specified in 4.3.3.5.3.

5.5.4.3.3.5.1.7.17 Reserved mode codes (10110 to 11111).

These mode codes are reserved for future use and shall not be used.

5.5.4.3.3.5.2 Data word.

A data word shall be comprised of a sync waveform, data bits, and a parity bit (see Figure 3).

5.5.4.3.3.5.2.1 Sync.

The data sync waveform shall be an invalid Manchester waveform as shown on Figure 5. The width shall be three bit times, with the waveform being negative for the first one and one-half bit times, and then positive for the following one and one-half bit times. Note that if the bits preceding and following the sync are logic ones, then the apparent width of the sync waveform will be increased to four bit times.

5.5.4.3.3.5.2.2 Data.

The sixteen bits following the sync shall be utilized for data transmission as specified in 4.3.2.

5.5.4.3.3.5.2.3 Parity.

The last bit shall be utilized for parity as specified in 4.3.3.5.1.6.

5.5.4.3.3.5.3 Status word.

A status word shall be comprised of a sync waveform, RT address, message error bit, instrumentation bit, service request bit, three reserved bits, broadcast command received bit, busy bit, subsystem flag bit, dynamic bus control acceptance bit, terminal flag bit, and a parity bit. For optional broadcast operation, transmission of the status word shall be suppressed as specified in 4.3.3.6.7.

5.5.4.3.3.5.3.1 Sync.

The status sync waveform shall be as specified in 4.3.3.5.1.1.

5.5.4.3.3.5.3.2 RT address.

The next five bits following the sync shall contain the address of the RT which is transmitting the status word as defined in 4.3.3.5.1.2.

5.5.4.3.3.5.3.3 Message error bit.

The status word bit at bit time nine (see Figure 3) shall be utilized to indicate that one or more of the data words associated with the preceding receive command word from the bus controller has failed to pass the RT's validity tests as specified in 4.4.1.1. The bit shall also be set under the conditions specified in 4.4.1.2., 4.4.3.4 and 4.4.3.6. A logic one shall indicate the presence of a message error, and a logic zero shall show its absence. All RT's shall implement the message error bit.

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5.5.4.3.3.5.3.4 Instrumentation bit.

The status word at bit time ten (see Figure 3) shall be reserved for the instrumentation bit and shall always be a logic zero. This bit is intended to be used in conjunction with a logic one in bit time ten of the command word to distinguish between a command word and a status word. The use of the instrumentation bit is optional.

5.5.4.3.3.5.3.5 Service request bit.

The status word bit at bit time eleven (see Figure 3) shall be reserved for the service request bit. The use of the bit is optional. This bit when used, shall indicate the need for the bus controller to take specific predefined actions relative to either the RT or associated subsystem. Multiple subsystems, interfaced to a single RT, which individually require a service request signal shall logically OR their individual signals into the single status word bit. In the event this logical OR is performed, then the designer must make provisions in a separate data word to identify the specific requesting subsystem. The service request bit is intended to be used only to trigger data transfer operations which take place on an exception rather than periodic basis. A logic one shall indicate the presence of a service request, and a logic zero its absence. If this function is not implemented, the bit shall be set to zero.

5.5.4.3.3.5.3.6 Reserved status bits.

The status word bits at bit times twelve through fourteen are reserved for future use and shall not be used. These bits shall be set to logic zero.

5.5.4.3.3.5.3.7 Broadcast command received bit.

The status word at bit time fifteen shall be set to a logic one to indicate that the preceding valid command word was a broadcast command and a logic zero shall show it was not a broadcast command. If the broadcast command option is not used, this bit shall be set to a logic zero.

5.5.4.3.3.5.3.8 Busy bit.

The status word bit at bit time sixteen (see Figure 3) shall be reserved for the busy bit. The use of this bit is optional. This bit, when used, shall indicate that the RT or subsystem is unable to move data to or from the subsystem in compliance with the bus controller's command. A logic one shall indicate the presence of a busy condition, and a logic zero its absence. In the event the busy bit is set in response to a transmit command, then the RT shall transmit its status word only. If this function is not implemented, the bit shall be set to logic zero.

5.5.4.3.3.5.3.9 Subsystem flag bit.

The status word bit at bit time seventeen (see Figure 3) shall be reserved for the subsystem flag bit. The use of this bit is optional. This bit, when used, shall flag a subsystem fault condition, and alert the bus controller to potentially invalid data. Multiple subsystems, interfaced to a single RT, which individually require a subsystem flag bit signal shall logically OR their individual signals into the single status word bit. In the event the logical OR is performed, then the designer must take provisions in a separate data word to identify the specific reporting subsystem. A logic one shall indicate the presence of the flag, and a logic zero its absence. If not used, this bit shall be set to logic zero.

5.5.4.3.3.5.3.10 Dynamic bus control acceptance bit.

The status word bit at bit time eighteen (see Figure 3) shall be reserved for the acceptance of dynamic bus control. This bit shall be used if the RT implements the optional dynamic bus control function. This bit, when used, shall indicate acceptance or rejection of a dynamic bus control offer as specified in 4.3.3.5.1.7.1. A logic one shall indicate acceptance of control, and a logic zero shall indicate rejection of control. If this function is not used, this bit shall be set to a logic zero.

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5.5.4.3.3.5.3.11 Terminal flag bit.

The status word bit at bit time nineteen (see Figure 3) shall be reserved for the terminal flag function. The use of this bit is optional. This bit, when used, shall flag a RT fault condition. A logic one shall indicate the presence of the flag, and logic zero, its absence. If not used, this bit shall be set to logic zero.

5.5.4.3.3.5.3.12 Parity bit.

The least significant bit in the status word shall be utilized for parity as specified in 4.3.3.5.1.6.

5.5.4.3.3.5.4 Status word reset.

The status word bit, with the exception of the address, shall be set to logic zero after a valid command word is received by the RT with the exception as specified in 4.3.3.5.1.7. If the conditions which caused bits in the status word to be set (e.g., terminal flag) continue after the bits are reset to logic zero, then the affected status word bit shall be again set, and then transmitted on the bus as required.

5.5.4.3.3.6 Message formats.

The messages transmitted on the data bus shall be in accordance with the formats on Figure 6 and Figure 7. The maximum and minimum response times shall be as stated in 4.3.3.7 and 4.3.3.8. No message formats, other than those defined herein, shall be used on the bus.

5.5.4.3.3.6.1 Bus controller to remote terminal transfers.

The bus controller shall issue a receive command followed by the specified number of data words. The RT shall, after message validation, transmit a status word back to the controller. The command and data words shall be transmitted in a contiguous fashion with no interword gaps.

5.5.4.3.3.6.2 Remote terminal to bus controller transfers.

The bus controller shall issue a transmit command to the RT. The RT shall, after command word validation, transmit a status word back to the bus controller, followed by the specified number of data words. The status and data words shall be transmitted in a contiguous fashion with no interword gaps.

5.5.4.3.3.6.3 Remote terminal to remote terminal transfers.

The bus controller shall issue a receive command to RT A followed contiguously by a transmit command to RT B. RT B shall, after command validation, transmit a status word followed by the specified number of data words. The status and data words shall be transmitted in a contiguous fashion with no gap. At the conclusion of the data transmission by RT B, RT A shall transmit a status word within the specified time period.

5.5.4.3.3.6.4 Mode command without data word.

The bus controller shall issue a transmit command to the RT using a mode code specified in Table I. The RT shall, after command word validation, transmit a status word.

5.5.4.3.3.6.5 Mode command with data word (transmit).

The bus controller shall issue a transmit command to the RT using a mode code specified in Table I. The RT shall, after command word validation, transmit a status word followed by one data word. The status word and data word shall be transmitted in a contiguous fashion with no gap.

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5.5.4.3.3.6.6 Mode command with data word (receive).

The bus controller shall issue a receive command to the RT using a mode code specified in Table I, followed by one data word. The command word and data word shall be transmitted in a contiguous fashion with no gap. The RT shall, after command and data word validation, transmit a status word back to the controller.

5.5.4.3.3.6.7 Optional broadcast command.

See 10.6 for additional information on the use of the broadcast command.

5.5.4.3.3.6.7.1 Bus controller to remoter terminal(s) transfer (broadcast).

The bus controller shall issue a receive command word with 11111 in the RT address field followed by the specified number of data words. The command word and data words shall be transmitted in a contiguous fashion with no gap. The RT(s) with the broadcast option shall after message validation, set the broadcast command received bit in the status word as specified in 4.3.3.5.3.7 and shall not transmit the status word.

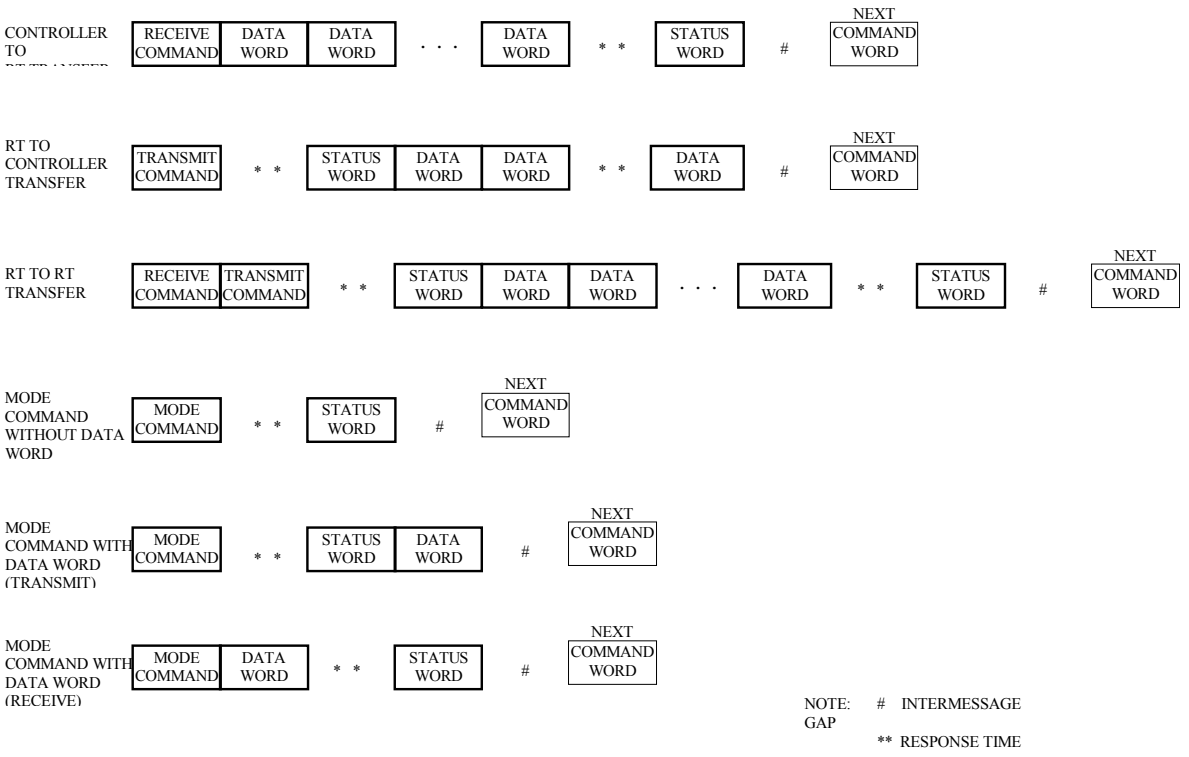


FIGURE 6. Information transfer formats.

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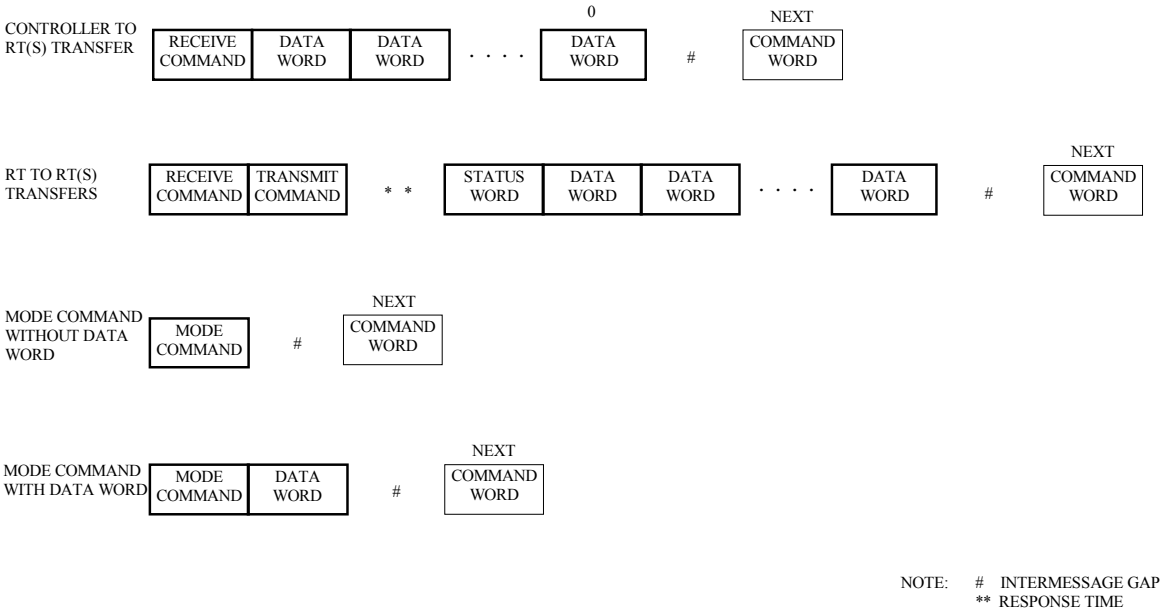


FIGURE 7. Broadcast information transfer formats.

5.5.4.3.3.6.7.2 Remote terminal to remote terminal(s) transfers (broadcast).

The bus controller shall issue a receive command word with 11111 in the RT address field followed by a transmit command to RT A using the RT's address. RT A shall, after command word validation, transmit a status word followed by the specified number of data words. The status and data words shall be transmitted in a contiguous fashion with no gap. The RT(s) with the broadcast option, excluding RT A, shall after message validation, set the broadcast received bit in the status word as specified in 4.3.3.5.3.7 and shall not transmit the status word.

5.5.4.3.3.6.7.3 Mode command without data word (broadcast).

The bus controller shall issue a transmit command word with 11111 in the RT address field, and a mode code specified in Table I. The RT(s) with the broadcast option shall after command word validation, set the broadcast received bit in the status word as specified in 4.3.3.5.3.7 and shall not transmit the status word.

5.5.4.3.3.6.7.4 Mode command with data word (broadcast).

The bus controller shall issue a receive command word with 11111 in the RT address field and a mode code specified in Table I, followed by one data word. The command word and data word shall be transmitted in a contiguous fashion with no gap. The RT(s) with the broadcast option shall after message validation, set the broadcast received bit in the status word as specified in 4.3.3.5.3.7 and shall not transmit the status word.

5.5.4.3.3.7 Intermessage gap.

The bus controller shall provide a minimum gap time of 4.0 microseconds (μs) between messages as shown on Figure 6 and Figure 7. This time period, shown as T on Figure 8, is measured at point A of the bus controller as shown on Figure 9 or Figure 10. The time is measured from the mid-bit zero crossing of the last bit of the preceding message to mid-zero crossing of the next command word sync.

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5.5.4.3.3.8 Response time.

The RT shall respond, in accordance with 4.3.3.6, to a valid command word within the time period of 4.0 to 12.0 μs. This time period, shown as T on Figure 8, is measured at point A of the RT as shown on Figure 9 or Figure 10. The time is measured from the mid bit zero crossing of the last word specified in 4.3.3.6 and as shown on Figure 6 and Figure 7 to the mid-zero crossing of the status word sync.

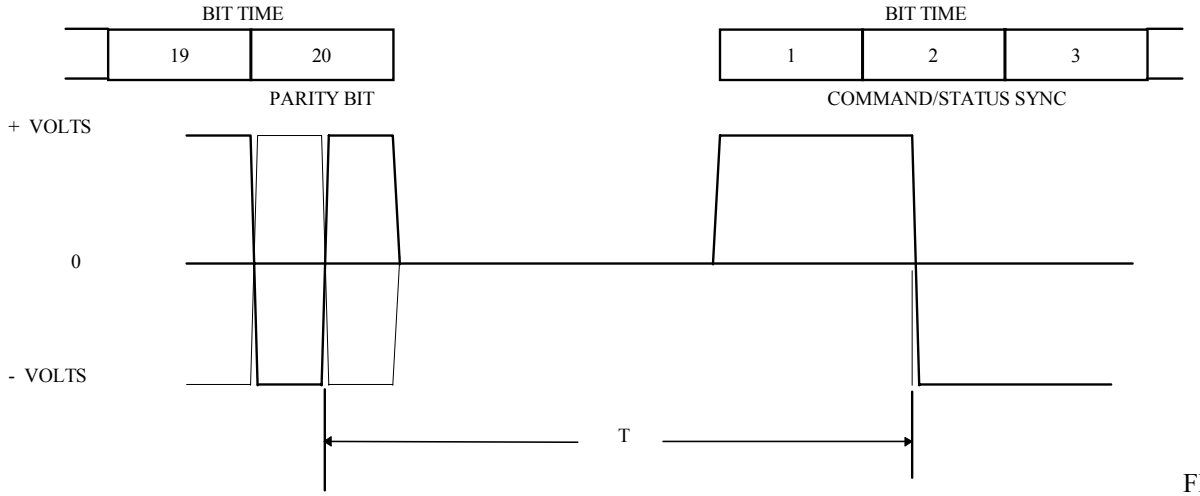
5.5.4.3.3.9 Minimum no-response time-out.

The minimum time that a terminal shall wait before considering that a response as specified in 4.3.3.8 has not occurred shall be 14.0 μs. The time is measured from the mid-bit zero crossing of the last bit of the last word to the mid-zero crossing of the expected status word sync at point A of the terminal as shown in Figure 9 or Figure 10.

5.5.4.4 Terminal operation.

5.5.4.4.1 Common operation.

Terminals shall have common operating capabilities as specified in the following paragraphs.



FI

FIGURE 8. Intermessage gap and response time.

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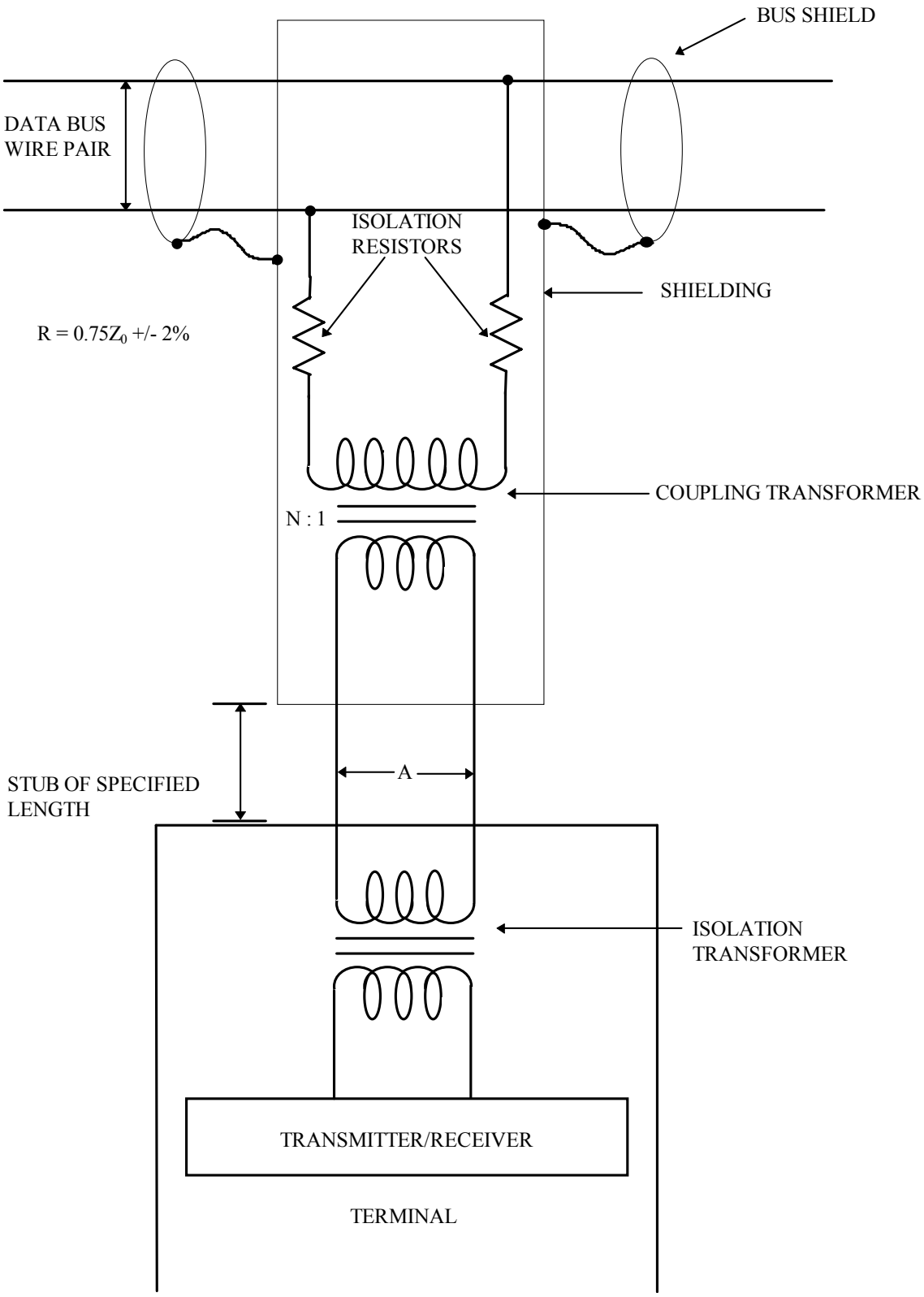


FIGURE 9. Data bus interface using transformer coupling.

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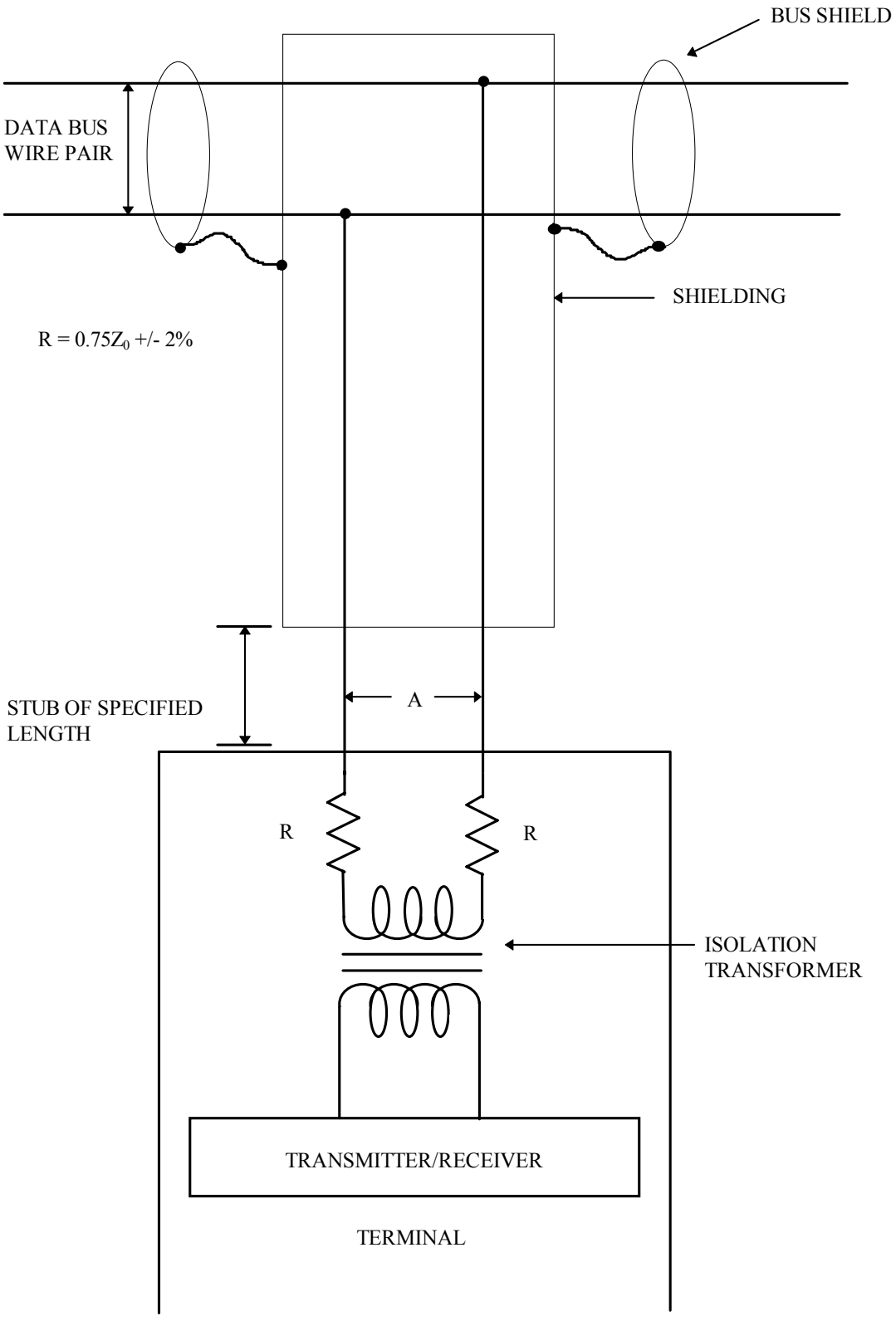


FIGURE 10. Data interface using direct coupling.

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5.5.4.4.1.1 Word validation.

The terminal shall insure that each word conforms to the following minimum criteria:

a. The word begins with valid sync field.

b. The bits are in a valid Manchester II code.

c. The information field has 16 bits plus parity.

d. The word parity is odd.

When a word fails to conform to the preceding criteria, the word shall be considered invalid.

5.5.4.4.1.2 Transmission continuity.

The terminal shall verify that the message is contiguous as defined in 4.3.3.6. Improperly timed data syncs shall be considered a message error.

5.5.4.4.1.3 Terminal fail-safe.

The terminal shall contain a hardware implemented time-out to preclude a signal transmission of greater than 800.0 μs. This hardware shall not preclude a correct transmission in response to a command. Reset of this time-out function shall be performed by the reception of a valid command on the bus on which the time-out has occurred.

5.5.4.4.2 Bus controller operation.

A terminal operating as a bus controller shall be responsible for sending data bus commands, participating in data transfers, receiving status responses, and monitoring system status as defined in this standard. The bus controller function may be embodied as either a stand-alone terminal, whose sole function is to control the data bus(s), or contained within a subsystem. Only one terminal shall be in active control of a data bus at any one time.

5.5.4.4.3 Remote terminal.

5.5.4.4.3.1 Operation.

A remote terminal (RT) shall operate in response to valid commands received from the bus controller. The RT shall accept a command word as valid when the command word meets the criteria of 4.4.1.1, and the command word contains a terminal address which matches the RT address or an address of 11111, if the RT has the broadcast option. No combination of RT address bits, T/R bit, subaddress/mode bits, and data word count/mode code bits of a command word shall result in invalid transmissions by the RT. Subsequent valid commands shall be properly responded to by the RT.

5.5.4.4.3.2 Superseding valid commands.

The RT shall be capable of receiving a command word on the data bus after the minimum intermessage gap time as specified in 4.3.3.7 has been exceeded, when the RT is not in the time period T as specified in 4.3.3.8 prior to the transmission of a status word, and when it is not transmitting on that data bus. A second valid command word sent to an RT shall take precedence over the previous command. The RT shall respond to the second valid command as specified in 4.3.3.8.

5.5.4.4.3.3 Invalid commands.

A remote terminal shall not respond to a command word which fails to meet the criteria specified in 4.4.3.1.

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5.5.4.4.3.4 Illegal command.

An illegal command is a valid command as specified in 4.4.3.1, where the bits in the subaddress/mode field, data word count/mode code field, and the T/R bit indicate a mode command, subaddress, or word count that has not been implemented in the RT. It is the responsibility of the bus controller to assure that no illegal commands are sent out. The RT designer has the option of monitoring for illegal commands. If an RT that is designed with this option detects an illegal command and the proper number of contiguous valid data words as specified by the illegal command word, it shall respond with a status word only, setting the message error bit, and not use the information received.

5.5.4.4.3.5 Valid data reception.

The remote terminal shall respond with a status word when a valid command word and the proper number of contiguous valid data words are received, or a single valid data word associated with a mode code is received. Each data word shall meet the criteria specified in 4.4.1.1.

5.5.4.4.3.6 Invalid data reception.

Any data word(s) associated with a valid receive command that does not meet the criteria specified in 4.4.1.1 and 4.4.1.2 or an error in the data word count shall cause the remote terminal to set the message error bit in the status word to a logic one and suppress the transmission of the status word. If a message error has occurred, then the entire message shall be considered invalid.

5.5.4.4.4 Bus monitor operation.

A terminal operating as a bus monitor shall receive bus traffic and extract selected information. While operating as a bus monitor, the terminal shall not respond to any messages except one containing its own unique address if one is assigned. All information obtained while acting as a bus monitor shall be strictly used for off-line applications (e.g., flight test recording, maintenance recording or mission analysis) or to provide the back-up bus controller sufficient information to take over as the bus controller.

5.5.4.5 Hardware characteristics.

5.5.4.5.1 Data bus characteristics

5.5.4.5.1.1 Cable.

The cable used for the main bus and all stubs shall be a two conductor, twisted, shielded, jacketed cable. The wire-to-wire distributed capacitance shall not exceed 30.0 picofarads per foot. The cables shall be formed with not less than four twists per foot where a twist is defined as a 360 degree rotation of the wire pairs; and, the cable shield shall provide a minimum of 75 percent coverage.

5.5.4.5.1.2 Characteristic impedance.

The nominal characteristic impedance of the cable (Z_0) shall be within the range of 70.0 ohms to 85.0 ohms at a sinusoidal frequency of 1.0 megahertz (MHz).

5.5.4.5.1.3 Cable attenuation.

At the frequency of 4.5.1.2, the cable power loss shall not exceed 1.5 decibels (dB) / 100 feet (ft).

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5.5.4.5.1.4 Cable termination.

The two ends of the cable shall be terminated with a resistance, equal to the selected cable nominal characteristic impedance (Z_0) ± 2.0 percent.

5.5.4.5.1.5 Cable stub requirements.

The cable shall be coupled to the terminal as shown on Figure 9 or Figure 10. The use of long stubs is discouraged, and the length of a stub should be minimized. However, if installation requirements dictate, stub lengths exceeding those lengths specified in 4.5.1.5.1 and 4.5.1.5.2 are permissible.

5.5.4.5.1.5.1 Transformer Coupled stubs.

The length of a Transformer Coupled stub should not exceed 20 feet. If a Transformer Coupled stub is used, then the following shall apply.

5.5.4.5.1.5.1.1 Coupling transformer.

A coupling transformer, as shown on Figure 9, shall be required. This transformer shall have a turns ratio of 1:1.41 ± 3.0 percent, with the higher turns on the isolation resistor side of the stub.

5.5.4.5.1.5.1.1.1 Transformer input impedance.

The open circuit impedance as seen at point B on Figure 11 shall be greater than 3000 ohms over the frequency range of 75.0 kilohertz (kHz) to 1.0 megahertz (MHz), when measured with a 1.0 V root-mean-square (RMS) sin wave.

5.5.4.5.1.5.1.1.2 Transformer waveform integrity.

The droop of the transformer using the test configuration shown of Figure 11 at point B, shall not exceed 20.0 percent. Overshoot and ringing as measured at point B shall be less than ± 1.0 V peak. For this test, R shall equal 360.0 ohms ± 5.0 percent and the input A of Figure 11 shall be a 250.0 kHz square wave, 27.0 V peak-to-peak, with a rise and fall time no greater than 100 nanoseconds (ns).

5.5.4.5.1.5.1.1.3 Transformer common mode rejection.

The coupling transformer shall have a common mode rejection ratio greater than 45.0 dB at 1.0 MHz.

5.5.4.5.1.5.1.2 Fault isolation.

An isolation resistor shall be placed in series with each connection to the data bus cable. This resistor shall have a value of 0.75 Z_0 ohms plus or minus 2.0 percent, where Z_0 is the selected cable nominal characteristic impedance. The impedance placed across the data bus cable shall be no less than 1.5 Z_0 ohms for any failure of the coupling transformer, cable stub, or terminal transmitter/receiver.

5.5.4.5.1.5.1.3 Cable coupling.

All coupling transformers and isolation resistors, as specified in 4.5.1.5.1.1 and 4.5.1.5.1.2, shall have continuous shielding which will provide a minimum of 75 percent coverage. The isolation resistors and coupling transformers shall be placed at minimum possible distance from the junction of the stub to the main bus.

5.5.4.5.1.5.1.4 Stub voltage requirements.

Every data bus shall be designed such that all stubs at point A of Figure 9 shall have a peak-to-peak amplitude, line-to-line within the range of 1.0 and 14.0 V for a transmission by any terminal on the bus. This shall include the

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maximum reduction of data bus signal amplitude in the event that one of the terminals has a fault which causes it to reflect a fault impedance specified in 4.5.1.5.1.2 on the data bus. This shall also include the worse case output voltage of the terminals as specified in 4.5.2.1.1.1 and 4.5.2.2.1.1.

5.5.4.5.1.5.2 Direct coupled stubs.

The length of a Direct Coupled stub should not exceed 1 foot. Refer to 10.5 for comments concerning Direct Coupled stubs. If a Direct Coupled stub is used, then the following shall apply.

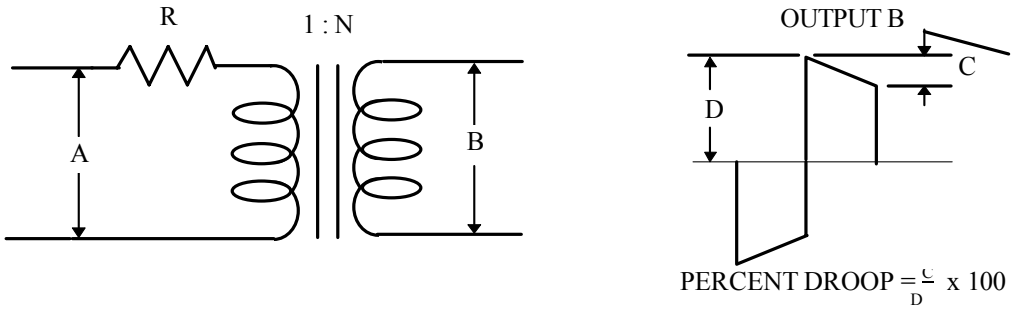


FIGURE 11. Coupling transformer.

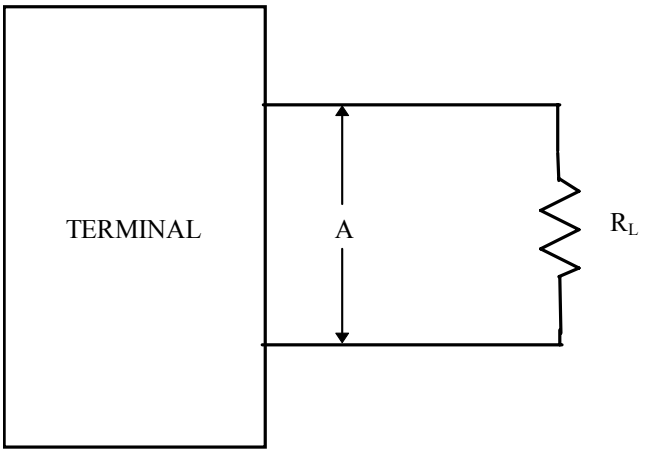


FIGURE 12. Terminal I/O characteristics for transformer coupled and direct coupled stubs.

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5.5.4.5.1.5.2.1 Fault isolation.

An isolation resistor shall be placed in series with each connection to the data bus cable. This resistor shall have a value of 55.0 ohms plus or minus 2.0 percent. The isolation resistors shall be placed within the RT as shown on Figure 10.

5.5.4.5.1.5.2.2 Cable coupling.

All bus-stub junctions shall have continuous shielding which will provide a minimum of 75 percent coverage.

5.5.4.5.1.5.2.3 Stub voltage requirements.

Every data bus shall be designed such that all stubs at point A of Figure 10 shall have a peak-to-peak amplitude, line-to-line within the range of 1.4 and 20.0 V for a transmission by any terminal on the bus. This shall include the maximum reduction of data bus signal amplitude in the event that one of the terminals has a fault which causes it to reflect a fault impedance of 110 ohms on the data bus. This shall also include the severe output voltage of the terminals as specified in 4.5.2.1.1.1 and 4.5.2.2.1.1.

5.5.4.5.1.5.3 Wiring and cabling for EMC.

For purposes of electromagnetic capability (EMC), the wiring and cabling provisions of MIL-STD-464 shall apply.

5.5.4.5.2 Terminal characteristics.

5.5.4.5.2.1 Terminals with Transformer Coupled stubs.

5.5.4.5.2.1.1 Terminal output characteristics.

The following characteristics shall be measured with R_L, as shown on Figure 12, equal to 70.0 ohms ± 2.0 percent.

5.5.4.5.2.1.1.1 Output voltage levels.

The terminal output voltage levels shall be measured using the test configuration shown on Figure 12. The terminal output voltage shall be within the range of 18.0 to 27.0 V, peak-to-peak, line-to-line, when measured at point A on Figure 12.

5.5.4.5.2.1.1.2 Output waveform.

The waveform, when measure at point A on Figure 12 shall have zero crossing deviations which are equal to, or less than, 25.0 ns from the ideal crossing point, measured with respect to the previous zero crossing (i.e., .5 ± .025 µs, 1.0 ± .025 µs, 1.5 ± .025 µs, and 2.0 ± .025 µs). The rise and fall time of this waveform shall be from 100.0 to 300.0 ns when measured from levels of 10 to 90 percent of full waveform peak-to-peak, line-to-line, voltage as shown on Figure 13. Any distortion of the waveform including overshoot and ringing shall not exceed ± 900.0 millivolts (mV) peak, line-to-line, as measured at point A, Figure 12.

5.5.4.5.2.1.1.3 Output noise.

Any noise transmitted when the terminal is receiving or has power removed, shall not exceed a value of 14.0 mV, RMS, line-to-line, as measured at point A, Figure 12.

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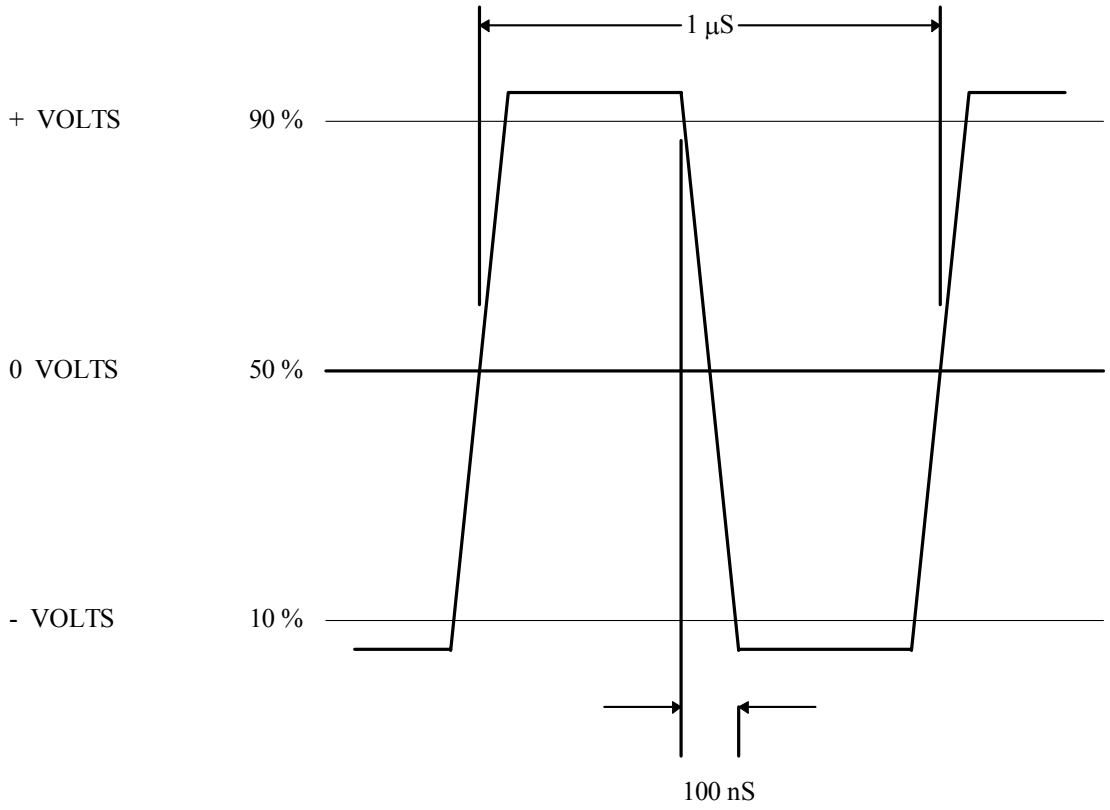


FIGURE 13. Output waveform.

5.5.4.5.2.1.1.4 Output symmetry.

From the time beginning 2.5 µs after the mid-bit crossing of the parity bit of the last word transmitted by a terminal, the maximum voltage at point A of Figure 12 shall be no greater than ± 250.0 mV peak, line-to-line. This shall be tested with the terminal transmitting the maximum number of words it is designed to transmit, up to 33. This test shall be run six times with each word in a contiguous block of words having the same bit pattern. The six word contents that shall be used are 8000_{16}, $7FFF_{16}$, 0000_{16}, $FFFF_{16}$, 5555_{16}, $AAAA_{16}$. The output of the terminal shall be as specified in 4.5.2.1.1.1 and 4.5.2.1.1.2.

5.5.4.5.2.1.2 Terminal input characteristics.

The following characteristics shall be measured independently.

5.5.4.5.2.1.2.1 Input waveform compatibility.

The terminal shall be capable of receiving and operating with the incoming signals specified herein, and shall accept waveform varying from square wave to sine wave with a maximum zero crossing deviation from the ideal with respect to the previous zero crossing of ± 150 ns, (i.e., 2.0 ± .15 µs, 1.5 ± .15 µs, 1.0 ± .15 µs, .5 ± .15 µs). The terminal shall respond to an input signal whose peak-to-peak amplitude, line-to-line, is within the range of .86 to 14.0 V. The terminal shall not respond to an input signal whose peak-to-peak amplitude, line-to-line, is within the range of 0.0 to .20 V. The voltages are measured at point A on Figure 9.

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5.5.4.5.2.1.2.2 Common mode rejections.

Any signals from direct current (DC) to 2.0 MHz, with amplitudes equal to or less than ± 10.0 V peak, line-to-ground, measured at point A on Figure 9, shall not degrade the performance of the receiver.

5.5.4.5.2.1.2.3 Input impedance.

The magnitude of the terminal input impedance, when the RT is not transmitting, or has power removed, shall be a minimum of 1000.0 ohms within the frequency range of 75.0 kHz to 1.0 MHz. This impedance is that measured line-to-line at point A on Figure 9.

5.5.4.5.2.1.2.4 Noise rejection.

The terminal shall exhibit a maximum word error rate of one part in 10^7, on all words received by the terminal, after validation checks specified in 4.4, when operating in the presence of AWGN distributed over a bandwidth of 1.0 kHz to 4.0 MHz at an RMS amplitude of 140 mV. A word error shall include any fault which causes the message error bit to be set in the terminal's status word, or one which causes a terminal to not respond to a valid command. The word error rate shall be measured with a 2.1 V peak-to-peak, line-to-line, input to the terminal as measured at point A on Figure 9. The noise tests shall be run continuously until, for a particular number of failures, the number of words received by the terminal, including both command and data words, exceeds the required number for acceptance of the terminal, or is less than the required number for rejection of the terminal, as specified in Table II. All data words used in the tests shall contain random bit patterns. These bit patterns shall be unique for each data word in a message, and shall change randomly from message to message.

5.5.4.5.2.2 Terminals with Direct Coupled stubs.

5.5.4.5.2.2.1 Terminal output characteristics.

The following characteristics shall be measured with R_L, as shown on Figure 12, equal to 35.0 ohms ± 2.0 percent.

5.5.4.5.2.2.1.1 Output voltage levels.

The terminal output voltage levels shall be measured using the test configuration shown on Figure 12. The terminal output voltage shall be within the range of 6.0 to 9.0 V, peak-to-peak, line-to-line, when measured at point A on Figure 12.

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TABLE II. Criteria for acceptance or rejection for the noise rejection test.

No. of Errors	TOTAL WORDS RECEIVED BY THE TERMINAL (in multiples of 10^7) Reject (Equal or less)	Accept (Equal or more)
0	N/A	4.40
1	N/A	5.21
2	N/A	6.02
3	N/A	6.83
4	N/A	7.64
5	N/A	8.45
6	.45	9.27
7	1.26	10.08
8	2.07	10.89
9	2.88	11.70
10	3.69	12.51
11	4.50	13.32
12	5.31	14.13
13	6.12	14.94
14	6.93	15.75
15	7.74	16.56
16	8.55	17.37
17	9.37	18.19
18	10.18	19.00
19	10.99	19.81
20	11.80	20.62
21	12.61	21.43
22	13.42	22.24
23	14.23	23.05
24	15.04	23.86
25	15.85	24.67
26	16.66	25.48
27	17.47	26.29
28	18.29	27.11
29	19.10	27.92
30	19.90	28.73
31	20.72	29.54
32	21.53	30.35
33	22.34	31.16
34	23.15	31.97
35	23.96	32.78
36	24.77	33.00
37	25.58	33.00
38	26.39	33.00
39	27.21	33.00
40	28.02	33.00
41	33.00	N/A

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5.5.4.5.2.2.1.2 Output waveform.

The waveform, when measured at point A on Figure 12, shall have zero crossing deviations which are equal to, or less than, 25.0 ns from the ideal crossing point, measured with respect to the previous zero crossing (i.e., .5 ± .025 µs, 1.0 ± .025 µs, 1.5 ± .025 µs, and 2.0 ± .025 µs). The rise and fall time of this waveform shall be from 100.0 to 300.0 ns when measured from levels of 10 to 90 percent of full waveform peak-to-peak, line-to-line, voltage as shown on Figure 13. Any distortion of the waveform including overshoot and ringing shall not exceed ± 300.0 mV peak, line-to-line, as measured at point A, Figure 12.

5.5.4.5.2.2.1.3 Output noise.

Any noise transmitted when the terminal is receiving or has power removed, shall not exceed a value of 5.0 mV, RMS, line-to-line, as measured at point A on Figure 12.

5.5.4.5.2.2.1.4 Output symmetry.

From the time beginning 2.5 µs after the mid-bit crossing of the parity bit of the last word transmitted by a terminal, the maximum voltage at point A of Figure 12 shall be no greater than ± 90.0 mV peak, line-to-line. This shall be tested with the terminal transmitting the maximum number of words it is designed to transmit, up to 33. This test shall be run six times with each word in a contiguous block of words having the same bit pattern. The six word contents that shall be used are 8000_{16}, $7FFF_{16}$, 0000_{16}, $FFFF_{16}$, 5555_{16}, $AAAA_{16}$. The output of the terminal shall be as specified in 4.5.2.2.1.1 and 4.5.2.2.1.2.

5.5.4.5.2.2.2 Terminal input characteristics.

The following characteristics shall be measured independently.

5.5.4.5.2.2.2.1 Input waveform compatibility.

The terminal shall be capable of receiving and operating with the incoming signals specified herein, and shall accept waveform varying from square wave to sine wave with a maximum zero crossing deviation from the ideal with respect to the previous zero crossing of plus or minus 150 ns, (i.e., 2.0 ± .15 µs, 1.5 ± .15 µs, 1.0 ± .15 µs, .5 ± .15 µs). The terminal shall respond to an input signal whose peak-to-peak amplitude, line-to-line, is within the range of 1.2 to 20.0 V. The terminal shall not respond to an input signal whose peak-to-peak amplitude, line-to-line, is within the range of 0.0 to .28 V. The voltages are measured at point A on Figure 10.

5.5.4.5.2.2.2.2 Common mode rejections.

Any signals from DC to 2.0 MHz, with amplitudes equal to or less than ± 10.0 V peak, line-to-ground, measured at point A on Figure 10, shall not degrade the performance of the receiver.

5.5.4.5.2.2.2.3 Input impedance.

The magnitude of the terminal input impedance, when the RT is not transmitting, or has power removed, shall be a minimum of 2000.0 ohms within the frequency range of 75.0 kHz to 1.0 MHz. This impedance is that measured line-to-line at point A on Figure 10.

5.5.4.5.2.2.2.4 Noise rejection.

The terminal shall exhibit a maximum word error rate of one part in 10^7, on all words received by the terminal, after validation checks specified in 4.4, when operating in the presence of AWGN distributed over a bandwidth of 1.0 kHz to 4.0 MHz at an RMS amplitude of 200 mV. A word error shall include any fault which causes the message error bit to be set in the terminal's status word, or one which causes a terminal to not respond to a valid command. The word error rate shall be measured with a 3.0 V peak-to-peak, line-to-line, input to the terminal as measured at

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point A on Figure 10. The noise tests shall be run continuously until, for a particular number of failures, the number of words received by the terminal, including both command and data words, exceeds the required number for acceptance of the terminal, or is less than the required number for rejection of the terminal, as specified in Table II. All data words used in the tests shall contain random bit patterns. These bit patterns shall be unique for each data word in a message, and shall change randomly from message to message.

5.5.4.6 Redundant data bus requirements.

If redundant data buses are used, the requirements as specified in the following paragraphs shall apply to those data buses.

5.5.4.6.1 Electrical isolation.

All terminals shall have a minimum of 45 dB isolation between data buses. Isolation here means the ratio in dB between the output voltage on the active data bus and the output voltage on the inactive data bus. This shall be measured using the test configuration specified in 4.5.2.1.1 or 4.5.2.2.1 for each data bus. Each data bus shall be alternately activated with all measurements being taken at point A on Figure 12 for each data bus.

5.5.4.6.2 Single event failures.

All data buses shall be routed to minimize the possibility that a single event failure to a data bus shall cause the loss of more than that particular data bus.

5.5.4.6.3 Dual standby redundant data bus.

If a dual redundant data bus is used, then it shall be a dual standby redundant data bus as specified in the following paragraphs.

5.5.4.6.3.1 Data bus activity.

Only one data bus can be active at any given time except as specified in 4.6.3.2.

5.5.4.6.3.2 Superseding valid commands.

If while operating on a command, a terminal receives another valid command, from the other data bus, it shall reset and respond to the new command on the data bus on which the new command is received. The terminal shall respond to the new command as specified in 4.3.3.8.

5.5.5 DETAIL REQUIREMENTS

(Not Applicable)

APPENDIX

5.5.10. General.

The following paragraphs in this appendix are presented in order to discuss certain aspects of the standard in a general sense. They are intended to provide a user of the standard more insight into the aspects discussed.

5.5.10.1 Redundancy.

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It is intended that this standard be used to support rather than to supplant the system design process. However, it has been found, through application experience in various aircraft, that the use of a dual standby redundancy technique is very desirable for use in integrating mission avionics. For this reason, this redundancy scheme is defined in 4.6 of this standard. None the less, the system designer should utilize this standard as the needs of a particular application dictate. The use of redundancy, the degree to which it is implemented, and the form which it takes must be determined on an individual application basis. Figures 10.1 and 10.2 illustrate some possible approaches to dual redundancy. These illustrations are not intended to be inclusive, but rather representative. It should be noted that analogous approaches exist for the triple and quad redundant cases.

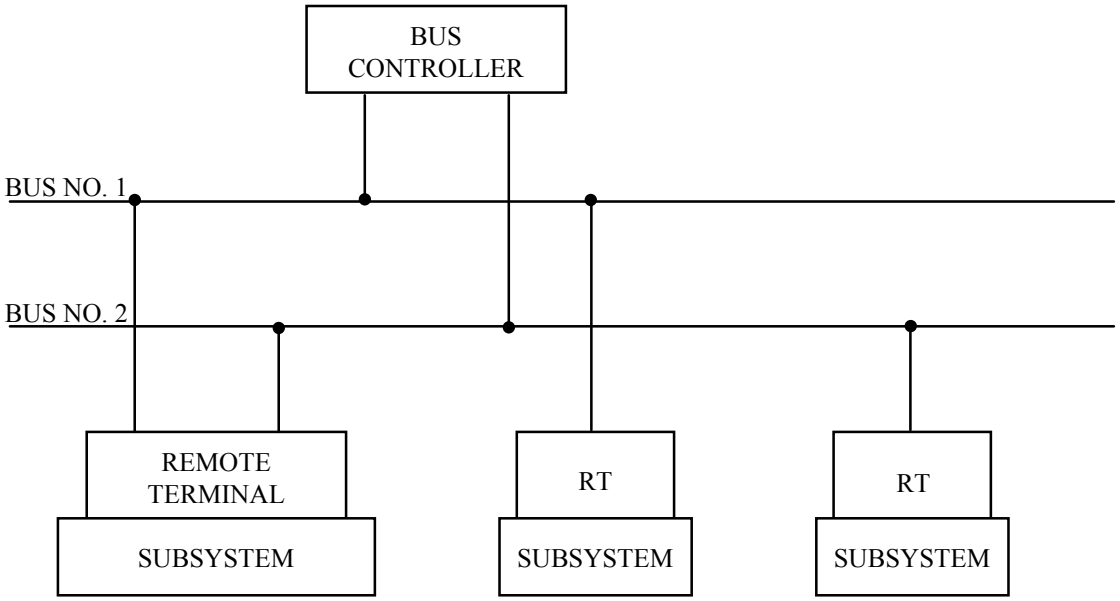
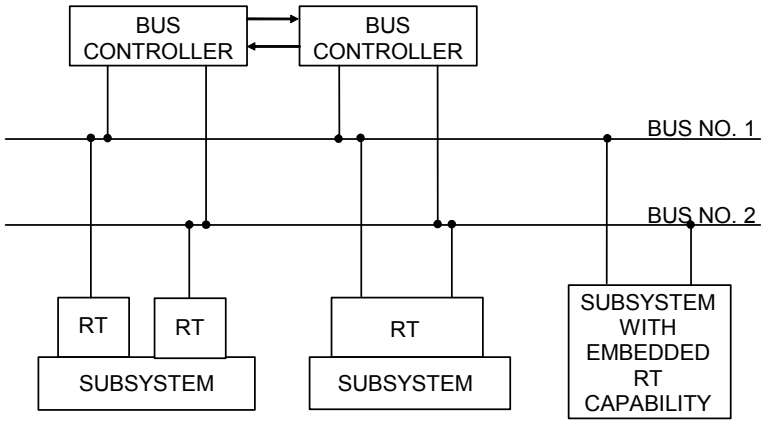


FIGURE 10.1. Illustration of possible redundancy.



NOTE: RT - Remote Terminal

FIGURE 10.2. Illustration of possible redundancy.

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5.5.10.2 Bus controller.

The bus controller is a key part of the data bus system. The functions of the bus controller, in addition to the issuance of commands, must include the constant monitoring of the data bus and the traffic on the bus. It is envisioned that most of the routine minute details of bus monitoring (e.g., parity checking, terminal non-response time-out, etc.) will be embodied in hardware, while the algorithms for bus control and decision making will reside in software. It is also envisioned that, in general, the bus controller will be a general purpose computer with a special input/output (I/O) to interface with the data bus. It is of extreme importance in bus controller design that the bus controller be readily able to accommodate terminals of differing protocol's and status word bits used. Equipment designed to MIL-STD-1553A will be in use for a considerable period of time; thus, bus controllers must be capable of adjusting to their differing needs. It is also important to remember that the bus controller will be the focal point for modification and growth within the multiplex system, and thus the software must be written in such a manner as to permit modification with relative ease.

5.5.10.3 Multiplex selection criteria.

The selection of candidate signals for multiplexing is a function of the particular application involved, and criteria will in general vary from system to system. Obviously, those signals which have bandwidths of 400 Hz or less are prime candidates for inclusion on the bus. It is also obvious that video, audio, and high speed parallel digital signals should be excluded. The area of questionable application is usually between 400 Hz and 3 KHz bandwidth. The transfer of these signals on the data bus will depend heavily upon the loading of the bus in a particular application. The decision must be based on projected future bus needs as well as the current loading. Another class of signals which in general are not suitable for multiplexing are those which can be typified by a low rate (over a mission) but possessing a high priority or urgency. Examples of such signals might be a nuclear event detector output or a missile launch alarm from a warning receiver. Such signals are usually better left hardwired, but they may be accommodated by the multiplex system if a direct connection to the bus controller's interrupt hardware is used to trigger a software action in response to the signal.

5.5.10.4 High reliability requirements.

The use of simple parity for error detection within the multiplex bus system was dictated by a compromise between the need for reliable data transmission, system overhead, and remote terminal simplicity. Theoretical and empirical evidence indicates that an undetected bit error rate of 10^{-12} can be expected from a practical multiplex system built to this standard. If a particular signal requires a bit error rate which is better than that provided by the parity checking, then it is incumbent upon the system designer to provide the reliability within the constraints of the standard or to not include this signal within the multiplex bus system. A possible approach in this case would be to have the signal source and sink provide appropriate error detection and correction encoding/decoding and employ extra data words to transfer the information. Another approach would be to partition the message, transmit a portion at a time, and then verify (by interrogation) the proper transfer of each segment.

5.5.10.5 Stubbing.

Stubbing is the method wherein a separate line is connected between the primary data bus line and a terminal. The direct connection of a stub line causes a mismatch which appears on the waveforms. This mismatch can be reduced by filtering at the receiver and by using bi-phase modulation. Stubs are often employed not only as a convenience in bus layout but as a means of coupling a unit to the line in such a manner that a fault on the stub or terminal will not greatly affect the transmission line operation. In this case, a network is employed in the stub line to provide isolation from the fault. These networks are also used for stubs that are of such length that the mismatch and reflection degrades bus operation. The preferred method of stubbing is to use Transformer Coupled stubs, as defined in 4.5.1.5.1. This method provides the benefits of DC isolation, increased common mode protection, a doubling of effective stub impedance, and fault isolation for the entire stub and terminal. Direct coupled stubs, as defined in 4.5.1.5.2 of this standard, should be avoided if at all possible. Direct coupled stubs provide no DC isolation or common mode rejection for the terminal external to its subsystem. Further, any shorting fault between the subsystems internal isolation resistors (usually on a circuit board) and the main bus junction will cause failure of

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that entire bus. It can be expected that when the Direct Coupled stub length exceeds 1.6 feet, that it will begin to distort the main bus waveforms. Note that this length includes the cable runs internal to a given subsystem.

5.5.10.6 Use of broadcast option.

The use of a broadcast message as defined in 4.3.3.6.7 of this standard represents a significant departure from the basic philosophy of this standard in that it is a message format which does not provide positive closed-loop control of bus traffic. The system designer is strongly encouraged to solve any design problems through the use of the three basic message formats without resorting to use of the broadcast option. If system designers do choose to use the broadcast command, they should carefully consider the potential effects of a missed broadcast message, and the subsequent implications for fault or error recovery design in the remote terminals and bus controllers.

5.5.10.7 Other related documents.

Several documents exist which are related to this standard. MIL-HDBK-1553 describes implementation practices for this standard and other related data. This standard is embodied in or referenced by the following international documents: NATO STANAG 3838, ASCC Air Standard 50/2, and UK DEF STAN 00-18 (PART 2)/Issue 1.

5.5.20. Referenced documents.

Not applicable.

5.5.30. General requirements.

5.5.30.1 Option selection.

This section of the appendix shall select those options required to further define portions of the standard to enhance tri-service interoperability. References in parentheses are to paragraphs in this standard which are affected.

5.5.30.2 Application.

Section 30 of this appendix shall apply to all dual standby redundant applications for the Army, Navy, and Air Force. All Air Force aircraft internal avionics applications shall be dual standby redundant, except where safety critical or flight critical requirements dictate a higher level of redundancy.

5.5.30.3 Unique address (4.3.3.5.1.2).

All remote terminals shall be capable of being assigned any unique address from decimal address 0 (00000) through decimal address 30 (11110). The address shall be established through an external connector, which is part of the system wiring and connects to the remote terminal. Changing the unique address of a remote terminal shall not require the physical modification or manipulation of any part of the remote terminal. The remote terminal shall, as a minimum, determine and validate its address during power-up conditions. No single point failure shall cause a terminal to validate a false address. The remote terminal shall not respond to any messages if it has determined its unique address is not valid.

5.5.30.4 Mode codes (4.3.3.5.1.7).

5.5.30.4.1 Subaddress/mode (4.3.3.5.1.4).

An RT shall have the capability to respond to mode codes with both subaddress/mode of 00000 and 11111. Bus controllers shall have the capability to issue mode commands with both subaddress/mode of 00000 and 11111. The subaddress/mode of 00000 and 11111 shall not convey different information.

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5.5.30.4.2 Required mode codes (4.3.3.5.1.7).

5.5.30.4.2.1 Remote terminal required mode codes.

An RT shall implement the following mode codes as a minimum:

Mode Code	Function
00010	Transmit status word
00100	Transmitter shutdown
00101	Override transmitter shutdown
01000	Reset remote terminal

5.5.30.4.2.2 Bus controller required mode codes.

The bus controller shall have the capability to implement all of the mode codes as defined in 4.3.3.5.1.7. For Air Force applications, the dynamic bus control mode command shall never be issued by the bus controller.

5.5.30.4.3 Reset remote terminal (4.3.3.5.1.7.9).

An RT receiving the reset remote terminal mode code shall respond with a status word as specified in 4.3.3.5.1.7.9 and then reset. While the RT is being reset, the RT shall respond to a valid command with any of the following: no response on either data bus, status word transmitted with the busy bit set, or normal response. If any data is transmitted from the RT while it is being reset, the information content of the data shall be valid. An RT receiving this mode code shall complete the reset function within 5.0 milliseconds following transmission of the status word specified in 4.3.3.5.1.7.9. The time shall be measured from the mid-bit zero crossing of the parity bit of the status word to the mid-sync zero crossing of the command word at point A on Figures 9 and 10.

5.5.30.4.4 Initiate RT self test (4.3.3.5.1.7.4).

If the initiate self test mode command is implemented in the RT, then the RT receiving the initiate self test mode code shall respond with a status word as specified in 4.3.3.5.1.7.4 and then initiate the RT self test function. Subsequent valid commands may terminate the self-test function. While the RT self test is in progress, the RT shall respond to a valid command with any of the following: no response on either data bus, status word transmitted with the busy bit set, or normal response. If any data is transmitted from the RT while it is in self test, the information content of the data shall be valid. An RT receiving this mode code shall complete the self test function and have the results of the self test available within 100.0 milliseconds following transmission of the status word specified in 4.3.3.5.1.7.4. The time shall be measured from the mid-bit zero crossing of the parity bit of the status word to the mid-sync zero crossing of the command word at point A on Figures 9 and 10.

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5.5.30.5 Status word bits (4.3.3.5.3).

5.5.30.5.1 Information content.

The status word transmitted by an RT shall contain valid information at all times, e.g., following RT power up, during initialization, and during normal operation.

5.5.30.5.2 Status bit requirements (4.3.3.5.3).

An RT shall implement the status bits as follows:

Message error bit (4.3.3.5.3.3) - Required

Instrumentation bit (4.3.3.5.3.4) - Always logic zero

Service request bit (4.3.3.5.3.5) - Optional

Reserved status bits (4.3.3.5.3.6) - Always logic zero

Broadcast command received bit (4.3.3.5.3.7) - If the RT implements the broadcast option, then this bit shall be required.

Busy bit (4.3.3.5.3.8) - As required by 30.5.3

Subsystem flag bit (4.3.3.5.3.9) - If an associated subsystem has the capability for self test, then this bit shall be required.

Dynamic bus control acceptance bit (4.3.3.5.3.10) - If the RT implements the dynamic bus control function, then this bit shall be required.

Terminal flag bit (4.3.3.5.3.11) - If an RT has the capability for self test, then this bit shall be required.

5.5.30.5.3 Busy bit (4.3.3.5.3.8).

The existence of busy conditions is discouraged. However, any busy condition, in the RT or the subsystem interface that would affect communication over the bus shall be conveyed via the busy bit. Busy conditions, and thus the setting of the busy bit, shall occur only as a result of particular commands/ messages sent to an RT. Thus for a non-failed RT, the bus controller can, with prior knowledge of the remote terminal characteristics, determine when the remote terminal can become busy and when it will not be busy. However, the RT may also set the busy bit (in addition to setting the terminal flag bit or subsystem flag bit) as a result of failure/fault conditions within the RT/subsystem.

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5.5.30.6 Broadcast (4.3.3.6.7).

The only broadcast commands allowed to be transmitted on the data bus by the bus controller shall be the broadcast mode commands identified in Table I. The broadcast option may be implemented in remote terminals. However, if implemented, the RT shall be capable of distinguishing between a broadcast and a non-broadcast message to the same subaddress for non-mode command messages. The RT address of 11111 is still reserved for broadcast and shall not be used for any other purpose.

5.5.30.7 Data wrap-around (4.3.3.5.1.4).

Remote terminals shall provide a receive subaddress to which one to N data words of any bit pattern can be received. Remote terminals shall provide a transmit subaddress from which a minimum of N data words can be transmitted. N is equal to the maximum word count from the set of all messages defined for the RT. A valid receive message to the data wrap-around receive subaddress followed by a valid transmit command to the data wrap-around transmit subaddress, with the same word count and without any intervening valid commands to that RT, shall cause the RT to respond with each data word having the same bit pattern as the corresponding received data word. A data wrap-around receive and transmit subaddress of 30 (11110) is desired.

5.5.30.8 Message formats (4.3.3.6).

Remote terminals shall, as a minimum, implement the following non-broadcast message formats as defined in 4.3.3.6: RT to BC transfers, BC to RT transfers, RT to RT transfers (receive and transmit), and mode command without data word transfers. For non-broadcast messages, the RT shall not distinguish between data received during a BC to RT transfer or data received during a RT to RT transfer (receive) to the same subaddress. The RT shall not distinguish between data to be transmitted during an RT to BC transfer or data to be transmitted during an RT to RT transfer (transmit) from the same subaddress. Bus controllers shall have the capability to issue all message formats defined in 4.3.3.6.

5.5.30.9 RT to RT validation (4.3.3.9).

For RT to RT transfers, in addition to the validation criteria specified in 4.4.3.6, if a valid receive command is received by the RT and the first data word is received after 57.0 plus or minus 3.0 microseconds, the RT shall consider the message invalid and respond as specified in 4.4.3.6. The time shall be measured from the mid-bit zero crossing of the parity bit of the receive command to the mid-sync zero crossing of the first expected data word at point A as shown on Figures 9 and 10. It is recommended that the receiving RT of an RT to RT transfer verify the proper occurrence of the transmit command word and status word as specified in 4.3.3.6.3.

5.5.30.10 Electrical characteristics (4.5).

5.5.30.10.1 Cable shielding (4.5.1.1).

The cable shield shall provide a minimum of 90.0 percent coverage.

5.5.30.10.2 Shielding (4.5.1).

All cable to connector junctions, cable terminations, and bus-stub junctions shall have continuous 360 degree shielding which shall provide a minimum of 75.0 percent coverage.

5.5.30.10.3 Connector polarity.

For applications that use concentric connectors or inserts for each bus, the center pin of the connector or insert shall be used for the high (positive) Manchester bi-phase signal. The inner ring shall be used for the low (negative) Manchester bi-phase signal.

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5.5.30.10.4 Characteristic impedance (4.5.1.2).

The actual (not nominal) characteristic impedance of the data bus cable shall be within the range of 70.0 ohms to 85.0 ohms at a sinusoidal frequency of 1.0 megahertz.

5.5.30.10.5 Stub coupling (4.5.1.5).

For Navy applications, each terminal shall have both transformer and Direct Coupled stub connections externally available. For Navy systems using these terminals, either transformer or Direct Coupled connections may be used. For Army and Air Force applications, each terminal shall have Transformer Coupled stub connections, but may also have Direct Coupled stub connections. For Army and Air Force systems, only Transformer Coupled stub connections shall be used. Unused terminal connections shall have a minimum of 75 percent shielding coverage.

5.5.30.10.6 Power on/off noise.

A terminal shall limit any spurious output during a power-up or power-down sequence. The maximum allowable output noise amplitude shall be ±250 mV peak, line-to-line for Transformer Coupled stubs and ±90 mV peak, line-to-line for Direct Coupled stubs, measured at point A of Figure 12.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Patent Notice.

Attention is called to the possibility that implementation of the Standard may require use of subject matter covered by patent rights. By publication of this standard, The Department of the Air Force takes no position with respect to the existence or validity of any patent rights in connection therewith. Furthermore, the Department of the Air Force will not be responsible for identifying patents for which a license may be required or for conducting inquiries into the legal validity or scope of those patents that are brought to its attention.

6.2 Subject term (key word) listing.

Multi carrier waveform
Open systems interconnection
Trapezoidal waveform

6.3 Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:

Air Force – 11

Preparing activity:

Air Force - 11

(Project MCCR-2006-002)

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